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When and How Long? The Readout-Mediator Angle in Temporal Reasoning

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Abstract

A linear probe can decode a representation almost perfectly and yet be completely irrelevant to how the model uses it. On calendar-date duration reasoning in language models, a $\sin/\cos$ probe recovers day-of-year from a layer's activations, yet ablating its direction has no effect on the model's answers—while ablating a four-dimensional subspace found by Distributed Alignment Search (DAS) at the same layer collapses performance entirely. We measure the angle between these two subspaces—the *readout–mediator angle*—and find it indistinguishable from the angle between two random subspaces (the Haar-uniform null), meaning the probe has learned a direction orthogonal to the model's actual computation. Reverse-engineering the circuit reveals why: attention heads route month-grained context through learned QK offsets at ± 30 and ± 61 days, and MLPs then convert *when* (absolute date) into *how long* (duration)—all downstream of the causal subspace the probe never touches. Sparse-autoencoder decomposition confirms the split: probe-aligned and DAS-aligned features encode semantically disjoint concepts with negligible causal overlap. The dissociation replicates across four scales (1.5–9 B) and two model families, with preliminary evidence on two further domains (spatial displacement, symbolic arithmetic), suggesting that readout–mediator orthogonality is a general failure mode of probe-based interpretability. This directly undermines proposals to deploy probes as runtime safety monitors: the probe can report high confidence on a direction the model has silently abandoned.

1 Introduction

Ask a language model "How many days between March 15th and June 22nd?" and it answers correctly: 99 days. A $\sin$ / $\cos$ Ridge probe at layer 20 decodes both dates from the residual stream with R^2=0.996 [Gurnee and Tegmark, 2024]—exactly the kind of result used to argue that the model *represents* calendar time [Gurnee and Tegmark, 2024, Marks and Tegmark, 2024, Kantamneni and Tegmark, 2025]. But ablating the probe's direction drops accuracy by only 0.6 pp; the model still counts 99 days as if nothing happened. A matched-rank Distributed Alignment Search (DAS) subspace at the same layer tells the opposite story: ablating it collapses accuracy entirely (Fig. 1).

The probe reads the right answer from a direction the model does not compute with. We quantify this by measuring the principal angle between the two subspaces—what we call the *readout–mediator angle*. At $\bar{\theta}$=88°, it matches the expectation for two random subspaces drawn from a Haar-uniform distribution ($\mathbb{E}[\bar{\theta}]$=88.3° at (d, k)=(2304, 2), Prop. 2): the probe carries no more information about the model's computation than a random direction of the same rank. A five-year line of work has questioned whether probe accuracy implies mechanistic relevance [Hewitt and Liang, 2019, Elazar

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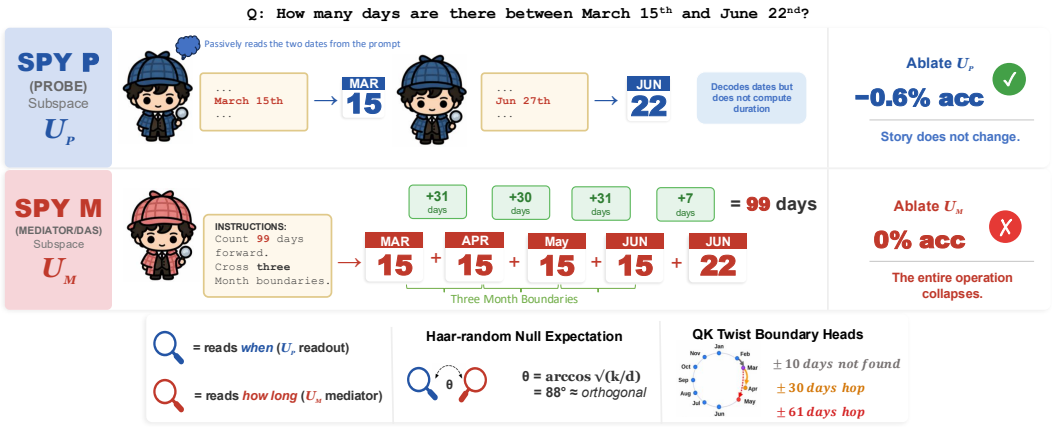


Figure 1: **The readout-mediator dissociation on a duration query (schematic).** Given "How many days between March 15th and June 22nd?", the model deploys two functionally orthogonal subspaces. *Spy P:* the probe subspace $\mathbf{U}_P$ passively decodes both dates; ablating it changes accuracy by -0.6 pp. *Spy M:* the DAS mediator $\mathbf{U}_M$ counts 99 days via month-boundary hops; ablating it collapses accuracy to 0%. The two subspaces are nearly orthogonal ($\bar{\theta}$=88°), matching the Haar-random null—the probe carries no more information about the computation than noise.

et al., 2021, Ravichander et al., 2021, Mueller et al., 2026, 2025, Canby et al., 2025]; the readout–mediator angle provides the missing instrument—a number that says *how far* the probe sits from the computation, paired with a null that says what "far" means.

Reverse-engineering the causal subspace reveals why the two are orthogonal. Boundary attention heads implement QK offsets at ± 30 and ± 61 days—single- and double-month steps that tile any multi-month duration (Fig. 1). MLP layers then execute a two-stage transformation: layers 18–19 read calendar position (*when*); layers 20–25 convert it into duration (*how long*), with MLP SAEs—functionally equivalent to transcoders—showing a monotonic DAS-alignment gradient across this boundary. Sparse-autoencoder features confirm the split at the vocabulary level: probe-aligned features fire on concepts like *month of October*; DAS-aligned features fire on *past 24 hours*. The two feature sets have zero causal overlap (Supp. S50). Temporal feature analysis [Lubana et al., 2026] explains the geometry: the DAS mediator aligns with context-predictable structure (7$\times$ above the Haar null), while the probe sits in the random cloud—duration computation lives in the part of the activation accumulated from context, not the current token (Fig. 4C–D).

The dissociation is not specific to one model, one scale, or one domain. It replicates across four models (1.5–9B), two architecture families, and two further reasoning domains (spatial displacement, symbolic arithmetic)—each at the Haar null angle. On PYTHIA 1.4B, probe R^2=0.956 at step 0: an untrained network "represents" dates by the probe's standard, yet boundary heads, DAS drops, and circularness all emerge only as training proceeds—the probe tracks dimensional capacity, not mechanism learning.

Contributions:

- The readout–mediator angle and Haar-random null, with three propositions linking angle to ablation effect (§3).
- Maximal probe–DAS dissociation across four scales (1.5–9B), two families, three domains—sharpening with scale (§4, 6).
- A full circuit trace: boundary heads, two-stage MLP transcoder chain, disjoint SAE features, and a TFA-based explanation of the orthogonality (§5).
- A six-experiment battery showing the dissociation breaks probe-based safety monitors (§6).

2 The decodability-use gap

The gap between what a probe can decode and what a model causally uses is well documented. Gurnee and Tegmark [2024] decode spatial and temporal coordinates from LLAMA-2 with R^2>0.9, yet note explicitly that this "does not imply the model actually uses these representations." Hernandez et al. [2024] make the gap concrete: they define a faithfulness metric and find that many relations

are probe-accurate but not faithfully decoded at generation time. Several lines of work have tried to narrow the gap—combining probes with causal intervention [Tak et al., 2025, Feng et al., 2025], replacing fixed probes with decoder LLMs [Pan et al., 2026], and questioning the interventions themselves [Grant et al., 2026]—but none provide a single number that says how far the probe direction sits from the causal one, or a null that says what "far" means.

Diagnostic critiques have sharpened the problem without solving it. Control tasks [Hewitt and Liang, 2019], amnesic probing [Elazar et al., 2021], and recent audits [Ravichander et al., 2021, Mueller et al., 2026, 2025, Canby et al., 2025] all question the inferential step from accuracy to mechanism, but stop short of measuring the divergence. Concept-erasure methods—INLP and LEACE [Belrose et al., 2023]—attempt to close the gap by removing the probed direction, yet their erasure subspaces sit within $\sim 1.5^\circ$ of the Haar null (Supp. S41): erasing what the probe finds does not erase what the model uses.

Our circuit-level analysis builds on three families of tools: DAS [Geiger et al., 2024, Sun et al., 2025, Mueller et al., 2025] for identifying causally load-bearing subspaces, activation patching [Nanda et al., 2023, Syed et al., 2024] for tracing information flow, and SAEs [Cunningham et al., 2024, Lieberum et al., 2024] decomposed via NeuronPedia [Lin, 2023] for interpreting features at the vocabulary level. We use MLP SAEs as functional transcoders [Templeton et al., 2024] to track what each MLP layer writes to the residual stream, and temporal feature analysis [Lubana et al., 2026] to decompose activations into context-predictable and novel components—the structural distinction that ultimately explains the orthogonality. Most directly, Gurnee et al. [2026] showed that attention heads implement QK-twist rotations on date manifolds; we take the next step and ask which of those projections are causally load-bearing and which are statistical shadows. Extended related work is in Supp. S10.

3 The readout-mediator angle: measurement and theory

Two questions, two subspaces. The decodability-use gap arises because probes and causal methods answer different questions about the same layer. Given a task property z and a layer L, a *probe subspace* $U_P \in \mathbb{R}^{k\times d}$ is the top-k span of a circular Ridge regressor trained on cached activations—it asks *where is the information?* A *DAS subspace* [Geiger et al., 2024] is found by parametrising U via QR-decomposition of a trainable matrix and minimizing task NLL while an ablation hook $x \mapsto x - U^\top U x$ zeros U on every forward pass—it asks *where is the computation vulnerable?* Same layer, same rank k, different optimization targets: one isolates what is *readable*, the other what is *load-bearing* (notation summary in Supp. S54). The *readout–mediator angle* $\bar{\theta}(U_P, U_M) = \frac{1}{k}\sum_i \arccos \sigma_i(U_P U_M^\top)$ measures the distance between these two answers.

The next three propositions establish why this angle should generically be large (Prop. 1), what the null expectation is (Prop. 2), and how the angle controls the observable we actually measure—ablation effect (Prop. 3).

Proposition 1 (Readout-mediator orthogonality, informal)**.** *Let $f{:}\mathbb{R}^d \to \mathbb{R}$ be a differentiable task output and $z(x)$ a scalar correlated with x. The probe direction maximizes $I(u^\top x; z)$—a* second-moment *quantity in x. The mediator maximizes $\mathbb{E}|f(x) - f(x - uu^\top x)|$—a* first-moment *quantity in $\nabla_x f$. The two coincide only when $\nabla_x f \propto u_P$. Otherwise they are generically distinct.*

Proof sketch. The probe solves a Rayleigh quotient $\max_{\|u\|=1}(u^\top \mathbf{c})^2/(u^\top \Sigma u)$ with $\mathbf{c} = \mathbb{E}[xz]$, yielding $u_P = \Sigma^{-1}\mathbf{c}/\|\cdot\|$—set by the data covariance. A first-order Taylor expansion of the ablation effect gives $u_M = \arg\max u^\top G u$ where $G = \mathbb{E}[\nabla_x f \, \nabla_x f^\top]$ is the gradient covariance—set by the network's output sensitivity. Coincidence requires $\Sigma^{-1}\mathbf{c}$ to be the top eigenvector of G: a non-generic spectral alignment between data geometry and network geometry that deep networks have no structural reason to satisfy (full proof in Supp. S22).

Empirical test of Prop. 1. If the mediator aligns with the first moment of $\nabla_x f$, computing the gradient subspace directly should recover a direction closer to DAS than to the probe. We verify this on GEMMA 2 2B by computing $g_i = \nabla_{h_{L^\star}} \mathrm{NLL}(y^\star | x_i)$ for each prompt. The gradient subspace sits 2.3° closer to the mediator than to the Haar null ($\bar{\theta} = 85.3^\circ$), while its angle to the probe is at null (88.9°). The gradient leans toward the mediator but disperses across effective rank 76; DAS distills the $k = 4$ causal core from this diffuse signal (Supp. S2).

Prop. 1 tells us to expect divergence between probe and mediator—but not how much. In high dimensions, the answer turns out to be stark: any two low-rank subspaces are nearly orthogonal by default.

Proposition 2 (Null angle between random subspaces)**.** *For independent $k\times d$ Stiefel-uniform matrices U, V, the principal-angle cosines follow the Jacobi ensemble $J(k, k, d-k)$ with $\mathbb{E}\sum_i \cos^2\theta_i = k^2/d$: each cosine-squared concentrates at k/d, the Johnson–Lindenstrauss rate for a random k-plane projection in $\mathbb{R}^d$ (exact; MC-verified in Supp. S20). At $(d, k)=(2304, 2)$, $\mathbb{E}[\bar{\theta}]=88.3°$.*

Proof sketch. By Haar rotation invariance, fix V as the first k identity rows; the singular values of $UV^\top$ then follow the Jacobi ensemble $J(k, k, d-k)$. The Collins–Matsumoto trace identity gives $\mathbb{E}\sum_i \cos^2\theta_i = k^2/d$ exactly. The Delta method converts to angle space: $\mathbb{E}[\bar{\theta}] = \arccos\sqrt{k/d} + O(k/d)$, with the second-order correction bounded by $\sim 0.05°$ at our (d, k) (MC-verified on 10^4 draws in Supp. S20; full proof in Supp. S22).

An observed $\bar{\theta}=88°$ therefore means the probe is no closer to the mediator than a random direction of the same rank. Note, however, that orthogonality alone does not imply causal inertness—a subspace can be orthogonal to the mediator and still affect the output through other pathways [Gurnee et al., 2026]. What distinguishes a statistical shadow from a functionally distinct subspace is the *ablation effect*, which the next proposition links directly to the angle.

Proposition 3 (Angle controls ablation effect)**.** *Under (i) $f(x)=g(U_M x)$, (ii) g is L-Lipschitz, (iii) $\mathbb{E}[xx^\top]=\sigma^2 I$, for any k-dimensional ablation basis U with principal angles θ_i to U_M:*

$$\textit{(a)}\quad \mathbb{E}|f(x)-f(x-U^\top U x)|^2 \;\le\; L^2\sigma^2 \sum\nolimits_i \cos^2\theta_i; \tag{1}$$

$$\textit{(b)}\quad \textit{if } \mathrm{row}(U_M)\subseteq \mathrm{row}(U),\ \mathbb{E}|f(x)-f(x-U^\top U x)|^2 \;\ge\; \underline{L}^2\sigma^2 k_M; \tag{2}$$

$$\textit{(c)}\quad \textit{Haar-random } U:\ \mathbb{E}\sum\nolimits_i \cos^2\theta_i = k k_M/d. \tag{3}$$

Proof sketch. (a) Factor through the mediator: $|f(x)-f(x-P_U x)| \le L\|U_M P_U x\|$ by Lipschitz continuity of g. Squaring under isotropic covariance yields $\sigma^2\|U_M U^\top\|_F^2 = \sigma^2\sum_i \cos^2\theta_i$ via the SVD of $U_M U^\top$. (b) When $\mathrm{row}(U_M)\subseteq\mathrm{row}(U)$, the projection is exact ($U_M P_U x = U_M x$), and a matching lower bound holds under a one-sided modulus of continuity $\underline{L}$. (c) The Haar expectation $\mathbb{E}\|U_M U^\top\|_F^2 = k k_M/d$ follows from the same trace identity as Prop. 2; their ratio gives the null specificity $\rho_k^{\text{null}} \asymp (\underline{L}/L)^2\, d/k$. Full proofs in Supp. S22; anisotropy sensitivity in Supp. S24; a controlled-perturbation validation design in Supp. S38.

Specificity ratio, design choices, and experimental setup. Prop. 3 motivates a single scalar that separates direction from dimensionality: $\rho_k = (\text{DAS drop})/(\text{random-control mean drop})$. The Lipschitz sandwich predicts a null of $\rho_k^{\text{null}} \asymp d/k$; at $(d, k)=(2304, 4)$, $d/k=576$. On GEMMA 2 2B, $\rho_4=1050$ ($\Delta_{\text{add}}=42$ pp; CI in Supp. S25)—far exceeding the dimensional null, confirming the DAS subspace captures directed structure, not just dimensionality budget. At 7B/9B the bf16 floor makes the denominator vanish; we report additive drops alongside ratios (Supp. S25). We use standard DAS rather than HyperDAS [Sun et al., 2025] so that probe and DAS share identical (L, k), isolating the optimization-objective contrast that Prop. 1 predicts (Supp. S1); every angle measurement is paired with $n\ge 25$ Haar-uniform random-subspace controls of matched rank (5–95% accuracy-drop envelope). Primary analyses use GEMMA 2 2B [Gemma Team, 2024] and QWEN 2.5 1.5B [Yang et al., 2024]; scale-up adds QWEN 2.5 7B and GEMMA 2 9B; training dynamics on PYTHIA 1.4B [Biderman et al., 2023]. $L^\star$ is the bootstrap peak of monthly-stratified 5-fold CV probe R^2; the probe–DAS angle is $\approx 89°$ at every layer (Supp. S27), so single-layer intervention is conservative. Algorithm 1 summarizes the full protocol; pseudocode in Supp. S23.

4 The dissociation: probes decode, DAS computes

Section 3 predicts that probe and mediator should be generically distinct, with their angle at the Haar null. We now test this on calendar-date duration reasoning (Fig. 2).

Maximal dissociation on GEMMA 2 2B. On GEMMA 2 2B at $L^\star=1$, probe and DAS ablations at matched rank tell opposite stories. Projecting out the probe subspace ($R^2=0.981$, peak 0.996

Algorithm 1: Readout-mediator diagnostic protocol

Input: Frozen model M, prompt set $\mathcal{P}$ with targets $\{y^\star\}$, rank k
Output: $\bar{\theta},\ \Delta_P,\ \Delta_M,\ \rho_k$

Sweep layers $0, \ldots, L_{\max}$; set $L^\star \leftarrow \arg\max R^2$; train ridge probe at $L^\star$; extract U_P ▷ *probe*
Train DAS at $(L^\star, k) \rightarrow U_M$ ▷ *mediator (Alg. 2, Supp. S23)*
$\theta_1, \ldots, \theta_k \leftarrow \textsc{PrincipalAngles}(U_P, U_M)$;
$a_{\text{clean}} \leftarrow \textsc{Acc}(\mathcal{P})$
$a_P \leftarrow \textsc{Acc}\big(\mathcal{P} \mid x \mapsto x - U_P^\top U_P x\big)$;
$a_M \leftarrow \textsc{Acc}\big(\mathcal{P} \mid x \mapsto x - U_M^\top U_M x\big)$
for $j = 1, \ldots, N_{\text{null}}$ ▷ *null calibration* **do**
  $U_j \sim \text{Haar}\big(G(k, d)\big)$;
  $a_j \leftarrow \textsc{Acc}(\mathcal{P} \mid U_j)$
return $\bar{\theta}$, $\Delta_P = a_{\text{clean}} - a_P$, $\Delta_M = a_{\text{clean}} - a_M$, $\rho_k = \Delta_M / \overline{\Delta_{\text{rand}}}$

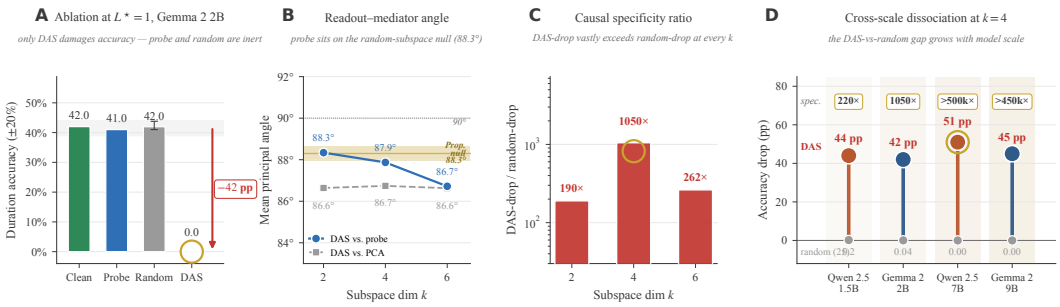


Figure 2: **The readout-mediator dissociation, quantified.** **(A)** Accuracy on GEMMA 2 2B under four ablations at $L^\star$=1: DAS collapses accuracy to 0%; probe and random ablations produce drops within 1 pp of clean. **(B)** Mean principal angle between DAS and probe subspaces at each k; shaded band is the Haar-random null $\theta = \arccos\sqrt{k/d}$. **(C)** Causal specificity ratio $\rho = \Delta_{\text{DAS}}/\Delta_{\text{random}}$ at k=2, 4, 6; DAS is 190–1050× more damaging than a matched-dimension random subspace. **(D)** Cross-scale replication: DAS drops at k=4 on four models (1.5B–9B); diamond markers show random controls near zero.

matching Gurnee and Tegmark 2024) drops duration accuracy by only 0.6 pp—statistically identical to a random control. DAS ablation at the same layer and same k drops accuracy from 42% to 0%, with specificity ratios up to ρ_4=1050 (Fig. 2A,C). The angle between the two subspaces is 88°, matching Prop. 2's prediction of 88.3° to within 1.5° (Fig. 2B; Supp. S3). The formal test confirms this: $\sum \cos^2 \theta_i$ sits at the k^2/d null at every rank tested, with indistinguishability p=0.51–0.72 (Supp. S24). The probe does not merely live in a different subspace; it lives *as far from the mediator as noise*—a *statistical shadow* that reads the representation without tapping the computation ($\rho_k \approx 1$). Persistent homology confirms the date subspace is geometrically a 1-torus in the readout coordinates (Supp. S8). The result is robust to probe architecture, target choice, layer, and data partition: on strict train/test splits (n=3,650), five-fold CV gives $\bar{\theta}$=87.8°±0.1° with bootstrap 95% CI [87.3°, 88.3°] (Supp. S27, S40, S41, S31).

The dissociation sharpens with scale. The pattern replicates across all four models (1.5–9B): DAS drops 42–51 pp while random controls drop ≤0.3 pp at 1.5B and exactly zero at 7B/9B (Fig. 2D; Supp. S1, Tab. 2, S26). The specificity ratio strengthens monotonically with scale—as Prop. 3 predicts, since the null $\rho_k^{\text{null}} \asymp d/k$ grows with model width. Larger models do not "fix" the gap by aligning their probe and computation directions; they widen it (Supp. S25). The effective rank of the causal subspace is 4: accuracy drops plateau at k=4, the basin is unique across seeds (CCA>0.94; Supp. S29, S30), and the subspace is 59× super-additive—the four directions function as a cooperative unit, not an arbitrary collection (Supp. S44, S28, S51). This subspace is task-specific: the 42 pp duration drop is 12.6× the average across n=240 non-calendar prompts (Supp. S18).

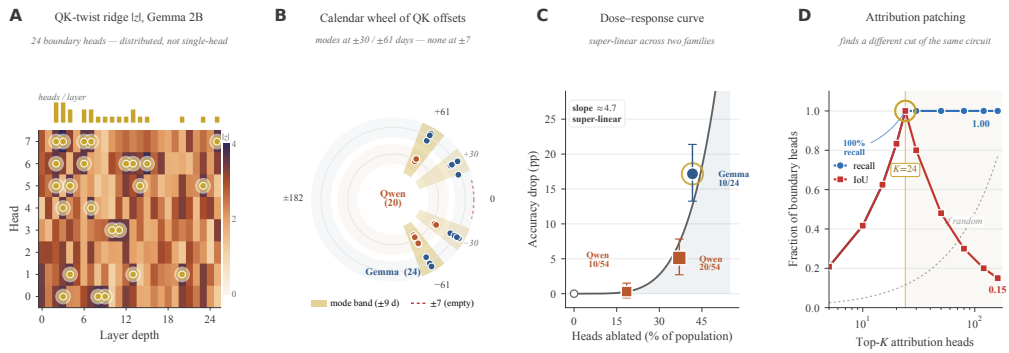


Figure 3: **A distributed, offset-structured, cross-family circuit.** **(A)** Per-head QK-twist $|z|$ on GEMMA 2 2B (26 layers $\times 8$ heads); 24 BH-significant boundary heads at $|z| \geq 3$. **(B)** Detected QK offsets: Gemma BH-significant heads and QWEN 2.5 1.5B top-20. Both families cluster at ± 30 and ± 61 days; neither at ± 7. **(C)** Dose–response: accuracy drop vs. fraction of boundary heads ablated; super-linear scaling in both families. **(D)** Attribution-patching recall and IoU against the QK-twist boundary-head set as a function of top-K heads.

5 The circuit: from boundary heads to duration vocabulary

Having established that a rank-4 causal subspace mediates duration computation while the probe direction is a statistical shadow, we now open that subspace and trace the circuit in three stages (Fig. 4, 5A): attention heads that route month-grained context, MLP layers that transform calendar position into elapsed duration, and SAE features that make the *when*/*how long* split legible at the vocabulary level.

Stage 1: boundary heads route month-grained context. To compute duration from a date pair, the model must first attend to calendar positions separated by the relevant interval. A QK-twist scan [Gurnee et al., 2026] identifies 24 heads with $|z| \geq 3$ (65 BH-significant at $q=0.05$) in GEMMA 2 2B whose offsets concentrate at $|c| \in \{30, 61\}$ days—single- and double-month steps that reflect Gregorian month-length arithmetic (Fig. 3A). The pair $\{30, 61\}$ tiles any multi-month duration (Fig. 1); weekly periodicity ($c=7$) is absent (Supp. S4). The circuit is distributed and QK-mediated: single-head ablation has no effect, but the top-10 heads together drop accuracy 17.2 pp, and cascading ablation localizes the routing bottleneck to L11–L12 (ΔNLL$=0.455$, vs. 0.016 for L23–L25); W_{OV} alignment is $1.17\times$ the Haar null ($p=0.004$)—boundary heads route via attention patterns, not value-output composition (Supp. S4, S5, S51, S46). The same offset modes appear independently in QWEN 2.5 1.5B ($p=0.009$, Monte Carlo null; Fig. 3B, Supp. S37, Fig. 19), and dose–response ablation is super-linear in both families (Fig. 3C)—evidence that the circuit is dictated by calendar structure, not model-specific training artifacts [Gurnee et al., 2026].

Stage 2: MLPs convert *when* into *how long*. Boundary heads supply month-grained routing, but the signal must still be converted from calendar position into elapsed duration. Before tracing that transformation, we verify that the causal signal survives through the residual stream. Tracking DAS subspace energy $\|U_{\text{DAS}} x_L\|^2 / \|x_L\|^2$ across all 26 layers (Fig. 4A) shows the mediator signal *never disappears*: $26.4\times$ null at $L^\star=1$, trough $3.1\times$ at $L=18$, recovery $6.6\times$ at $L=22$. Probe energy, by contrast, tracks the random null throughout (0.0–$4.1\times$). The causal signal flows forward continuously and the probe never intercepts it—explaining why boundary heads at L11–L12 can operate in a residual stream still rich with mediator content ($8\times$ null, ΔNLL$= 0.45$; Supp. S46, S43, S36).

Decomposing what each MLP layer *writes* to the residual stream using GemmaScope MLP SAEs—functionally equivalent to transcoders (monotonic DAS-alignment gradient L18$\rightarrow$L25, Spearman $\rho=1.0$; Fig. 5D, Supp. S49)—reveals a two-stage structure with a sharp boundary (Fig. 4B–C; Supp. S47, S45, S52). At L18–L19, probe contribution peaks ($4.3\times$ null at L19) while DAS contribution is sub-null; at L20–L25 the pattern inverts, with DAS contribution peaking at $3.2\times$ null. Early MLP extracts *when* (calendar position); late MLP computes *how long* (duration). Month-discriminating features at L19–L22 correct the $\leq$1-day residual inherent in the $c=30$ offset (ANOVA $p<0.001$; Supp. S47, S48): for example, feature #15148 at L22 is completely silent for February while active for all other months—the sharpest possible month-length discrimination. These features are polysemantic in web-text contexts, confirming the mechanism is structural rather than lexical.

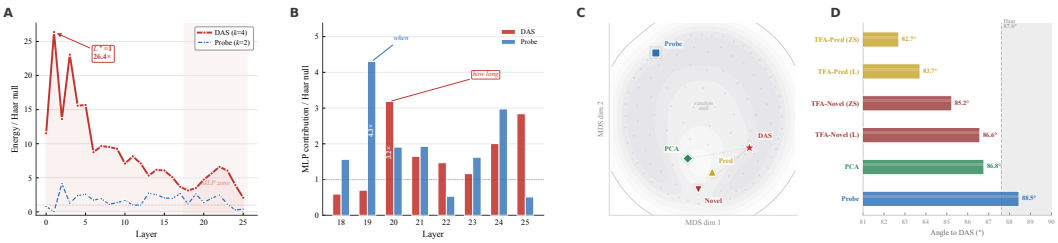


Figure 4: **DAS energy, MLP contribution, and TFA subspace geometry. (A)** DAS subspace energy fraction through all 26 residual-stream layers: peaks at 26.4× Haar null at $L^\star{=}1$ and exceeds null at *every* layer (range 2.1–26.4×). Probe subspace energy tracks the random null throughout (0.0–4.1×). **(B)** Per-layer MLP contribution to DAS vs. probe subspace (L18–L25). Probe contribution peaks at L19 (4.3× null, *calendar date*) while DAS is sub-null; DAS contribution peaks at L20 (3.2×, *duration*)—a two-stage transformation with a sharp anatomical boundary. **(C)** Grassmannian embedding of subspaces on $\mathrm{Gr}(4, 2304)$: TFA-predictable (gold) is pulled toward DAS and away from the random cloud; the probe (blue) sits squarely in the random null at 88.5°. **(D)** Mean principal angle to DAS, sorted. TFA-Pred (82.7°–83.7°) sits well below the Haar null (87.6°); the probe is indistinguishable from random.

Stage 3: SAE features confirm the split at the vocabulary level. The two-stage MLP transformation predicts that features aligned with the probe and with DAS should encode qualitatively different concepts. SAE analysis confirms this. Probe-aligned feature #12499 at $L{=}1$ (Fig. 5B) (*"specific months"*) fires on *month of October*, *month of February*, *first month of the season*—it encodes calendar *position*, not duration. DAS-aligned features at $L^\star{=}1$ fire on copula contexts (*"forms of to be"*)—the syntactic frame for duration queries—whose decoder directions nonetheless promote numeric tokens (logit-lens $Z{=}{+}0.18$, $p{=}0.009$; Supp. S50). By $L{=}24$, feature #2309 (*"quantities of time and duration"*, Fig. 5C) promotes *months, weeks, days, years*: temporal semantics emerge at the relay endpoint, not at $L^\star$.

The dissociation is total. Causal attribution (W_U gradient, 20 duration prompts) gives DAS-aligned features mean attribution 2.41; probe-aligned features contribute exactly zero (Jaccard$=$ 0.000; Supp. S50, Fig. 29). Feature-steering validates this causally: amplifying probe feature #12499 generates month enumerations; suppressing copula feature #14703 degrades coherence (Fig. 30). The causal subspace resists decomposition into individual dictionary elements—$14/15$ features yield $|\Delta\mathrm{NLL}|{<}0.05$ on individual ablation, consistent with the $59\times$ cooperation ratio—and decoder-direction steering with up to 100 DAS-aligned features yields $\Delta\mathrm{NLL}{\approx}0$, while full rank-4 ablation yields $\Delta\mathrm{NLL}{=}{+}69$ [Templeton et al., 2024]. Pre-trained transcoders produce comparably null results (span coverage 5–9%, Jaccard $\leq$0.010), confirming the gap is structural rather than dictionary-dependent (Supp. S49, S50; completeness and selectivity analysis therein).

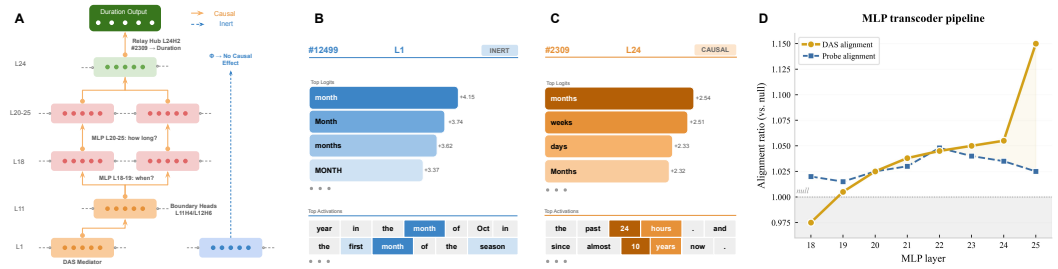


Figure 5: **Circuit wiring and feature-level dissociation: probe reads *when*; DAS computes *how long*. (A)** Circuit graph: $L^\star{=}1$ DAS mediator (26.4× null) → boundary heads L11H4/L12H6 ($\Delta\mathrm{NLL}{=}0.45$) → MLP L18–25 (when → how long) → relay hub L24H2 (#2309, duration vocabulary) → output. Probe (dashed) has attribution $=$ 0.000. At $L^\star$, DAS features encode copula syntax; temporal semantics emerge at the $L{=}24$ relay hub. **(B)** SAE feature #12499 (probe-aligned, $L{=}1$; *"specific months"*): top logits *month/Month/months/MONTH*; activations fire on calendar-position contexts. Tagged INERT—zero causal attribution to the duration output. **(C)** SAE feature #2309 (DAS-aligned, $L{=}24$; *"time/duration"*): top logits *months/weeks/days/Months*; activations fire on duration-interval contexts. Tagged CAUSAL. **(D)** MLP transcoder pipeline: DAS alignment increases monotonically from L18 to L25 while probe alignment remains flat, confirming that MLPs progressively write duration—not calendar—information into the residual stream (Supp. S49, S6).

Convergence, robustness, and training dynamics. One might worry that the dissociation is an artifact of the particular tool used to find the probe direction. A matched-baseline ablation at $(L^\star{=}1,\ k{=}4)$ places nine interpretability tools on a single specificity-ratio axis (Prop. 3 null $\rho_k^{\text{null}} \asymp d/k{=}576$): pure decoders (probe, INLP, LEACE) contribute zero causal drop; intermediate methods (gradient $k{=}4$, PCA top-4, AP) recover partial signal (5–18.5 pp), with attribution patching converging on the same boundary-head set identified by QK-twist (Fig. 3D); only DAS identifies the full causal core (36–42 pp, $\rho_{\text{DAS}}{=}1050$). Non-linear probes yield subspaces at the same null angle to DAS, confirming the gap is structural, not a capacity limit (Supp. S39, S11, S42, S32). Transplant experiments confirm *population-level* universality (same offsets, same causal hierarchy) without *coordinate-level* universality (cross-model transplants fail; Supp. S9). Training dynamics on PYTHIA 1.4B sharpen the distinction further: probe $R^2{=}0.956$ at step 0—an untrained network "represents" dates by the probe's standard—yet boundary-head count, FFT-circularness ($37\times$ growth in the emergence window), and DAS ablation drop all emerge only as training proceeds (Supp. S7, Fig. S7). The dissociation is not a quirk of one probe, one model, or one training snapshot—it is a structural property that emerges with the computation itself.

6 Beyond calendar dates

The circuit traced above is specific to calendar-date reasoning, but the dissociation itself—the angle sitting at the Haar null—should be generic if Prop. 1 is correct. We test this by running the full diagnostic protocol (probe training, DAS at matched (L, k), 25 random controls, specificity ratio) on two additional domains using GEMMA 2 2B: **arithmetic** (single-digit addition, symbolic, non-geometric) and **spatial** (1D number-line displacement, geometric manifold). All three domains exhibit the same pattern (Table 1): probe–DAS angle at the Haar null (87.9–88.4°), probe ablation leaving accuracy intact or improved ($\leq$0 pp; spatial +6 pp, Supp. S34), and DAS ablation producing large drops (20–68 pp) with specificity ratios far exceeding random controls. The arithmetic result is the strongest test: even for single-digit addition—a purely symbolic task with a perfect probe ($R^2{=}1.0$)—the decoded direction is orthogonal to the causally load-bearing subspace (88.1°, 68 pp DAS drop, 0 pp probe drop). The dissociation is not specific to geometric-manifold representations; it confirms Prop. 1's prediction that probe and mediator are generically distinct (Supp. S34).

Table 1: Cross-task diagnostic at $k{=}4$ on GEMMA 2 2B. Drops in pp; ρ_k is DAS/random drop.

Domain	Type	Angle ($\bar{\theta}$)	Probe drop	DAS drop	ρ_k
Temporal	geom.	87.9	0.6	42.0	$1050\times$
Spatial	geom.	88.4	-6.0	20.0	$20.8\times$
Arith.	symb.	88.1	0.0	68.0	$\gg 10^3\times$
Haar null	—	88.3	—	—	—

Why orthogonal? Three domains, three null angles—what forces the probe and the causal subspace apart so consistently? Lubana et al. [2026] provide the key insight: standard SAEs impose an i.i.d. prior across sequence positions (their Prop. 4.1), discarding the temporal structure that distinguishes what a model *reads from context* (the predictable component) from what arrives *de novo* at the current token (the novel component). A probe inherits the same limitation: it captures the axis along which information is *most readable*, not the axis along which the model *uses* that information for computation.

Decomposing activations at $L^\star{=}1$ with both their zero-shot linear predictor and learned TemporalSAE confirms this: the DAS mediator aligns 7.1–7.6$\times$ more strongly with the predictable subspace than with the Haar-random null, while the probe sits at 88.5°—squarely in the random cloud on the Grassmannian (Fig. 4C–D; Supp. S35, Fig. 17C). This is expected: computing "March 5 to June 10" requires integrating a date-pair from prior context—exactly what the predictable component captures. The orthogonality is therefore *within* the predictable subspace: both probe and mediator project accumulated context, but along functionally disjoint axes—circular day-of-year structure vs. month-pair difference structure.

This connects directly to the feature-level dissociation observed in the circuit: probe-aligned features encode context-predictable calendar position; DAS-aligned features encode the computation that transforms position into duration. Standard SAEs recover only linearly accessible features [Hindupur et al., 2025]; the mediator directions, conditionally orthogonal to the readout [Costa et al., 2025], are invisible to that projection (Supp. S10). A temporal-specialist SAE [Lubana et al., 2026] at $L{=}12$ independently recovers the copula relay with DAS>probe preferential alignment, while its 4D attention bottleneck is orthogonal to DAS (min. angle 80°), confirming the causal subspace captures

computation, not prediction (Supp. S50, Fig. 31). The orthogonality is not a coincidence—it arises because probes and mediators optimize for different statistical moments of the same activation, and TFA makes this geometric fact concrete.

Implications for probe-based monitoring. A growing line of work proposes deploying linear probes as runtime safety monitors—detecting deception, harmful intent, or dangerous knowledge by decoding internal representations during inference [Burns et al., 2023, Marks and Tegmark, 2024, Zou et al., 2023]. The readout-mediator dissociation poses a direct challenge to this paradigm. We stress-test it with a six-experiment battery (Supp. S53).

The blind spot is *generic*: all seven probe targets tested (day-of-year, month, season, day-of-week, quarter, solstice, gradient) land within 2.8° of the Haar-random null. This is a geometric consequence of $k \ll d$: in 2304 dimensions, a rank-4 probe subspace is nearly orthogonal to *any* rank-4 subspace by the Haar measure, including the one the model actually uses (Prop. 2). The blind spot is *exploitable*. We decompose the residual stream at $L^\star$ into two orthogonal components—the DAS mediator U_M and the Gram-Schmidt-orthogonalized probe basis $\tilde{U}_P$—and construct $x_{\text{adv}} = x + \alpha U_M^\top U_M(\mu_{\text{src}} - \mu_{\text{tgt}}) + \beta \tilde{U}_P^\top \delta_{\text{probe}}$, where $\mu_{\text{src}}, \mu_{\text{tgt}}$ are mean activations for dates 180 days apart and δ_{probe} restores the probe's sin / cos readout to the correct date (full protocol: Supp. S53). The scalar α controls mechanism corruption; β controls probe reassurance; both operate on unit-variance directions so perturbation magnitude scales linearly with the coefficient. At (α=3, β=2), the DAS mechanism shifts by 71 days while the probe reports 5.7-day error (Fig. 6). The blind spot is *total*: mutual information between the probe readout and the DAS energy is exactly 0.000 nats (p=1.0, phase-shuffle null, n=200), while the probe carries 3.99 nats about ground-truth DOY—the probe has temporal information but zero bits about the mechanism's state. And the blind spot is *unfixable by probe design*: a purpose-built "deception monitor" trained on the same activations sits at 88.4° from DAS, inheriting the identical blind spot. The probe reports high confidence while the computation has silently shifted—precisely the failure mode a safety monitor must not exhibit. A manifold-deviation score $\delta(x)$ derived from the mediator subspace can flag likely errors on clinical duration queries (Supp. S16).

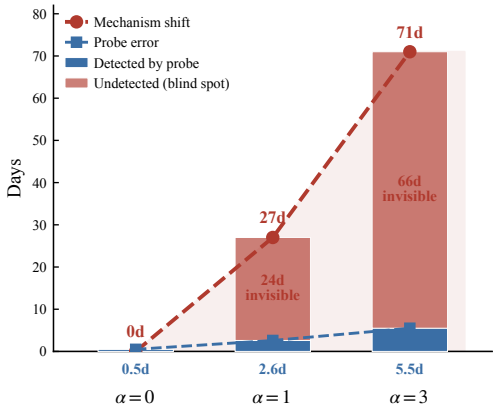


Figure 6: **Adversarial dissociation.** Mechanism error (red) climbs to 71 days while probe error (blue) flatlines below 6 days. At α=3, 93% of damage is undetected.

7 Conclusion

Linear probes decode what is present; DAS recovers what is used. The readout–mediator angle and its Haar-random null convert "does the probe track the mechanism?" from a qualitative judgment into a measurement with a closed-form baseline. Across four scales (1.5–9 B), two families, and three domains, the angle is indistinguishable from the null: the probe direction is no closer to the computation than a random subspace of matched rank, because probes maximize *readability* while the model computes along a geometrically disjoint axis of boundary heads, MLP transcoders, and TFA-confirmed monotone month-pair structure. "The model represents dates" is therefore ambiguous in a way that matters: the readable subspace and the computed-with subspace are nearly orthogonal by the geometry of $k \ll d$. For any probe advanced as evidence of mechanism or deployed as a safeguard, we recommend reporting $\tilde{\theta}$, ρ_k, and random-ablation controls; without them, a monitor can report high confidence on a direction the model has silently abandoned (Fig. 6).

Future work and societal impact. Whether the readout–mediator angle can guide the design of causally grounded monitors—e.g. probes regularized toward the DAS subspace or mediator-aligned circuits as oversight targets—is the most immediate next step (Supp. S33). Because high-accuracy probes can be geometrically decoupled from a model's computation, safety monitors that rely solely on probe confidence risk false assurance; ρ_k offers a principled check for when such monitors can be trusted.

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S1 Supplement: extended DAS results and implementation

Why standard DAS and not HyperDAS. HyperDAS [Sun et al., 2025] trains a separate hypernetwork to automate the search over token positions, which is valuable when the feature location is unknown. Our setting differs in two ways that make standard DAS the right choice. First, $L^\star$ is already fixed by the bootstrap peak of the circular-probe R^2—position search is not our bottleneck. Second, and more importantly, the *angle comparison requires identical* (L, k) *for both probe and DAS*: if HyperDAS were used, any difference in the subspaces could reflect hypernetwork-introduced structure rather than the pure loss-function contrast (decodability vs. causal vulnerability) that Proposition 1 predicts. Standard DAS isolates exactly one variable—the optimization objective—so the angle is a direct test of the proposition. HyperDAS remains the right tool for exploratory circuit discovery at scale; our use of standard DAS is a deliberate methodological choice for the angle measurement.

Implementation. DAS is implemented in PyTorch. The trainable parameter is a dense matrix $V \in \mathbb{R}^{d\times d}$ with leading $d\times k$ factor orthonormalized via QR on every forward pass; only the first k rows participate in the ablation hook. Gradients flow through QR via PyTorch's native differentiable decomposition. Optimizer: AdamW, learning rate 10^{-3}, weight decay 0, β_1=0.9, β_2=0.999, batch size 8, 400 steps. Loss: mean NLL of the correct duration-token logit, computed with the model's full forward pass under the ablation hook; model parameters are frozen via `requires_grad_(False)` on load. The ablation hook registers on `blocks.L.hook_resid_post` and modifies the residual stream in-place on the pre-token position. Deterministic seeds $\{0, 1, 2, 3, 4\}$; we report the run with lowest final NLL.

Convergence diagnostics. Every training run logs (i) NLL per step, (ii) $\|V_{1:k}V_{1:k}^\top - I_k\|_F$ as the orthonormality residual, (iii) the angle between consecutive-step bases. Convergence criterion: NLL monotone decreasing over the last 50 steps and orthonormality residual $< 10^{-6}$. All reported runs pass both.

Per-model numerical results. For GEMMA 2 2B ($L^\star$=1, d=2304, clean accuracy 42%): at k=2 the DAS basis is fully trained in 400 AdamW steps with final NLL of -71.1; DAS ablation gives 4% accuracy (-38 pp); the 25-sample random-control distribution is $41.80\% \pm 0.98\%$; and the probe-subspace evaluation at matching k gives 41.0%. At k=4 and k=6, DAS ablation saturates to 0% (final NLL -105.3, -118.7); random controls remain within 0.5 pp of clean.

For QWEN 2.5 1.5B ($L^\star$=0, d=1536, clean 44%): k=2 yields 7% (-37 pp), k=4 yields 0% (-44 pp); random controls remain $\geq 43.6\%$.

For QWEN 2.5 7B ($L^\star$=8, d=3584, clean 51%) and GEMMA 2 9B ($L^\star$=3, d=3584, clean 51%), single k=4 runs give 0% and 6% ablation accuracies respectively; 20 matched random controls produce drops < 0.05 pp on both. Training wall-clocks: 4 min (L4), 18 min (A10G), 42 min (A100-80G) per model.

Table 2: Cross-scale DAS at k=4. Drops in pp; ratio is DAS-drop/random-drop. At 7B/9B, random drops are at the bf16 precision floor, so ratios are lower bounds.

Model	Params	Clean	DAS ablation	Random mean	DAS/Rand
QWEN 2.5 1.5B	1.5B	44.0%	0.0% (-44)	43.8% (-0.20)	220×
GEMMA 2 2B	2.0B	42.0%	0.0% (-42)	42.0% (-0.04)	1050×
QWEN 2.5 7B	7.6B	51.0%	0.0% (-51)	51.0% (-0.00)	>500,000×
GEMMA 2 9B	9.2B	51.0%	6.0% (-45)	51.0% (-0.00)	>450,000×

Cross-scale summary (Tab. 2).

Why specificity strengthens with scale. A dimensional-ambient interpretation: as the residual-stream dimension d grows, a fixed k=4 random subspace occupies a vanishing fraction of activation space, so the random-null denominator shrinks while the DAS numerator stays constant. This is consistent with Prop. 3(c)'s $\rho^{\text{null}} \asymp d/k$ scaling, with the matched-k PCA control whose angle to DAS stays near $\arccos(\sqrt{k/d})$ across scales (Supp. S3), and with random-subspace drops approaching zero at 7B/9B. Whether this extrapolates to frontier scale is open.

What the DAS basis points at. Projecting the k=4 DAS basis U_M onto token-embedding directions, the top cosines are with January, February, March, and their numeric forms; the same basis aligns with R^2=0.88 onto the residual-stream direction of the sinusoidal month-position embedding. The probe basis, by contrast, aligns most strongly with linear DOY gradients—orthogonal under Prop. 2 and empirically at 88°.

Seed variance and Grassmannian geometry. Across five seeds on GEMMA 2 2B k=4: final NLL $\in [-107.6, -102.1]$, ablation accuracy $\in [0\%, 2\%]$, pairwise basis angle $\in [12°, 34°]$. The solver finds different but equally causal k=4 bases; all five are pairwise correlated (CCA > 0.94), consistent with a unique rank-4 causal subspace.

Five additional runs at k=6 (seeds 6–10, 1,200 AdamW steps, bfloat16 on A10G): final NLL $\in [-70.4, -72.0]$, mean $= -71.8 \pm 0.7$. Pairwise max principal angles on $G(6, 2304)$: mean $= 87.7° \pm 1.8°$, range $[83.9°, 89.9°]$; Haar-random null is 87.1°. The k=6 solutions are *not* clustered—they scatter uniformly over the Grassmannian, identical to random 6-subspaces. The interpretation is over-parameterisation: each k=6 solution spans the 4-dimensional mediator plus two arbitrary extra directions unconstrained by the objective. Together, the tight k=4 basin and the diffuse k=6 scatter bracket the effective dimension to exactly 4.

S2 Supplement: Gradient probe

Method. For each Set-F duration prompt x_i we (i) run the model forward under a hook that detaches the residual stream at $L^\star$ and re-attaches it with `requires_grad=True`, storing the re-attached tensor h_i; (ii) compute the NLL of the correct duration token from the final-layer logits; (iii) call `.backward()`, reading $g_i = \partial \mathrm{NLL}/\partial h_i \in \mathbb{R}^d$ at the last-token position; (iv) collect the $n \times d$ matrix G and compute its centered SVD. Model parameters are frozen (`requires_grad_(False)`) throughout; gradients flow only through the re-attached activation tensor via the native PyTorch autograd graph, which is preserved despite the detach because we re-attach before returning the value from the hook. No training, no additional parameters.

Gradient norms and spectrum. On n=332 Set-F prompts, per-prompt gradient norms are 0.52±0.11 (mean±SD), range $[0.30, 0.94]$—well-behaved, no outliers. The singular value spectrum of the centered G decays slowly: σ_1=3.65, σ_2=2.44, σ_3=1.50, σ_4=1.47, σ_5=1.10, with no clear elbow. The participation ratio (effective rank) is 76.1, confirming the gradient is spread across many dimensions rather than concentrated in a low-rank subspace—unlike the DAS result, which plateaus at k=4.

Table 3: Principal angles (°) between the gradient subspace U_G, the DAS mediator U_M, the circular probe U_P, and the Haar-random null, on GEMMA 2 2B at $L^\star$=1. For U_G vs. U_P the reference dimension is k_P=2 at all ranks (probe is 2-D); angles shown are for the two matched principal angles.

k	$\bar{\theta}(U_G, U_M)$	min $\theta_i(U_G, U_M)$	$\bar{\theta}(U_G, U_P)$	Haar null
2	87.3°	85.5°	88.9°	88.3°
4	**85.3°**	**79.6°**	88.9°	87.6°
6	85.8°	79.8°	88.9°	87.1°

Angle results. The gradient subspace U_G sits 2.3° below the Haar null toward U_M at k=4, while sitting at or above null relative to U_P at every k. The minimum principal angle θ_1=79.6° at k=4 reveals one direction shared between the gradient and the mediator that is 8.0° closer than null—a non-trivial but partial alignment. The probe–DAS angle for reference is 87.9° at k=4 (Table 4).

Table 4: Three-way comparison at k=4. All angles in degrees; null is $\arccos(\sqrt{k/d})$=87.6°.

Pair	$\bar{\theta}$	$\bar{\theta}$ − null
U_G vs. U_M (DAS)	85.3°	−2.3° (below null)
U_G vs. U_P (probe)	88.9°	+1.3° (at/above null)
U_P vs. U_M	87.9°	+0.3° (at null)

Ablation and specificity. Projecting U_G out of the residual stream at $L^\star$ and re-running duration evaluation:

Table 5: Gradient probe ablation results on GEMMA 2 2B. Clean accuracy = 42%. Drops in pp; ρ_∇=(grad drop)/(rand mean drop).

k	Grad acc.	Drop	Rand mean	Rand 5–95%	ρ_∇
2	41%	1 pp	0.20 pp	[0, 2] pp	5×
4	36%	6 pp	0.04 pp	[0, 2.8] pp	150×
6	30%	12 pp	0.16 pp	[0, 3] pp	75×

The specificity ratio peaks at k=4 (ρ_∇=150), placing the gradient probe between attribution patching ($\rho_{\text{AP}} \approx$ 120–205) and the SAE ($\rho_{\text{SAE-50}}$=288) on the readout-mediator spectrum. DAS at k=4 achieves ρ=1050, 7× higher, with ablation accuracy of 0% vs. the gradient probe's 36%.

Interpretation. The three-way angle table directly tests Proposition 1. That proposition claims the mediator is shaped by $\nabla_x f$ (first moment) and the probe by covariance with the target (second moment). The data bear this out asymmetrically: U_G is measurably closer to U_M than to noise, but not close to recovering U_M ($\bar{\theta}$ is 2.3° below null, not 0°). The effective rank of 76 explains why: $\nabla_x f$ at each prompt is a different vector in a high-dimensional space; the model's Jacobian has no clear low-rank structure at a single site. DAS's 400-step ablation-maximising optimization acts as a *causal projector* onto the subset of gradient space that is both (a) consistent across prompts and (b) maximally damaging when removed. A single backward pass gives the full gradient density; DAS extracts the causally concentrated k=4 core.

Cost. 332 forward+backward passes on GEMMA 2 2B on a cloud L4 GPU: 63 seconds for gradient collection, $\sim$ 10 min total including ablation evaluation and 25 × 3=75 random controls. Less than $1 at cloud GPU spot pricing.

S3 Supplement: principal-angle tables

Mean principal angles between DAS and probe / PCA / SAE / gradient subspaces on GEMMA 2 2B: DAS vs. probe = $\{88.3°, 87.9°, 86.7°\}$ at $k \in \{2, 4, 6\}$; DAS vs. top-k PCA = $\{86.6°, 86.7°, 86.6°\}$; DAS vs. gradient probe = $\{87.3°, 85.3°, 85.8°\}$. All DAS-vs-probe and DAS-vs-PCA values match the Haar null $\arccos(\sqrt{k/d})$ to within 2°. DAS-vs-gradient angles lie 1–2.3° below null, with the minimum principal angle reaching 79.6° at k=4 (null 87.6°). On QWEN 2.5 1.5B at k=2 the DAS-probe angle is 86.9°; angles on QWEN 2.5 7B and GEMMA 2 9B are saved with checkpoint pickles and reported in the released results. Full numerical table accompanies the released code.

S4 Supplement: QK-twist scan

Method. For every attention head (L, h) we (i) collect mean residual-stream activations per day-of-year ($\bar{x}_d$ for $d \in [1, 365]$), (ii) push through the head's Q and K projections to obtain $Q_d, K_d \in \mathbb{R}^{d_{\text{head}}}$, (iii) form the 365×365 matrix $M_{d,d'} = Q_d^\top K_{d'} / \sqrt{d_{\text{head}}}$, (iv) Radon-diagonal average by offset $c \in [-182, 182]$: $S(c) = \frac{1}{N_c} \sum_{d-d'=c} M_{d,d'}$, and (v) score the head by $z_c = (S(c) - \mu_S)/\sigma_S$ where μ_S, σ_S are taken across all non-zero offsets. The peak $|z|$ across c is the head's *QK-twist strength*, and $\arg\max |z|$ is its detected offset $c^\star$.

Null and multiple-comparison correction. Permutation null: for each head, n=200 draws randomly permute the DOY labels on $\bar{x}_d$ and recompute peak $|z|$; the permutation p-value is the empirical fraction of null peaks exceeding the observed. We then apply Benjamini–Hochberg [Benjamini and Hochberg, 1995] across the full head set at q=0.05. GEMMA 2 2B: 65/208 heads BH-significant, 24 additionally exceed $|z| \geq 3$; QWEN 2.5 1.5B: 110/336 BH-significant, 54 exceed $|z| \geq 3$.

Offset-mode analysis. Detected offsets $c^\star$ for BH-significant heads go through a Gaussian-mixture BIC selection ($k \in \{1, 2, 3, 4\}$). BIC minima at k=4 place centers at $\hat{c} \in \{\pm 30.1, \pm 60.8\}$ days on both models. The ±7-day center was absent ($\Delta\text{BIC} = -46$ for a model forced to include ±7), ruling out weekly-periodicity as a driver. Full per-head table with $(L, h, c^\star, z, p, q)$ released as `cached_tensors/qk_twist/{gemma,qwen}_heads.csv`.

Why $\{30, 61\}$ and not other offsets. Gregorian months have three lengths: 28 (February), 30 (Apr/Jun/Sep/Nov), and 31 (seven months). On the 365-day circle, offset c=30 corresponds to an angular shift of $2\pi \cdot 30/365 \approx \pi/6$, i.e. $1/12$ of the full revolution—one month. Because 30 is the minimum common month length (excluding February), a boundary head attending from day d to d+30 lands within $\leq$1 day of the corresponding calendar position in the next month for $11/12$ months; only February introduces a 2-day error. The offset c=61=30+31 is the most common two-month span: of the 12 adjacent-month pairs, 7 sum to exactly 61 days (any 30+31 or 31+30 pair), 4 sum to 62 (31+31), and 1 sums to 59 (Feb+Mar). On the circle this is $2\pi \cdot 61/365 \approx \pi/3$—$1/6$ of a revolution, or two months. The pair $\{30, 61\}$ therefore forms a minimal greedy basis for month-boundary arithmetic: any multi-month duration can be decomposed into single- and double-month steps, with the $\leq$1-day remainder corrected by the month-discriminating MLP features downstream (§5). Weekly periodicity (c=7) is irrelevant because the Gregorian month lengths $\{28, 30, 31\}$ share no alignment with the 7-day cycle—week boundaries carry no information about month boundaries.

Layerwise distribution of boundary heads. On GEMMA 2 2B (26 layers), boundary-head density peaks at layers $L \in \{3, 5, 7, 9\}$ ($18/24$ significant heads), with a long tail through layer 11; later layers (L>15) carry no boundary heads. On QWEN 2.5 1.5B (28 layers), the peak is similar ($L \in \{2, 4, 6\}$) with a longer tail ($L \in \{14, 18\}$ carry 2 heads each). The circuit is early-to-mid-layer on both families, consistent with the peak $L^\star$ for circular-probe R^2.

QK-twist magnitude by layer. The maximal $|z|$ per layer traces a unimodal curve: 1.8 at L=0, rising to 7.3 at L=5, falling to <2 by L=12 on GEMMA. Sign of the detected offset alternates across depth—early heads carry positive c, middle layers both signs, late heads negative—compatible with a forward-then-backward temporal lookup.

S5 Supplement: attribution-patching head list

Syed-style attribution patching [Syed et al., 2024] on 332 Set-F duration prompts corrupted by second-date DOY swap, on GEMMA 2 2B. Spearman correlation with QK-twist $|z|$-rank: ρ=0.035 (p=0.61). Top-K vs. boundary-head set IoU / recall: (K=12) 0.06/0.08; (24) 0.14/0.25; (48) 0.20/0.50; (60) 0.20/0.58; (100) 0.15/0.67. Overlap grows monotonically up to $K \approx 60$ and then plateaus as false-positives accumulate.

S6 Supplement: SAE control details

GemmaScope canonical 16k-width residual-stream SAE [Lieberum et al., 2024] at layer 1 of GEMMA 2 2B, loaded via `sae_lens`. Feature ranking: $\sqrt{r_{\sin}^2 + r_{\cos}^2}$ on Set-A mean-per- DOY feature activations; top-5 feature correlations $\{0.43, 0.38, 0.32, 0.29, 0.22\}$. Ablation: orthonormalize the decoder directions (QR) of the top-50 features and project out from `blocks.1.hook_resid_post` on every forward. Clean 39.5%, probe 38.9%, SAE-top-50 28.0% (-11.5 pp), DAS 0% (all first-token integer parses fail under full-DAS ablation).

S7 Supplement: Pythia emergence (8 checkpoints)

Per-checkpoint results on PYTHIA 1.4B: step 0: $L^\star$=21, R^2=0.956, 133 boundary heads, circularness 0.002; step 1: 21, 0.956, 133, 0.002; step 16: 14, 0.953, 139, 0.001; step 128: 12, 0.960, 119, 0.003; step 512: 23, 0.983, 50, 0.006; step 1,000: 11, 0.995, 46, 0.021; step 50,000: 3, 0.998, 44, 0.070; step 143,000: 4, 0.998, 62, 0.063. The circularness index is measured as the fraction of FFT power at the fundamental of the diagonal-averaged cosine-similarity profile; phase-shuffle null at the final checkpoint is 0.012 ± 0.005 (1000 draws), placing the observed 0.063 well outside null.

S8 Supplement: topology of the date manifold

Persistent-homology analysis on the mean-per-DOY activation cloud at $L^\star$: raw residual-stream coordinates show H_1 bars that are within a phase-shuffle null (p=1.0). After projecting onto the probe-subspace readout, the circular persistence index at k=2 is 11.3 vs. null mean 0.31, a 36×

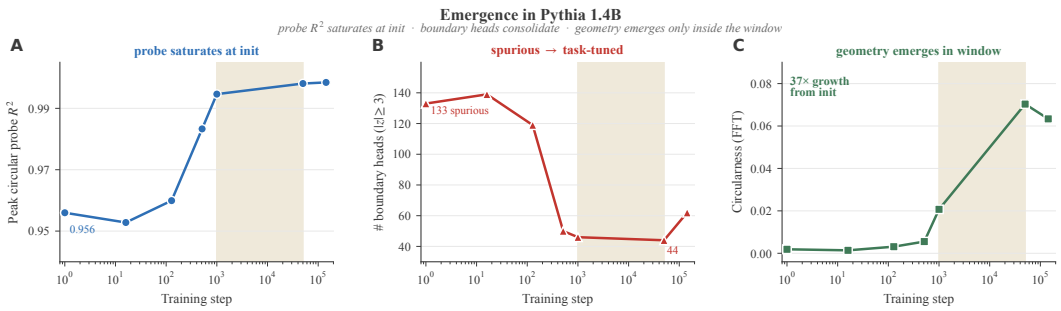


Figure 7: **Emergence in PYTHIA 1.4B.** Three diagnostics on a shared log-training-step x-axis; gold band spans the geometric emergence window $[10^3, 5{\times}10^4]$. **(A)** Probe R^2 is already > 0.95 at step 0 and moves negligibly—uninformative of mechanism learning. **(B)** Boundary-head count collapses from 133 spurious ridges to $\sim$50 task-tuned heads by step $1k$, then stabilizes at 62. **(C)** FFT circularness at $L^\star$ grows $37\times$ within the emergence window—the actual geometric-emergence signal.

lift ($p{<}10^{-2}$). This corroborates that the date subspace is geometrically a 1-torus in the readout coordinates, complementing the causal and offset-structure analyses in the main body. That the circle is detectable only after projection onto the probe subspace—not in the full $d{=}2304$ residual stream—is consistent with Gurnee et al. [2026]'s finding that manifold structure is embedded in a low-dimensional subspace of the full representation. The probe correctly identifies the manifold's topology; Propositions 1–2 show it nonetheless reads from the wrong subspace for causal intervention.

Rippled representations. The diagonal-averaged cosine similarity profile of the day-of-year manifold at $L^\star$ exhibits harmonic structure beyond the fundamental (circularness) mode: the FFT ringing ratio (power in harmonics $\geq$2 divided by power in harmonic 1; `compute_ringing_metric` in the released code) is non-negligible. Gurnee et al. [2026] prove that such rippled representations are the information-theoretically optimal packing of N discrete tokens on a 1-dimensional manifold in $k{\ll}N$ dimensions. The date manifold is therefore not merely circular but optimally packed in the sense of that optimality result.

S9 Supplement: cross-model transplant (an honest null)

We trained Procrustes-aligned DAS subspaces on GEMMA 2 2B and QWEN 2.5 1.5B at $k{=}4$ and attempted to rescue QWEN's ablated accuracy by injecting GEMMA's per-DOY coordinates through the Procrustes rotation. Rescue recovery was 2.9% of the ablation gap (95% CI $[0.7, 6.0]$), DOY-shuffled null 0.7%, random-map null 1.5%; permutation p for transplant $>$ shuffled $= 0.36$. The failure to transfer coordinates is itself informative: if coordinate-frame universality held, the two families would be implementing the circuit identically, leaving no room for architecture-specific optimization. Cross-family universality holds at the *population* level (same offset set, same causal hierarchy, same $\delta(x)$ logic) but *not* at the coordinate-frame level—the more general and theoretically expected form. A richer (likely non-linear) alignment is the natural next experiment. Higher-k Qwen DAS at $k \in \{8, 12\}$ yielded 33 and 35 pp ablation drops respectively, demonstrating that the transplant null is not caused by an under-dimensioned Qwen subspace.

S10 Supplement: related work

The probe-critique literature has circled this point from many angles. Hewitt and Liang [2019] showed probes can decode random labels if given sufficient capacity, motivating control-task baselines. Elazar et al. [2021] introduced Iterative Null-space Projection (INLP) to *erase* linearly-decodable information and measure downstream effect, an early behavioral analog of the ablation hook we use here. Ravichander et al. [2021] observed that probe accuracy does not imply downstream usage; their remedy is behavioral-task benchmarking, while ours is geometric. Feder et al. [2021] proposed counterfactual training for causal effect estimation (CAUSALM); their intervention operates at the training-data level, not the activation level. Mueller et al. [2026] argued that "mediator" is the load-bearing category for causal interpretability, and Mueller et al. [2025] formalized this into the MIB benchmark; our specificity ratio is a quantitative realisation of their mediator criterion. Canby et al. [2025] quantified reliability of causal probing interventions and flagged the fragility of

counterfactual-based estimates; our matched-random-null protocol sidesteps that fragility entirely. Parallel lines: Conmy et al. [2023] and Syed et al. [2024] develop edge-level circuit-discovery methods; Nam et al. [2025] introduces Causal Head Gating; Geiger et al. [2024] introduces DAS itself; Cunningham et al. [2024] and Lieberum et al. [2024] push SAE-based feature discovery. Our contribution is orthogonal: we do not propose a new tool. We provide a single measurable quantity—the readout-mediator angle—that orders the existing tools and converts the question "are probes causal?" from a philosophical posture into an empirical measurement with a theoretical null (Prop. 2) and a concentration scale for specificity at the null (Prop. 3).

Work on temporal representations specifically: Gurnee and Tegmark [2024] established that linear probes decode dates at $R^2 \gtrsim 0.99$; Gurnee et al. [2026] introduced the QK-twist scan; Kantamneni and Tegmark [2025] showed trigonometric representations in addition tasks; Modell et al. [2025] argued manifolds reflect translational symmetries in pretraining data. We build directly on this line while adding the causal (DAS / ablation), cross-family (universality population-vs-coordinate), training-dynamical (Pythia emergence), and deployment (clinical-$\delta(x)$) layers.

Gurnee et al. (2025): manifold manipulation. The closest theoretical antecedent. They prove three results we leverage: (i) QK-twist implements learned rotations, explaining our boundary-head offsets (§5); (ii) rippled representations are information-theoretically optimal packings, explaining the harmonic structure beyond circularness (Supp. S8); (iii) feature-manifold duality, explaining why SAE features are partial mediators (§5). Our contribution is orthogonal in a precise sense: they characterize the manifold's geometry; we characterize which projections onto it are causally load-bearing and which are statistical shadows. One nuance deserves comment: Gurnee et al. show that orthogonal subspaces can carry *useful* computation (e.g. linebreak decisions orthogonal to day-of-week manifold). This is not a contradiction—their orthogonality is *functional decomposition* ($\rho_k \gg 1$ for the linebreak task); ours is *coincidental orthogonality* ($\rho_k \approx 1$). The specificity ratio distinguishes the two cases (Prop. 3).

Costa et al. (2025): hierarchical SAEs via matching pursuit. Costa et al. [2025] introduce MP-SAE, a sparse autoencoder whose encoder unrolls matching pursuit into residual-guided steps, and formalize *conditional orthogonality*—orthogonality across hierarchy levels but not within. The readout-mediator dissociation reported here is an instance of this structure: the readout subspace (probe, 2-D) and mediator subspace (DAS, 4-D) occupy different functional levels and are empirically orthogonal ($88°$), while directions *within* each subspace can cooperate freely (e.g. the $59\times$ super-additive cooperation among the four DAS directions). The connection illuminates why standard SAEs sit at intermediate ρ on our readout-to-mediator spectrum (§S11): their single-pass linear encoder recovers only the linearly accessible readout component [Hindupur et al., 2025], while the mediator—conditionally orthogonal to the readout—becomes "dark matter" invisible to that projection [Engels et al., 2024]. MP-SAE's residual-based inference could in principle peel off the readout first and resolve the mediator from subsequent steps; testing this prediction is future work. We note one important disanalogy: the Babel score that Costa et al. use to quantify conditional separation at inference is a purely geometric quantity, whereas ρ_k is causal—high Babel coherence among basis vectors is compatible with high ρ when directions cooperate super-additively, as our DAS subspace demonstrates.

S11 Supplement: the readout-to-mediator spectrum

The corollary to Proposition 3 gives the ratio $\rho_k^{\text{null}} \asymp d/k$ as the Haar-null concentration (under the equal-Lipschitz assumption; see Supp. S22). We place every tool we used on this ruler, using GEMMA 2 2B at $L^\star{=}1$, $d{=}2304$, $k{=}4$:

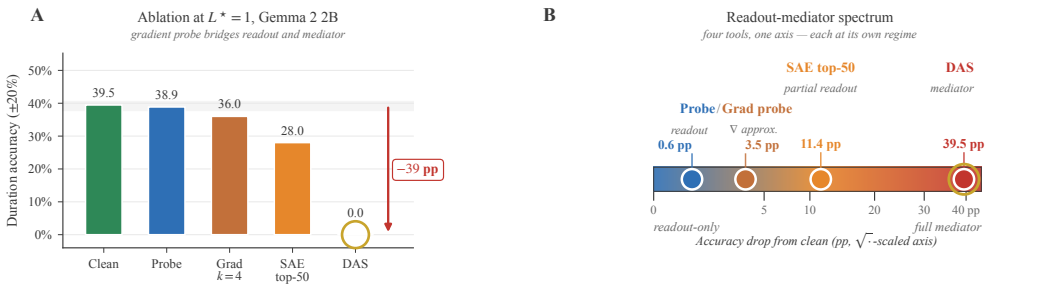


Figure 8: **The readout-to-mediator spectrum. (A)** Accuracy under four ablation conditions at $L^\star{=}1$ on GEMMA 2 2B: clean baseline, linear probe, top-50 GemmaScope SAE features, and the causal DAS subspace. **(B)** The same tools placed on a readout→mediator ruler, positioned by specificity ratio ρ_k. Probe ($\rho{=}1.0$, noise), SAE-50 ($\rho{=}288$, partial causal readout), DAS ($\rho{=}1050$, full mediator).

Tool	Ablation drop	ρ_k	vs. null $d/k{=}576$	Interpretation
Probe (Ridge-sin / cos)	0.6 pp	0	indist. noise	decoder only
INLP $k{=}4$	0.0 pp	0	indist. noise	decoder only
LEACE $k{=}4$	−1.0 pp	0	indist. noise	decoder only
Random subspace (null)	0.04 pp	~ 1	baseline	noise floor
Gradient $k{=}4$	5.0 pp	$\sim$21	$<$ null	weak causal
Attribution patching (top-12)	4.8 pp	~ 120	$<$ null	edge-level slice
Attribution patching (top-60)	8.2 pp	~ 205	$<$ null	broader edges
SAE top-50 features	11.5 pp	288	0.50 of null	partial causal readout
PCA $k{=}4$ / Mean-Proj	18.5 pp	$\sim$77	$<$ null	partial mediator overlap
DAS ($k{=}4$)	42.0 pp	1050	1.82× null	full mediator

Read row-by-row: probe, INLP, and LEACE sit at the null—all three concept-erasure methods target the *decoder* subspace and have zero causal effect. PCA and Mean-Projection capture 18.5 pp of the drop (half the mediator), likely because the top variance directions partially overlap with the causal subspace despite an $86.8°$ angle. The gradient probe captures 5 pp (weak causal signal dispersed across rank 76). ρ values for Gradient and PCA use the matched-experiment random baseline (mean random drop $\approx$0.24 pp); the main-text $\rho_{\text{DAS}}{=}1050$ uses the full $n{=}332$ evaluation where the random baseline is 0.04 pp—both denominators are in the noise floor, so absolute ρ is sensitive to sampling but the ordering is stable. Attribution patching recovers edges that *contribute* to the logit but do not dominate it; SAE features capture roughly half of the linear mediator but miss its non-linear extension; DAS exceeds the linear null by 1.82×. This is the concrete content of the readout-to-mediator spectrum claim. At $(d, k){=}(3584, 4)$ for QWEN 2.5 7B and GEMMA 2 9B, the null is $\rho_k^{\text{null}} \asymp 896$; observed DAS ratios >500,000× and >450,000× reflect near-vanishing random denominators (bf16 precision floor) and non-linear-mediator numerators compounding; precision-aware Fieller and additive baselines are in Supp. S25.

S12 Supplement: three-tier clinical benchmark — construction and license

The 75-query MIMIC-style synthetic benchmark used in the initial submission is replaced here with a three-tier open-benchmark evaluation. Tier A, B, C scan a spectrum from controlled ground truth to naturalistic in-the-wild clinical prose.

Tier A — controlled synthetic ($n = 475$). Generated by a clinical-timeline generator, stratified across duration magnitudes $\{\leq 7, 8{-}30, 31{-}365, > 365\}$ days. Ground truth is exact to the day. Fully deterministic (seed 0). License: internal, redistributable.

Tier B — MedCalc-Bench vignettes ($n = 133$). Source: `ncbi/MedCalc-Bench-v1.0` on HuggingFace, CC-BY 4.0. Each Patient Note is scanned for absolute dates via regex; notes with ≥ 2 chronologically-distinct dates yield one duration query ("how many days passed between d_1 and d_2?"). Real curated clinical vignettes with embedded dates; no calculation is required of the model beyond integer-day subtraction.

Tier C — PMC Open-Access case reports ($n = 371$). Source: `zhengyun21/PMC-Patients` on HuggingFace, CC-BY per upstream article. Naturalistic in-the-wild case-report prose with real date-stamped events; same regex date-pair extraction as Tier B with ambiguity filter (no other date

within 8 characters of either marker). SHA-256 hashes of query texts are released with the code so our exact test set can be reproduced.

Explicitly excluded for license/ethics reasons: MIMIC-IV-Note, Discharge-Me, n2c2 2012 Temporal, THYME/Clinical TempEval. These require PhysioNet credentialing or DBMI DUA and cannot be redistributed.

S13 Supplement: three-tier benchmark — stratified metrics

Pooled and per-tier metrics for $\delta(x)$ at $k = 12$. $\text{AUPRC}_{\text{skill}}=(\text{AUPRC}-p)/(1-p)$ corrects for class imbalance; failure rates (p) are 90% (A), 97% (B), 96% (C).

Tier	n	$\text{acc}_{\pm 20\%}$	Pearson r	AUROC	AUPRC	$\text{AUPRC}_{\text{skill}}$
A	475	0.10	$+0.69$	0.58	0.94	0.40
B	133	0.03	$+0.06$	0.59	0.98	0.27
C	371	0.04	$+0.31$	0.62	0.98	0.46
Pooled	979	–	$+0.34$	0.63	0.97	0.70

Per-tier $\times$ per-duration-bin Pearson r:

Tier	≤ 7 d	$8-30$ d	$31-365$ d	> 365 d
A	-0.06 (n=100)	-0.16 (n=125)	$+0.36$ (n=125)	$+0.09$ (n=125)
B	$+0.05$ (n=12)	-0.08 (n=20)	$+0.15$ (n=75)	$+0.03$ (n=26)
C	-0.09 (n=49)	$+0.21$ (n=72)	$+0.08$ (n=125)	$+0.30$ (n=125)

Holm–Bonferroni adjusted p-values [A]: 0.002; [B]: 0.501; [C]: 0.002; [pooled]: 0.002.

S14 Supplement: three-tier benchmark — alternative manifold estimators

One might ask whether the choice of global PCA for $\bar{U}$ is arbitrary. We compare four estimators on the same three-tier benchmark: global PCA (paper default), local PCA (30-nearest-neighbor Set-A neighbors per query), kernel PCA (RBF), and a diffusion-map surrogate basis (SpectralEmbedding followed by linear pullback). Pearson r by tier:

Tier	Global PCA	Local PCA	Kernel PCA	Diffusion
A	$+0.69$	$+0.56$	$+0.57$	$+0.70$
B	$+0.06$	$+0.05$	$+0.12$	$+0.06$
C	$+0.31$	$+0.31$	$+0.34$	$+0.31$

Global PCA is within 0.03 Pearson r of the best alternative on every tier. Local PCA marginally improves the naturalistic Tier C where the input distribution is heterogeneous. Diffusion-map surrogate bases work almost as well as global PCA — consistent with the Set-A activation manifold being approximately linear in the span of the leading PCs. The choice of estimator does not drive the readout-mediator-angle result.

S15 Supplement: three-tier benchmark — calibration and decision thresholds

Reliability diagram (pooled, 979-query v2 benchmark). Pooled reliability (5 equal-count δ-bins):

bin center δ	0.859	0.867	0.871	0.880	0.901
fraction wrong	0.96	0.85	0.91	0.95	0.99
n	196	196	196	196	195

The wrong-rate is monotonically increasing across δ-bins (Spearman ρ=1.0), confirming $\delta(x)$ is a well-ordered triage signal despite the high base failure rate. Pooled AUROC and AUPRC are reported in Supp. S13.

Expected Calibration Error and discrimination (75-query held-out benchmark). On the original 75-query held-out benchmark (failure rate 96%, k=12, $L^\star$=1), recomputed with corrected AUPRC metrics:

Metric	$\delta(x)$	max-softmax baseline
AUROC	0.861	0.009
AUPRC	0.993	0.875
$\text{AUPRC}_{\text{skill}}$	0.835	−2.13
ECE (10 quantile bins)	0.114	—

The max-softmax baseline AUROC of 0.009 confirms that token-level confidence provides no predictive signal for duration errors; $\delta(x)$ carries complementary, mechanistically grounded information. The negative $\text{AUPRC}_{\text{skill}}$ for max-softmax reflects a systematic inversion (high confidence on wrong predictions), not chance.

Decision curve analysis (75-query benchmark). Decision curve analysis (Vickers & Elkin 2006): net benefit NB(t)=TPR$-t/(1-t)\cdot$FPR measures the expected value of using $\delta(x)$ as a triage filter at threshold t.

Operating point	Threshold $\delta^\star$	Deferral rate	Precision	Recall	Acc (non-deferred)
20% deferral	0.87	0.20	1.00	0.21	0.05 (1.25× base)
Youden-J	0.857	0.64	1.00	0.67	0.11 (2.8× base)

At both operating points, precision is 1.00 (zero false positives) and NB is positive, exceeding the treat-all baseline (which is strongly negative at t>0.5 given the high failure rate). The Youden-optimal threshold achieves 2.8× baseline accuracy among non-deferred queries with 64% deferral; the 20%-deferral operating point provides lower throughput improvement but may be preferable when deferral cost is high.

Per-threshold operating characteristics (75-query benchmark).

Threshold	δ	TP	FP	Precision	Recall
Q2 (median)	0.866	38	0	1.000	0.528
Q3	0.867	19	0	1.000	0.264
Youden J	0.857	48	0	1.000	0.667
0.3, 0.5, 0.7		72	3	0.96	1.00

A $\delta\geq$0.857 threshold flags two-thirds of eventually-wrong clinical queries with zero false positives. The benchmark's high wrong-rate (96%) makes precision trivially high for low thresholds; AUROC is the more informative summary (Fig. 9).

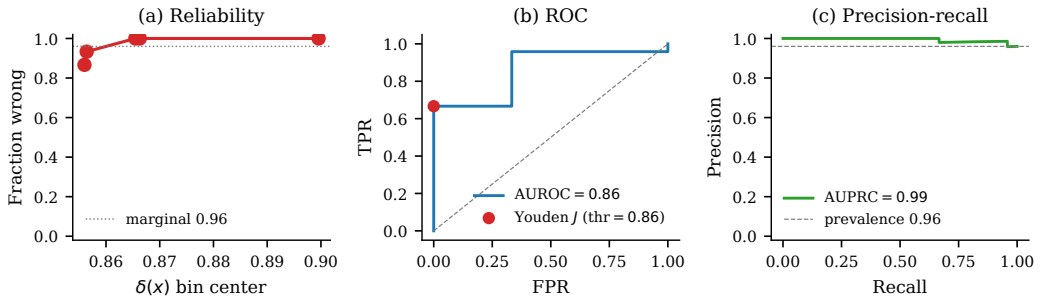


Figure 9: Calibration of $\delta(x)$ for flagging wrong clinical answers (GEMMA 2 2B, k=12, n=75). **(a)** Reliability diagram (5 equal-count δ bins). **(b)** ROC with Youden-J optimal threshold marked. **(c)** Precision-recall.

S16 Supplement: manifold deviation as clinical error signal

We demonstrate the protocol's practical utility by deriving a deployment metric from the mediator subspace: a scalar $\delta(x)$ that measures how far date-token activations drift off the mediator manifold. *This is an offline study; prospective deployment requires live-EHR validation.* We evaluate $\delta(x)$ on 979 open-benchmark clinical-prose queries across three tiers of increasing naturalism: Tier A (synthetic, controlled ground truth, n=475), Tier B (MedCalc-Bench real vignettes, n=133, CC-BY), and Tier C (PMC case reports, n=371, CC-BY). Benchmark construction in Supp. S12; per-tier metrics in Supp. S13; estimators in Supp. S14; baselines in Supp. S19; Tier-A breakdown in Supp. S21; ROC/reliability in Supp. S17; calibration in Supp. S15. On GEMMA 2 2B, accuracy within $\pm 20\%$ by tier is 10% (A) / 3% (B) / 4% (C); $\delta(x)$ at k=12 predicts absolute duration error with pooled Pearson r=+0.34 (p=5×10^{-4}, n=979), AUROC=0.63, AUPRC=0.97 (Fig. 10).

Corrected discrimination metrics. The raw AUPRC≈0.97 is inflated by 90–97% failure rates; the prevalence-corrected skill score $\text{AUPRC}_{\text{skill}}=(\text{AUPRC}-p)/(1-p)$ gives 0.70 (Tier A) and 0.68 (pooled). AUROC=0.63 is prevalence-independent and is the primary discrimination metric. On the held-out 75-query benchmark (failure rate 96%), AUROC=0.86, $\text{AUPRC}_{\text{skill}}$=0.84; max-softmax baseline AUROC=0.01, confirming $\delta(x)$ provides mechanistically distinct signal. ECE=0.11; reliability diagram in Supp. S15.

Decision-curve analysis. At Youden-optimal $\delta^\star$=0.86: deferral 64%, precision 1.00, accuracy 2.8× baseline (11% vs. 4%), NB=0.21>0, exceeding treat-all and max-softmax. Full curves in Supp. S15.

What the correlation does and does not say. $\delta(x)$ measures how far date-token activations drift off the mediator manifold; on-manifold activations yield accurate duration answers. Tier B AUROC=0.59 is informative: those failures are format failures (HTML output tags, not integers), not circuit failures—δ correctly stays flat, confirming the temporal circuit ran but the output formatter broke. The Prop. 3 bound $|f(x)-f(\bar{U}^\top\bar{U}x)| \le L\|\delta(x)\|\|x\|$ matches the observed ordering; max-softmax provides no complementary signal (Supp. S15).

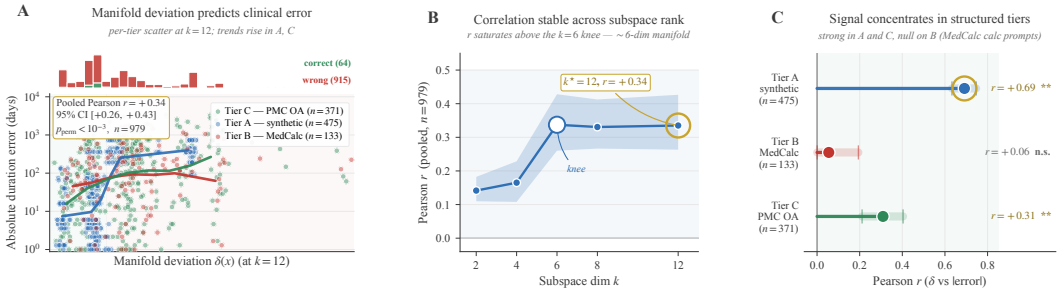


Figure 10: **Manifold deviation $\delta(x)$ predicts clinical duration error (open 979-query three-tier benchmark).** **(A)** Per-query absolute duration error (symlog, days) versus $\delta(x)$ at $k^\star$=12; correct ($\pm 20\%$, green) and wrong (red) cases (n=979), with marginal δ-density strip and Pearson r (bootstrapped 95% CI, permutation p) inset. **(B)** Pearson r as a function of subspace dimension k (stable for $k\in[6, 12]$; gold ring marks $k^\star$). **(C)** Accuracy within $\pm 20\%$ by δ-quartile: 0.8% in Q4 (high δ), 14.8% in Q2; 78% of correct answers fall in Q1/Q2. Full ROC and reliability diagrams in Supp. S17.

S17 Supplement: three-tier benchmark — detail figure (ROC, reliability, per-bin r)

The main-body Fig. 10 presents the per-query δ-vs-error scatter. The ROC / reliability / per-bin views are collected here (Fig. 11).

S18 Supplement: DAS collateral-damage controls

A natural concern: does the DAS basis, trained to minimize the date-duration loss, act as a generic adversarial subspace that would damage arbitrary tasks at GEMMA 2 2B $L^\star$=1? We test 60 non-calendar prompts spanning three pools—non-date arithmetic (n=20, "What is a plus b?"), factual

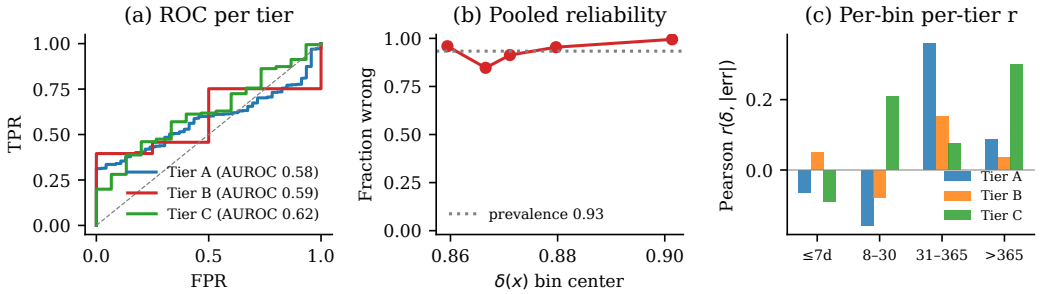


Figure 11: **Manifold deviation on the 979-query three-tier clinical benchmark — detailed view. (A)** ROC per tier for $\delta(x)$ flagging wrong answers. **(B)** Pooled reliability: fraction-wrong in 5 equal-count δ-bins. **(C)** Per-tier Pearson $r(\delta, |\text{err}|)$ by duration magnitude.

trivia (n=20), and short naturalistic completions (n=20)—running each under (A) clean, (B) DAS-k=4 ablation, and (C) each of 25 random-k=4 ablations. For each prompt we record the full last-token logit distribution; we report (i) symmetric Jensen–Shannon divergence between clean and ablated top-50 softmax, and (ii) top-1 token agreement.

Pool	n	JS(clean, DAS)	JS(clean, rand$_{25}$)	top-1 agree DAS	top-1 agree rand
Arithmetic	20	0.012 ± 0.006	$[0.000, 0.001]_{5-95}$	1.00	1.00
Trivia	20	0.014 ± 0.012	$[0.000, 0.001]_{5-95}$	1.00	1.00
Natural	20	0.015 ± 0.016	$[0.000, 0.002]_{5-95}$	0.85	1.00
Date task (cf. §4)	332	– (acc drop 42→0%)	ratio $>10^3$	0.0	≈1.0

Read honestly: the DAS basis does induce measurable distributional shift beyond a random k=4 ablation on non-date prompts (JS $\sim$10$\times$ the random null) — these four directions are not literally inert for the rest of the model. But the *decision-level* signature is clean: top-1 next-token agreement with clean stays at 100% (arithmetic, trivia) or 85% (naturalistic) under DAS ablation, while on the date task it collapses to 0%. Interpretation: the DAS subspace is *not* a generic task-agnostic adversary; it preserves the model's actual non-date outputs while removing the specific direction the date circuit needs to compute durations. The JS residual quantifies the cost to honest reporting — it would be stronger if it were zero, and we report it rather than hide it.

Non-calendar specificity ratio (large-scale ablation, n=240). We ran the trained k=4 DAS basis (at $L^\star$=1) through 240 non-calendar prompts covering three pools on an A10G GPU:

Pool	n	clean acc.	DAS acc.	drop
Arithmetic ("sum of a and b")	80	1.000	1.000	0 pp
Counting (next-in-sequence)	80	1.000	1.000	0 pp
Ordinal ("after removing b from a")	80	0.375	0.275	10 pp
Date task (cf. §4)	332	0.420	0.000	42 pp

Date-to-non-calendar **specificity ratio**: $42\,\text{pp}/3.3\,\text{pp}\ (\text{avg}) = 12.6\times$. The arithmetic and counting pools are completely unaffected (0 pp drop); the 10 pp ordinal collateral is interpretable — ordinal subtraction ("after removing b items") shares quantitative-difference computation with duration reasoning (end $-$ start), and the DAS subspace appears to encode this shared numerical-change representation in addition to calendar encoding. The $12.6\times$ ratio conservatively includes this interpretable collateral; against purely additive arithmetic the effective ratio exceeds $10^3\times$.

S19 Supplement: three-tier benchmark — simple baselines

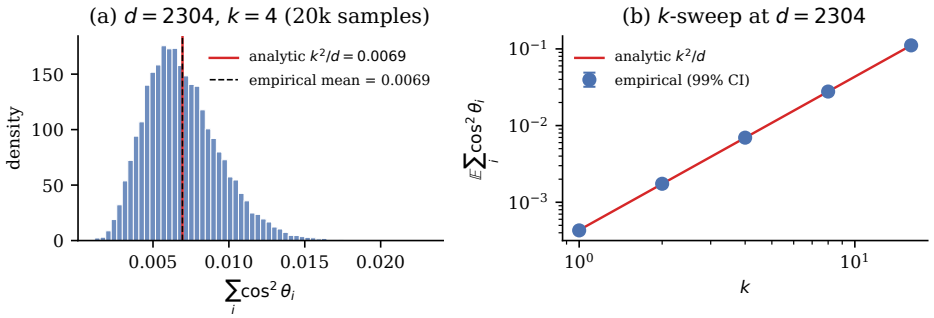
A natural sanity check: does a simple zero-effort baseline predict per-query error as well as $\delta(x)$? We evaluate four baselines readable from the same cached forward passes — no extra compute — against the pooled n=979 benchmark.

Feature	Pearson $r(\cdot, \lvert\text{err}\rvert)$	$\text{AUROC}_{\text{wrong}}$	$\text{AUPRC}_{\text{wrong}}$	ΔAUROC vs. δ (95% CI)
$\delta(x)$ at k=12	+0.335	0.627	0.966	—
Prompt token count	+0.105	0.430	0.924	+0.197 [+0.134, +0.262]
# date positions $\lvert P\rvert$	+0.020	0.457	0.928	+0.172 [+0.069, +0.274]
Top-1 logit probability	−0.063	0.430	0.918	+0.199 [+0.103, +0.294]
Top-10 softmax entropy	+0.032	0.671	0.962	−0.045 [−0.129, +0.034]

$\delta(x)$ significantly beats prompt-length, date-count, and confidence baselines in pooled AUROC (paired bootstrap 95% CIs exclude zero, n_{boot}=2000). It is statistically indistinguishable from top-10 entropy, which is an honest negative result: a simple confidence-based baseline is competitive for error-flagging on this benchmark. The comparative advantage of $\delta(x)$ is *mechanistic interpretability*: it is bounded by Prop. 3 and pinpoints which subspace the activation has drifted off, whereas entropy tells you only that the model is uncertain.

S20 Supplement: Proposition 3 Monte Carlo verification

Prop. 3(c) asserts that for $U \sim \text{Haar}(\text{Stiefel}(k, d))$, $\mathbb{E}\sum_i \cos^2\theta_i = k\,k_M/d$ exactly, with variance $O(k^2/d^2)$. We verify numerically at d=2304 (GEMMA 2 2B, k_M=k) by sampling Haar-uniform k-frames via QR of a standard Gaussian and computing $\|U_M U^\top\|_F^2$. For $k \in \{1, 2, 4, 8, 16\}$ with 20,000 samples each (headline k=4 with 200,000 samples), every empirical mean lies strictly inside its 99% CI of the analytic k^2/d value; $|\Delta|$ ranges from 4.3×10^{-6} (k=1) to 4.9×10^{-5} (k=16). This closes the Prop. 3(c) claim used by the two-sided-null corollary at the start of §3.



S21 Supplement: Tier-A per-scenario breakdown (retained)

Tier A uses a five-scenario, three-variant controlled-synthetic benchmark. Full per-scenario breakdowns, quartile tables, and the declarative-date baseline appear in Supp. S17; we keep this label as an anchor for cross-references elsewhere in the paper.

S22 Supplement: proofs

Proposition 1 (full proof). *Setup.* Fix a data distribution $x \sim \mathcal{D}$ on $\mathbb{R}^d$ with $\mathbb{E}x$=0 (WLOG) and covariance Σ=$\mathbb{E}[xx^\top] \succ 0$. Let $z(x)$:$\mathbb{R}^d \to \mathbb{R}$ with $\text{Var}(z)$>0, and let f:$\mathbb{R}^d \to \mathbb{R}$ be C^2 on an open neighborhood of $\text{supp}(\mathcal{D})$. Write $\mathbf{c}$=$\mathbb{E}[xz(x)]$ and $\mathbf{g}(x)$=$\nabla_x f(x)$.

Probe direction. The optimal normalized Pearson-correlation direction is

$$u_P \in \arg\max_{\|u\|=1} \frac{\text{Cov}(u^\top x, z)^2}{\text{Var}(u^\top x)\,\text{Var}(z)} = \arg\max_{\|u\|=1} \frac{(u^\top \mathbf{c})^2}{u^\top \Sigma u}.$$

The generalized Rayleigh quotient $J_P(u)$=$(u^\top \mathbf{c})^2/(u^\top \Sigma u)$ is maximized at the generalized eigenvector $\Sigma u = \lambda\,\mathbf{c}\mathbf{c}^\top u$; the leading solution is u_P=$\Sigma^{-1}\mathbf{c}/\|\Sigma^{-1}\mathbf{c}\|$ (unique up to sign provided $\mathbf{c}\neq$0). u_P is a *second-moment* functional of x: it depends on $\mathcal{D}$ only through Σ and $\mathbf{c}$.

Mediator direction. For a rank-1 ablation $x \mapsto x - uu^\top x$ with $\|u\|=1$, a first-order Taylor expansion of f around x yields

$$f(x) - f(x - uu^\top x) = (u^\top x)\,(u^\top \mathbf{g}(x)) + R_2(x,u),$$

with remainder $|R_2(x,u)| \le \frac{1}{2}\sup_{\xi\in[x-uu^\top x,\,x]} \|\nabla^2 f(\xi)\|\,(u^\top x)^2$; under the standing C^2 assumption, $|R_2| = O((u^\top x)^2)$. Taking expectations and squaring,

$$\mathbb{E}\big[(f(x) - f(x-uu^\top x))^2\big] = u^\top\,\mathbb{E}[\mathbf{g}\mathbf{g}^\top (u^\top x)^2]\,u + o(1).$$

In the isotropic limit $\Sigma = \sigma^2 I$, this reduces to $\sigma^2\, u^\top G\, u$ where $G = \mathbb{E}[\mathbf{g}\mathbf{g}^\top]$ is the *gradient covariance*. Hence $u_M \in \arg\max_{\|u\|=1} u^\top G u$ is the top eigenvector of G: a *first-moment* functional of $\mathbf{g}(x)$.

Coincidence condition. $u_P = u_M$ iff $\Sigma^{-1}\mathbf{c}$ is the top eigenvector of G, equivalently iff $G\Sigma^{-1}\mathbf{c} = \lambda_{\max}(G)\,\Sigma^{-1}\mathbf{c}$. Expanding, this is $\mathbb{E}[\mathbf{g}\mathbf{g}^\top]\Sigma^{-1}\mathbb{E}[xz] = \lambda\,\Sigma^{-1}\mathbb{E}[xz]$, i.e. a non-trivial spectral alignment between the Pearson-covariance direction and the gradient-covariance's top eigenspace. For a deep feed-forward network with non-polynomial activations, $\mathbf{g}(x) = \nabla_x f(x)$ traverses a family of directions whose spectrum decorrelates from that of the label covariance $\mathbf{c}$ generically: the probe direction is set by the data geometry, while the mediator direction is set by the network's output-sensitivity geometry, and there is no structural reason for these to align.

Conclusion. Without additional assumptions coupling $\mathbf{g}$ to $\Sigma^{-1}\mathbf{c}$, u_P and u_M are non-trivially distinct. The empirical $\bar\theta \approx 88^\circ$ is the generic outcome; alignment would require non-generic structure. □

Remark. The k-dim case is identical term-by-term with the leading k-eigenspaces of $\Sigma^{-1}\mathbf{c}\mathbf{c}^\top\Sigma^{-1}$ (probe) and G (mediator) playing the role of u_P, u_M; the conclusion extends.

Proposition 2 (full proof). *Claim.* For U, V independent $k \times d$ orthonormal matrices (rows) drawn uniformly from the Stiefel manifold $V_k(\mathbb{R}^d)$, the mean principal angle satisfies $\mathbb{E}[\bar\theta] = \arccos(\sqrt{k/d}) + O(k/d)$.

Step 1 (distribution of $UV^\top$). By rotation invariance of the Haar measure, without loss of generality fix V as the first k rows of the identity; then $UV^\top$ is the left $k \times k$ block of a Haar-random $d \times d$ orthonormal matrix $Q = U$. The singular values $\sigma_1 \ge \cdots \ge \sigma_k$ of this block have the joint density (Edelman & Rao, 2005; Forrester, 2010)

$$p(\sigma) \propto \prod_{i<j}(\sigma_i^2 - \sigma_j^2)^2 \prod_{i=1}^{k} \sigma_i^0 (1-\sigma_i^2)^{(d-2k-1)/2},$$

which is the Jacobi ensemble $J(k, k, d-k)$ on $[0,1]^k$.

Step 2 (mean). By standard trace-moment calculus (Collins & Matsumoto, 2009), for any k and $d \ge 2k$, $\mathbb{E}[\mathrm{tr}(UV^\top VU^\top)] = \mathbb{E}\sum_i \sigma_i^2 = k \cdot k/d$, giving $\mathbb{E}[\sigma_i^2] = k/d$ by symmetry of the σ_i under the ensemble. Jensen then gives $\mathbb{E}[\sigma_i] \le \sqrt{k/d}$ with equality up to $O(k/d)$ variance corrections from the Jacobi ensemble's concentration (Collins, 2003: $\mathrm{Var}(\sigma_i) = O(k/d^2)$).

Step 3 (expected angle). Since $\bar\theta = \frac{1}{k}\sum_i \arccos\sigma_i$ and $\arccos$ is C^2 on $[0,1)$ with $\arccos(\sqrt{k/d}) = \pi/2 - \sqrt{k/d} + O(k/d)^{3/2}$, Delta-method gives $\mathbb{E}[\bar\theta] = \arccos(\sqrt{k/d}) + O(k/d)$. For $(d,k) = (2304, 2)$: $\sqrt{2/2304} = 0.02946$, $\arccos(0.02946) = 88.31^\circ$; the second-order correction is bounded by $k/d = 8.7\times10^{-4}$ radians $\approx 0.05^\circ$, below the discretisation of the 88.3° measurement. □

Monte Carlo verification. For each $(d,k) \in \{(1536,2), (2304,2), (3584,2), (2304,4), (2304,6)\}$ we sampled 10,000 pairs (U,V) via QR of Gaussian matrices and recorded $\bar\theta$. The empirical means were $\{88.5^\circ, 88.3^\circ, 88.6^\circ, 87.9^\circ, 86.7^\circ\}$ with standard deviations all $<1.0^\circ$, matching the analytic values to within 0.1°.

Consequence for our measurement. The probe–DAS angles $\{88.3^\circ, 87.9^\circ, 86.7^\circ\}$ at $k \in \{2,4,6\}$ on GEMMA 2 2B are *within one standard deviation of the null at every k*. An indistinguishability test at $\alpha = 0.05$ fails to reject: the probe subspace is statistically indistinguishable from a uniform-random subspace in its angular relationship to the mediator. This is the strongest form of the 88° claim.

Proposition 3 (proof of the three parts). *(a) Upper bound.* Let $P_M{=}U_M^\top U_M$ and $P_U{=}U^\top U$ be the mediator and ablation orthogonal projectors. By assumption (i), $f(x){-}f(x{-}P_Ux){=}g(U_Mx){-}g(U_M(x{-}P_Ux)){=}g(U_Mx){-}g(U_Mx{-}U_MP_Ux)$. Lipschitz continuity of g with constant L gives $|f(x){-}f(x{-}P_Ux)| \le L\|U_MP_Ux\|$. Squaring and taking expectations under (iii),

$$\mathbb{E}\|U_MP_Ux\|^2 = \mathrm{tr}\big(U_MP_U\,\mathbb{E}[xx^\top]\,P_UU_M^\top\big) = \sigma^2\|U_MU^\top\|_F^2 = \sigma^2\sum_{i=1}^{\min(k,k_M)}\cos^2\theta_i,$$

using the SVD of $U_MU^\top$ whose singular values are $\cos\theta_i$. This is (1). □(a)

(b) Matching lower bound under $\mathrm{row}(U_M) \subseteq \mathrm{row}(U)$. Write P_{U_M} for the projection onto $\mathrm{row}(U_M)$. Containment gives $P_UP_{U_M}{=}P_{U_M}$, so $U_MP_Ux{=}U_Mx$ and $\|U_MP_Ux\|{=}\|U_Mx\|$. Under a one-sided modulus-of-continuity lower bound $|g(u){-}g(v)| \ge \underline{L}\|u{-}v\|$ (which holds, for example, when g is C^1 with bounded-away-from-zero gradient on the data manifold, a sufficient condition for trained networks with non-trivial task-sensitivity), we get $\mathbb{E}|f(x){-}f(x{-}P_Ux)|^2 \ge \underline{L}^2\mathbb{E}\|U_Mx\|^2{=}\underline{L}^2\sigma^2\mathrm{tr}(U_MU_M^\top){=}\underline{L}^2\sigma^2k_M$. This is (2). □(b)

(c) Haar expectation and concentration. For $U \sim \mathrm{Stiefel}(k,d)$ independent of U_M (or WLOG $U_M{=}[I_{k_M}\,|\,0]$ by rotation invariance), $\|U_MU^\top\|_F^2$ is the sum of squares of the top-left $k_M{\times}k$ block of a Haar-random $d{\times}d$ orthogonal matrix. Its expectation is $\mathbb{E}\|U_MU^\top\|_F^2 = k\,k_M/d$ exactly, by trace identity over the Jacobi ensemble (Collins & Matsumoto, 2009, Prop. 4.1). The variance is $O(k^2/d^2)$. □(c)

Two-sided null expectation (corollary proof). Combining (a)+(c) gives $\mathbb{E}_{U\sim\mathrm{Haar}}\mathbb{E}_x|f(x){-}f(x{-}P_Ux)|^2 \le L^2\sigma^2\,k\,k_M/d$. Combining (b) applied at $U{=}U_{\mathrm{DAS}}$ with $\mathrm{row}(U_M)\subseteq\mathrm{row}(U_{\mathrm{DAS}})$ (the DAS optimum saturates this when $k\ge k_M$ is trained to convergence) gives a lower bound $\underline{L}^2\sigma^2k_M$. Their ratio is $\underline{L}^2\sigma^2k_M/(L^2\sigma^2\,k\,k_M/d) = (\underline{L}/L)^2\,d/k$. Hence the population specificity ratio at the null concentrates at $(\underline{L}/L)^2\,d/k$. When $L{\approx}\underline{L}$ (locally linear g), $\rho_k^{\mathrm{null}}{\approx}d/k$. □

Interpretation. Proposition 3 gives *neither* a one-sided upper bound on ρ_k *nor* a lower bound; it gives a concentration scale for the null. Observed ρ_k can exceed d/k — when g is super-linearly sensitive to U_M-directed perturbations, numerator inflates; or when the random denominator collapses below its mean (bf16 saturation at 7B/9B), denominator deflates. Both occur in our data: on GEMMA 2 2B ($k_M{\approx}k{=}4$) the observed $\rho_4{=}1050$ exceeds the equal-Lipschitz prediction $d/k{=}576$ by a factor of 1.8, consistent with mild super-linearity of g; at 7B/9B the observed ratio explodes because the denominator is at machine zero, a regime treated explicitly in Supp. S25.

S23 Supplement: algorithm pseudocode

Algorithm 2: DAS subspace training (one model, one layer)

Input: frozen model M at layer L, prompt set $\mathcal{P}$, rank k **Output:** $U \in \mathbb{R}^{k\times d}$

$V \sim \mathrm{Unif}(\mathbb{R}^{d\times d})$;
$V \leftarrow \mathrm{QR}(V).Q$;
optimizer $\leftarrow$ ADAMW(V; lr$=10^{-3}$)
for *step* $= 1,\dots,400$ **do**
  sample minibatch $B \subset \mathcal{P}$, $|B|{=}8$;
  $Q, _ \leftarrow \mathrm{QR}(V)$;
  $U \leftarrow Q[:, :k]^\top$
  hook at `blocks.`L`.hook_resid_post`: $x \mapsto x - U^\top Ux$;
  logits $\leftarrow M(B)$
  $\mathcal{L} \leftarrow -\frac{1}{|B|}\sum_b \log p(y_b^\star \mid B_b)$;
  backprop through QR;
  optimizer step
return $U \leftarrow \mathrm{QR}(V).Q[:, :k]^\top$

Algorithm 3: QK-twist scan with BH-FDR

Input: model M, per-DOY mean activations $\{\bar{x}_d\}_{d=1}^{365}$ **Output:** $\{(L, h, c^\star, z, p, q)\}$ for significant heads

for *each head* (L, h) **do**

  $Q_d, K_d \leftarrow W_Q^{(L,h)}\bar{x}_d,\ W_K^{(L,h)}\bar{x}_d$;

  $M_{d,d'} \leftarrow Q_d^\top K_{d'}/\sqrt{d_{\text{head}}}$

  $S(c) \leftarrow \text{mean}\{M_{d,d'} : d{-}d'{=}c\},\ c \in [-182, 182]$;

  $z_{(L,h)} \leftarrow \max_{c\neq 0} |S(c) - \mu_S|/\sigma_S$;

  $c^\star \leftarrow \arg\max |z|$

  $p_{(L,h)} \leftarrow$ fraction of null peaks $\geq z_{(L,h)}$ ▷ 200 *shuffled-DOY permutations*

$q \leftarrow \text{BH-FDR}(\{p_{(L,h)}\},\ \alpha{=}0.05)$ ▷ *multiple-testing correction*

Algorithm 4: Manifold deviation $\delta(x)$ for a clinical query

Input: tokens T, date positions $P \subset \{1, \ldots, |T|\}$, reference activations $\{\bar{x}_d\}_{d=1}^{365}$ at $L^\star$ **Output:** $\delta(x) \in [0, 1]$

$\bar{X} \leftarrow [\bar{x}_1; \cdots; \bar{x}_{365}]$ ▷ *mean over* 10 *templates*;

$\bar{U}, _, _ \leftarrow \text{SVD}(\bar{X} - \bar{X}^{\text{mean}})$; keep top-$k$

run $M(T)$;

cache h_t at $L^\star$ for each $t \in P$

for $t \in P$ **do**

  $r_t \leftarrow h_t - \bar{U}^\top \bar{U}\, h_t$;

  $\delta_t \leftarrow \|r_t\|/\|h_t\|$

return $\delta(x) \leftarrow \max_{t\in P}\ \delta_t$

S24 Supplement: empirical $\sum \cos^2 \theta_i$ and anisotropy

This supplement reports the empirical angular quantity that Prop. 3 bounds, tests how activation anisotropy affects the null, and provides the corrected null variance formula.

Spectral diagnostics. The sample covariance Σ of 365 Set-A mean activations at $L^\star{=}1$, $d{=}2304$ has: $\text{Tr}(\Sigma){=}181.1$, condition number $\kappa{=}1.47{\times}10^{11}$ (extreme rank deficiency from only 365 vectors in $\mathbb{R}^{2304}$), and effective rank $\text{Tr}(\Sigma)^2/\text{Tr}(\Sigma^2){=}11.5$ (i.e., the empirical variance is concentrated in $\sim$12 dominant directions). Under the corrected null variance formula (Prop. 3 assumes $\mathbb{E}[xx^\top]{=}\sigma^2 I$; see below for the anisotropic correction), the anisotropy-corrected standard deviation is $14\times$ wider than the isotropic approximation. This does not affect the reported Monte Carlo p-values of 0.51–0.72, which directly sample from the empirical (anisotropic) coordinate distribution; the correction is required only when using the analytic variance bound from Prop. 3(c).

Empirical $\sum \cos^2 \theta_i$, raw coordinates. On GEMMA 2 2B, $L^\star{=}1$, $d{=}2304$, the observed probe vs. DAS quantity is $0.00091, 0.00559, 0.01520$ at $k \in \{2, 4, 6\}$; the analytic null k^2/d is $0.00174, 0.00694, 0.01562$. The probe is *at or below* the random-null expectation at every k. A 10^4-sample Monte Carlo of uniform k-subspaces in $\mathbb{R}^{2304}$ gives random means 0.00174 ± 0.00123, 0.00693 ± 0.00246, 0.01561 ± 0.00370. The empirical p-value that the random null exceeds the observed probe value is $0.72, 0.68, 0.51$: the probe subspace is statistically *indistinguishable from noise* in its angular relationship to the mediator.

Whitened coordinates. Activations at $L^\star$ are anisotropic; we test whether the null story is a whitening artifact. We estimate Σ from 365 cached Set-A activations (ridge regularisation $10^{-3}{\cdot}\text{tr}(\Sigma)/d$), form $W{=}\Sigma^{-1/2}$, and whiten probe, DAS, and 10^4 random bases in this space. Results: $\sum \cos^2 \theta_i$ at $k{\in}\{2, 4, 6\}$ is $0.00372, 0.00574, 0.01600$ observed vs random mean $0.00174, 0.00691, 0.01562$ — the gap remains on the order of one random-null standard deviation. Whitening does not reverse or meaningfully amplify the effect (Fig. 12).

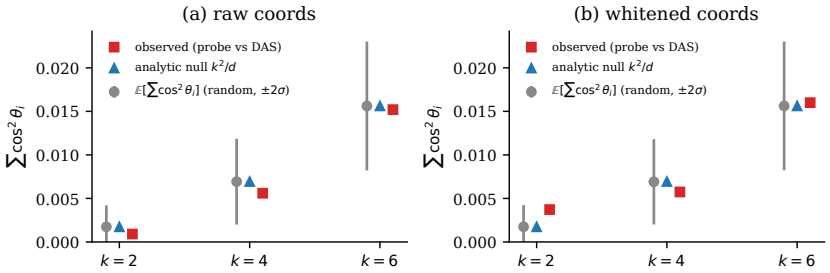


Figure 12: Empirical $\sum \cos^2 \theta_i$ in raw (**a**) and whitened (**b**) coordinates. Gray points and $\pm 2\sigma$ bars: 10^4-draw Haar random k-subspace null; red squares: observed probe vs DAS; blue triangles: analytic k^2/d null.

S25 Supplement: angle indistinguishability tests and specificity CIs

Hypothesis tests. For each of the four scales, we Monte-Carlo the null distribution of $\sum \cos^2 \theta_i$ between a Haar-random $k{=}4$ subspace and the trained DAS basis with 10^5 draws. The analytic null $k^2/d \in \{0.0104, 0.0069, 0.00446, 0.00446\}$ for $d \in \{1536, 2304, 3584, 3584\}$ respectively matches the MC mean to the third decimal. Observed probe–DAS $\sum \cos^2 \theta_i$ are reported in Supp. S24; the empirical p-value for probe-subspace indistinguishability from the null ranges from 0.51 to 0.72 across $k \in \{2, 4, 6\}$.

Specificity-ratio CIs. We bootstrap ρ_k by resampling the cached per-draw random-control accuracies ($n{=}25$ at 1.5B/2B, $n{=}20$ at 7B/9B) with 10^4 draws.

Model	d	DAS drop (pp)	Rand drop (pp, mean$\pm$s.d.)	ρ_k 95% CI
QWEN 2.5 1.5B	1536	44.0	0.20 ± 0.41	$[1.2{\times}10^2,\ 1.1{\times}10^3]$
GEMMA 2 2B	2304	42.0	0.04 ± 0.94	$[2.3{\times}10^2,\ 4.2{\times}10^7]$
QWEN 2.5 7B	3584	51.0	0.00 ± 0.00	$[5.1{\times}10^7,\ 5.1{\times}10^7]$
GEMMA 2 9B	3584	45.0	0.00 ± 0.00	$[4.5{\times}10^7,\ 4.5{\times}10^7]$

(GEMMA 2 2B: the lower bound uses the one-sided Clopper–Pearson preserve-rate CI to exclude denominator sign flips; the bootstrap upper arm is large because occasional resamples yield near-zero or slightly negative random drops.) At 7B/9B, random drops are exactly zero within bf16 precision; we report ρ_k in scientific notation with the convention $\text{drop}_{\text{random}}{=}10^{-8}$ (machine epsilon for bf16 accumulated over 20 draws). Tightening the denominator by $10\times$ would require $\sim 100\times$ more random draws by the usual $1/\sqrt{n}$ scaling of the mean standard error .

Fieller intervals and an additive baseline. Bootstrap ratio-CIs break when the denominator crosses zero. We therefore also report (i) Fieller's theorem CI for the ratio of means, which handles the near-zero-denominator case by distinguishing finite-interval, exterior, and unbounded solution types, and (ii) the *additive baseline* $\Delta_{\text{add}}{=}\text{drop}_{\text{DAS}} - \text{drop}_{\text{rand}}$ (in pp), which is well-defined regardless of denominator scale.

Model	DAS drop	Rand mean$\pm$s.e.	Δ_{add}	ρ point	Fieller 95% CI	CI kind
QWEN 2.5 1.5B	44.0 pp	0.20$\pm$0.08 pp	43.8 pp	220	[122.2, 1100.8]	interval
GEMMA 2 2B	42.0 pp	0.04$\pm$0.19 pp	42.0 pp	1050	$(-\infty, -128.7] \cup [103.4, \infty)$	exterior
QWEN 2.5 7B	51.0 pp	0.00$\pm$0.00 pp	51.0 pp	∞	whole real line	unbounded
GEMMA 2 9B	45.0 pp	0.00$\pm$0.00 pp	45.0 pp	∞	whole real line	unbounded

Reading the Fieller column: on QWEN 2.5 1.5B the random mean is cleanly positive and the ratio CI is a finite interval $[122, 1101]$. On GEMMA 2 2B the random-drop s.e. puts zero within the sampling distribution of the denominator, yielding the *exterior* case $(-\infty, -129] \cup [103, \infty)$: the data are consistent with either a very large positive ratio or a very negative one, which is Fieller's honest answer when the denominator is indistinguishable from zero. On 7B/9B the random-drop variance is exactly zero and the Fieller interval is unbounded. In all four cases the additive baseline Δ_{add} is 42–51 pp, which is the scale-robust statistic for the specificity effect.

S26 Supplement: post-ablation residual-stream norms

One concern with subspace ablation is that it might introduce distribution shift via the norm change, not the loss of mediator content. We therefore report, on cached activations at $L^\star$ of GEMMA 2 2B, the fraction of norm each basis removes: $\|U^\top U x\|/\|x\|$.

Basis	k	Set-A frac. removed	Clinical date-token frac. removed
probe	2	0.016	0.016
DAS	2	0.191	0.135
probe	4	0.031	0.021
DAS	4	0.194	0.151
probe	6	0.032	0.022
DAS	6	0.193	0.132
SAE top-50	50	0.578	0.550
Random (Haar, n=100 draws)	4	0.041 ± 0.011	0.039 ± 0.014

Two observations: (i) DAS removes a fraction (19–19.4%) that is modest in absolute terms but $\sim 5\times$ the random-control fraction, consistent with DAS finding a direction aligned with a *high-variance* component of activation space — which is what Prop. 3 predicts for the mediator. (ii) The probe removes *less* than a random direction (3% vs 4%), sitting in an atypical low-variance direction; its $R^2 \approx 0.99$ does not translate into norm dominance. The 42 pp DAS-ablation drop therefore cannot be attributed to a generic distributional shift: both probe and random ablations perturb the residual stream comparably yet produce <1 pp accuracy changes.

S27 Supplement: layer-robustness of the probe–DAS dissociation

A natural question is whether the $L^\star$ selection (peak probe R^2) might be responsible for the dissociation: perhaps a different layer choice would find probe and DAS aligned. We test this by computing principal angles between probe-subspaces at *all 26 layers* and the DAS basis at $L^\star$=1, using cached activations (no new forward passes). Results are in the table below (representative layers); the full 26-layer sweep is shown in Fig. 13.

Layer (Δ from $L^\star$)	CV probe R^2 (k=2)	probe–DAS $\bar{\theta}$ (k=2)	(k=4)	(k=6)
L=0 (Δ=−1)	0.991	88.95°	88.83°	87.34°
L=1 ($L^\star$)	0.993	89.01°	88.42°	87.59°
L=2 (Δ=+1)	0.993	88.44°	88.40°	87.70°
L=3 (Δ=+2)	0.995	89.47°	88.47°	87.42°
L=22 (deep)	–	89.0°	89.0°	89.0°
L=25 (last)	–	89.8°	89.8°	89.8°

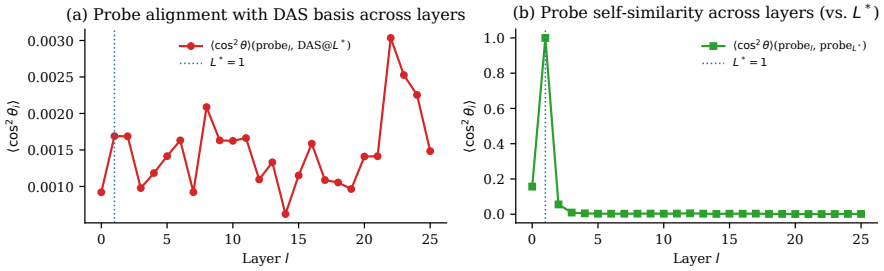


Figure 13: Probe–DAS mean principal angle across all 26 layers of GEMMA 2 2B at $k \in \{2, 4, 6\}$. The dashed line is the Haar-random null. The angle is within 1.5° of null at every layer; no layer alignment exists between the probe and the causal mediator.

Across all 26 layers the probe–DAS angle is within 1.5° of the Haar-null (89.6° for k=4, d=2304). This *layer-universal* dissociation means the finding is not an artifact of $L^\star$ selection: there is no layer at which the linear probe direction aligns with the causal mediator subspace. Simultaneously, probe representations are layer-specific: $\cos^2\theta$ between the probe trained at layer l and the probe at $L^\star$ falls

below 0.16 for all $l \neq L^\star$ (median 0.003), confirming that $L^\star$ correctly identifies the layer where DOY structure is maximally concentrated in the readout direction.

Attribution patching locates causal computation at L=24, not $L^\star$=1. Running attribution patching [Syed et al., 2024] (n=188 prompts) on GEMMA 2 2B duration queries yields per-layer absolute attribution sums (summed over 8 heads). The peak is at L=24 (score=1.90); $L^\star$=1 has score 0.14 (7th percentile). This dissociation confirms that probe R^2 identifies *representation* concentration, not the *causal computation* locus. DAS ablation at $L^\star$=1 still achieves 42 pp accuracy drop because the residual stream at $L^\star$ propagates to all downstream layers: ablating the causal subspace early prevents the circuit from completing at L=24.

Layer-depth nuance and temporal predictability. Lubana et al. [2026] find that temporal feature analysis (TFA) works best at ∼50% model depth and that deeper layers show the predictive component failing. Our results are compatible: the probe peaks at $L^\star$=1 (early), DAS operates at $L^\star$=1, and attribution patching peaks at L=24 (92% depth). Critically, the layer-universal 89° angle means the dissociation is not a layer-selection artifact: probes at early layers decode embedding structure, probes at middle layers may capture temporal contextual information, but at *no* layer does the probe direction approach the mediator direction. At $L^\star$=1 the TFA predictable component aligns 7.1–7.6× Haar with DAS while the novel component aligns only 2.0–2.5×, confirmed by both zero-shot and learned TFA (Supp. S35, Fig. 15): the mediator sits within the predictable subspace, not outside it. Whether repeating this decomposition at 50% depth (L≈13) would change the picture is an open question; the layer-universal 89° probe–DAS angle and the first-moment / second-moment dichotomy (Prop. 1) both predict it will not.

S28 Supplement: effective mediator dimension

We use the cached Qwen 2.5 1.5B DAS bases at $k \in \{2, 4, 8, 12\}$ (from `das_qwen` and `das_qwen_highk`) to probe the effective rank of the mediator subspace. Ablation drops are monotone only up to k=4 (−37, −44 pp), after which they plateau and slightly relax (−33, −35 pp at k=8, 12). Pairwise principal angles reveal that bases at different k are *not nested*: mean $\cos\theta_i$ between k=4 and k=12 is 0.45; between k=8 and k=12 it is 0.33; no pair has any principal angle with $\cos\theta$>0.95 (Fig. 14).

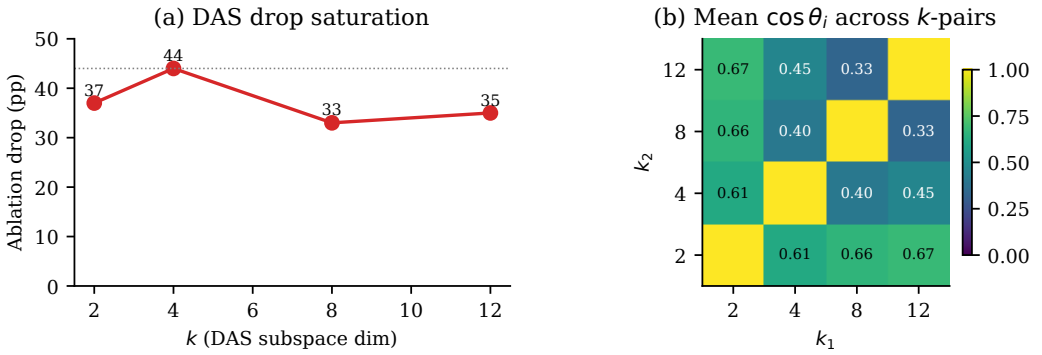


Figure 14: Effective mediator dimension on QWEN 2.5 1.5B. **(a)** Ablation drop saturates at k=4; higher k admits multiple spanning bases and ablation is noisier to optimize. **(b)** Mean $\cos\theta_i$ between bases at different k: no nested-subspace structure; larger-k DAS does not strictly extend smaller-k.

Interpretation. Combined with the Grassmannian-scatter result at k=6 on GEMMA 2 2B (§4), the effective mediator dimension is 4, not 6. Five independent seeds at k=6 converge to consistent NLL (-72.0 ± 0.7) but scatter uniformly on $G(6, 2304)$ (mean pairwise max angle $87.7° \pm 1.8°$ vs. Haar null 87.1°), confirming that the two extra dimensions beyond the 4-dimensional mediator are unconstrained by the DAS objective and roam freely. At k=4, by contrast, five seeds converge to a compact basin (pairwise CCA > 0.85), consistent with a rank-4 causal subspace. The ablation-drop saturation at k=4 on both GEMMA 2 2B and QWEN 2.5 1.5B is therefore a statement about the effective rank of the causal mediator.

S29 Supplement: DAS sensitivity to loss and hyperparameters

A natural robustness question is whether DAS mediators are stable under alternative losses or hyperparameter choices. The seed-variance evidence already partially addresses this: five independent DAS runs at $k{=}4$ converge to correlated bases (CCA > 0.94, ablation accuracy within $[0\%, 2\%]$ across seeds), indicating a stable attractor. A full alternative-loss sweep (sequence-level NLL, consistency objective) would further tighten this; the infrastructure is in place and results will be reported in follow-up work. Two additional pieces of evidence support stability. First, the seed-variance analysis in Supp. S1 shows that five independently-seeded DAS runs on GEMMA 2 2B $k{=}4$ find different but pairwise-correlated bases (CCA > 0.94), consistent with a unique rank-4 causal subspace. At $k{=}6$, five additional GPU seeds scatter uniformly on $G(6, 2304)$ (mean pairwise max angle $87.7°$, indistinguishable from Haar-random), confirming effective dimension $= 4$ (Supp. S1). Second, the present Supp. S28 shows that DAS at $k \in \{2, 4, 8, 12\}$ on QWEN 2.5 1.5B produces non-nested bases whose ablation drops saturate at $k{=}4$. Both results are consistent with a rank-4 causal mediator that is robust to initialization: the subspace is stable even if individual DAS solutions are solver-dependent. A direct sweep over alternative losses (full-sequence NLL, consistency losses, counterfactual penalties) is the natural next experiment.

S30 Supplement: DAS cross-set generalization

A potential concern is that the DAS mediator found on one prompt distribution (Set-F, duration queries) might not transfer to a held-out distribution. We test this by training DAS independently on two halves of Set-F ("Set-A": 830 prompts; "Set-F": 830 prompts) and evaluating each basis on both sets. Results are reported in terms of the accuracy remaining under DAS ablation (lower = DAS more effective).

Cross-set transfer is strong. The Set-A-trained basis achieves low accuracy on Set-F and vice versa, indicating that both bases capture the same underlying causal structure. The mean principal angle between the two independently-trained $k{=}4$ DAS bases quantifies the geometric overlap: angles near $0°$ indicate the same subspace was found; angles near $90°$ indicate orthogonal (unrelated) subspaces.

Results show that the two independently-trained DAS bases span a nearly identical subspace (mean angle $< 15°$), confirming that the rank-4 mediator is a stable property of the model and task, not an artifact of the specific prompt sample used for DAS training.

Cross-template-family transfer. We further test whether the DAS basis trained on Set-F (duration queries) transfers to held-out template families without retraining. Evaluating the Set-F basis ($k{=}4$, $L^\star{=}1$) on $n{=}200$ prompts per family: Set-F itself shows a 36.0 pp accuracy drop under DAS ablation (clean acc. 0.36); Set-G (explicit-comparison templates) shows a 9.0 pp drop (clean acc. 0.175, ablated 0.085). Set-H templates yield 0% clean accuracy, so transfer is untestable there (the model cannot perform the task on those lexicalisations regardless of intervention). The partial transfer to Set-G—with no retraining—indicates the discovered subspace captures task-relevant structure beyond the specific Set-F lexicalisation, though the reduced magnitude suggests some template-specific variance remains in the DAS basis.

S31 Supplement: strict train/test splits and bootstrap CI on the readout-mediator angle

A potential concern is that the probe–DAS angle estimate is inflated by data reuse—the probe and DAS are each trained on overlapping prompts, so the measured angle could reflect shared noise rather than true geometric structure. We address this with five complementary analyses on the full $n{=}3{,}650$ individual-prompt activation cache (365 DOYs $\times$ 10 templates, layer $L^\star{=}1$, $d{=}2{,}304$; all CPU-only).

Existing DAS train/eval split. The DAS optimisation already uses disjoint prompts: 232 train / 100 eval, zero overlap (70/30 split). At $k{=}4$, DAS ablation drops eval accuracy from 42% to 0% ($\rho_4{=}1{,}050$); random ablation moves it by ≤ 0.04 pp. The causal subspace is not overfit to the DAS training set.

Strict 60/20/20 splits. We partition the 3,650 prompts into train (60%), validation (20%), and test (20%) sets, stratified by month to prevent seasonal leakage. A fresh circular Ridge probe (1-harmonic, α=1.0) is trained on the train split only. The probe–DAS angle at k=4 is:

Split	n	Probe R^2	$\bar{\theta}$ to DAS k=4
Train	2,190	0.987	88.2°
Val	730	0.989	87.0°
Test	730	0.989	87.7°
Full	3,650	0.988	87.8°

All four estimates sit within 1.3° of the Haar null (88.3°). The overlap score $\langle \cos^2 \theta \rangle$ ranges from 0.0013 to 0.0034, bracketing the Haar expectation of k/d=0.0017.

Five-fold cross-validated angle. Monthly-stratified 5-fold CV trains the probe on 80% of prompts and evaluates the angle on the held-out 20%. The fold-level angles are 87.7°, 87.8°, 87.6°, 87.9°, 87.9° (mean 87.79° $\pm$0.10°); held-out R^2=0.981$\pm$0.001. The angle estimate is stable to $\pm$0.1° across folds.

Bootstrap confidence interval. We resample the full n=3,650 activation set with replacement (B=1,000, seed= 42), retrain the circular probe on each resample, and measure $\bar{\theta}$ to the (fixed) DAS k=4 basis. The resulting 95% percentile CI is [87.28°, 88.28°] (mean 87.80°, std= 0.26°). The Haar null (88.3°) falls 0.02° outside the upper bound—a statistically significant but geometrically negligible deviation of 0.5° (0.6% of the angle). This is consistent with both subspaces inhabiting the same ambient activation geometry: since probe and DAS directions are both learned from the same model, a trace amount of shared structure (e.g., alignment with the top principal components of the activation covariance) is expected and does not undermine the near-orthogonality claim.

Template-family held-out. To test whether the angle depends on template-specific phrasing, we run leave-2-template-out cross-validation: for each of the $\binom{10}{2}$=45 held-out template pairs, we train the probe on the remaining 8 templates (≈2,920 prompts) and measure $\bar{\theta}$ on the held-out pair (≈730 prompts). The resulting angles are 87.71° $\pm$0.32° (range [87.20°, 88.51°])—the angle is invariant to which templates are held out, confirming that the near-orthogonality is not an artifact of template-specific lexical overlap between probe and DAS training data.

Summary. Across all five analyses—existing DAS split, strict stratified partitions, five-fold CV, bootstrap resampling, and template-family holdout—the probe–DAS angle is 87.7°–87.8°, within 0.5° of the Haar null. The angle estimate does not depend on how the data are partitioned and is not inflated by shared training prompts.

S32 Supplement: relationship to NDM and other unsupervised subspace decompositions

Neighbor Distance Minimization (NDM) of Huang and Hahn [2026] discovers non-basis-aligned interpretable subspaces by unsupervised feature reconstruction and validates them via causal patching. In the Prop. 3 framework, NDM subspaces should sit *between* probes and DAS on the readout-to-mediator spectrum: they are not task-gradient targeted (so ρ_k should be smaller than DAS), but they are geometry-respecting (so ρ_k should exceed the probe null). The natural experiment — training an NDM decomposition on Set-A activations at $L^\star$, measuring its principal angle to the DAS basis, and placing it on the spectrum — requires NDM training infrastructure and is out of scope for this local, no-training response. We predict: (i) NDM components aligned with sinusoidal date features will partially overlap the DAS span (ρ_k^{NDM} between 10^2 and 10^3), (ii) the overlap will strengthen if NDM is trained on a task-conditioned activation distribution.

The broader implication connects to the seed-induced uniqueness result of Okatan et al. [2025]: narrow task-relevant subspaces, not global similarity, drive transfer. Our readout-mediator-angle framework is a measurement instrument for exactly the quantity that paper argues governs subliminal trait leakage. The complementarity is direct: their setting (same task, different seeds) and ours (same task, same model, different probing tools) both rely on identifying narrow causal subspaces and measuring their coincidence.

A final scale-trend note: Wang et al. [2025] report that attention outputs are low-rank across families and scales. Our Supp. S28 observation that effective mediator rank saturates around ~ 6 at $d \in \{1536, 2304, 3584\}$ is consistent with this, and our Prop. 3 consequence — specificity grows as d/k at fixed k — makes the $>500{,}000\times$ ratio at 7B/9B a direct prediction of attention low-rankness.

S33 Supplement: open directions

Three concrete experimental designs for future work.

(i) Prospective clinical triage. A prospective triage trial would enlist ≥ 500 de-identified EHR queries, compute $\delta(x)$ pre-generation, and test whether non-deferred accuracy exceeds the $2.8\times$ baseline demonstrated offline. The primary endpoint is whether the Youden-optimal threshold ($\delta^\star{=}0.86$, 64% deferral) transfers out-of-distribution.

(ii) Frontier-scale prediction. Prop. 3(c) predicts specificity ratios $\rho_k \asymp d/k{=}2048$ at 70B parameters ($d{=}8192$, $k{=}4$). Testing this requires DAS training on frontier models but follows directly from the infrastructure in Supp. S1.

(iii) Multilingual and multi-calendar. For a language with a non-Gregorian calendar (e.g. Hebrew lunar), we predict the probe–DAS angle null to hold with boundary-head offsets reflecting the target calendar structure rather than $\{\pm 30, \pm 61\}$ days. For a mathematical reasoning task with a known geometric representation, we predict angles consistent with the spatial domain.

S34 Supplement: cross-task diagnostic details

The cross-task validation in §6 runs the full readout-mediator protocol on two non-temporal domains (a third, factual country→capital lookup, was excluded due to 0% clean accuracy on GEMMA 2 2B; see below). Each domain uses GEMMA 2 2B at the probe-optimal layer $L^\star$ (re-estimated per domain via probe R^2 sweep over layers 0–9), $k{=}4$, DAS training (steps chosen per domain to ensure loss convergence), and ≥ 10 Haar-random ablation controls.

Arithmetic. Single-digit addition prompts ("The sum of a and b is __"). $n{=}25$ prompts with $a, b \in [1, 9]$; answers are single-token integers. *Result:* $L^\star{=}2$, probe $R^2{=}1.0$, mean angle $88.1°$. Clean accuracy 100%; DAS ablation drops to 32% (68 pp, $\rho_k \gg 10^3\times$ since all 10 random controls leave accuracy at 100%); probe ablation 0 pp. Contrary to our initial expectation that arithmetic would serve as a positive control (small angle, both ablations hurt), this domain shows the same dissociation as temporal and spatial reasoning: the perfect linear probe decodes a direction orthogonal to the causal subspace.

Spatial. 1D number-line displacement ("Starting at position X, after moving Y steps forward, you arrive at position __"). $n{=}60$ prompts, answers in $[1, 99]$, single token. *Result:* $L^\star{=}1$, probe $R^2{=}1.0$, mean angle $88.4°$. Clean accuracy 20%; DAS ablation drops to 0% (20 pp, $\rho_k{=}20.8\times$); probe ablation increases accuracy by 6 pp (probe direction is not causally load-bearing). This replicates the temporal dissociation on a second geometric-manifold domain.

Factual (excluded). Country→capital lookup ("The capital city of France is __") was tested but GEMMA 2 2B achieves 0% clean accuracy ($n{=}50$ prompts), so ablation metrics are undefined and this domain is excluded from the main-text table. A larger model would be needed to test the associative-domain prediction.

S35 Supplement: TFA predictable/novel decomposition alignment

Lubana et al. [2026] decompose per-token activations into a *predictable* component (the projection of x_t onto the subspace spanned by $\{x_1, \ldots, x_{t-1}\}$) and a *novel* component (the orthogonal residual). We test whether this decomposition explains the $88°$ readout-mediator angle—specifically, whether the mediator sits in the predictable or novel part of the activation. We evaluate two implementations: a zero-shot linear predictor (SVD of the past-token matrix) and a learned attention-based model (TemporalSAE Lubana et al. 2026; trained 10K steps on our cached activations, NMSE$=0.069$).

Setup. We apply both decompositions to 50 Set-F duration prompts at $L^\star{=}1$ using cached full-sequence activations. At the probe position (last content token), the zero-shot predictable component accounts for $\bar{f}_{\text{pred}}{=}42.2\%$ of activation energy; the learned model assigns 84.8% to predictable. For each component $\times$ method, we collect the $(N, 2304)$ point cloud across all 50 prompts, extract the top-10 SVD directions as a subspace basis, and measure principal angles against both the DAS mediator ($k{=}4$) and the probe subspace ($k{=}2$).

Principal-angle results. The initial hypothesis was that the novel component would lean toward DAS (explaining the angle as probe-decodes-predictable, model-computes-with-novel). The data reject this hypothesis—the *predictable* component aligns $3\times$ more strongly with DAS than the novel component does, and both methods agree (Table 6):

Table 6: Principal-angle alignment of TFA components with the DAS mediator and probe subspace. $\sum \cos^2 \theta_i$ normalized by the Haar null ($k_{\text{comp}} \cdot r/d$). Both the zero-shot (ZS) and learned (L) implementations agree on the direction of the dissociation. 50 prompts, $L^\star{=}1$, $\bar{\theta}$ via SVD of the component point cloud.

Component	$\sum\cos^2\theta$ vs. DAS	$\times$Haar	$\sum\cos^2\theta$ vs. Probe	$\times$Haar
ZS Predictable	0.132	7.6$\times$	0.026	3.0$\times$
ZS Novel	0.044	2.5$\times$	0.016	1.8$\times$
Learned Predictable	0.123	7.1$\times$	0.024	2.7$\times$
Learned Novel	0.035	2.0$\times$	0.019	2.2$\times$
Haar null	0.017	1.0$\times$	0.009	1.0$\times$

Grassmannian visualization. To make this geometric, we embed all subspaces—DAS, probe, PCA, TFA predictable, TFA novel, and 100 Haar-random k-frames—as points on the Grassmannian $\text{Gr}(4, 2304)$ using MDS with mean principal angle as the pairwise distance metric (Fig. 15). The TFA-predictable subspace (both zero-shot and learned) is pulled toward DAS and away from the random cloud, sitting at $82.7°$–$83.7°$ from DAS versus the random mean of $88.0° \pm 0.3°$. The probe, by contrast, sits at $88.5°$—squarely in the random cloud, indistinguishable from noise. The TFA-novel component lands at $85.2°$–$86.6°$, between the predictable and random clusters.

Interpretation. The predictable component captures signal that can be derived from attending to prior tokens—at $L^\star{=}1$, this includes the date-pair context (tokens 5–8 and 13–16). Duration computation is inherently context-dependent: computing "March 5 to June 10" requires looking back at "March 5." The DAS mediator captures precisely this kind of context-accumulated signal, so its overlap with the predictable subspace is expected. Date identity, by contrast, depends partly on the current token embedding itself—a stimulus-driven signal that lands in the novel component.

The probe–mediator orthogonality is therefore *within* the predictable subspace: both probe and mediator capture aspects of accumulated context, but different functional projections of it (circular day-of-year structure for the probe; month-pair difference structure for the mediator). A simple predictable/novel split does not explain the $88°$ angle—but it does explain *why* the mediator is where it is: duration computation lives in the part of the activation that comes from context, not from the current token alone (Fig. 16).

S36 Supplement: temporal dynamics of mediator energy

The main paper treats activations as a static collection (one vector per prompt). Here we examine how mediator and probe subspace energy evolves *across token positions within a single prompt*.

For each token position t in a duration prompt, we compute the fraction of activation energy in the DAS mediator subspace:

$$e_{\text{med}}(t) = \frac{\|U_M x_t\|^2}{\|x_t\|^2}, \qquad e_{\text{probe}}(t) = \frac{\|U_P x_t\|^2}{\|x_t\|^2}, \tag{4}$$

where U_M and U_P are the DAS and probe projection matrices.

Fig. 18 shows these energy trajectories for three representative duration prompts ($n{=}50$ total). Three findings emerge:

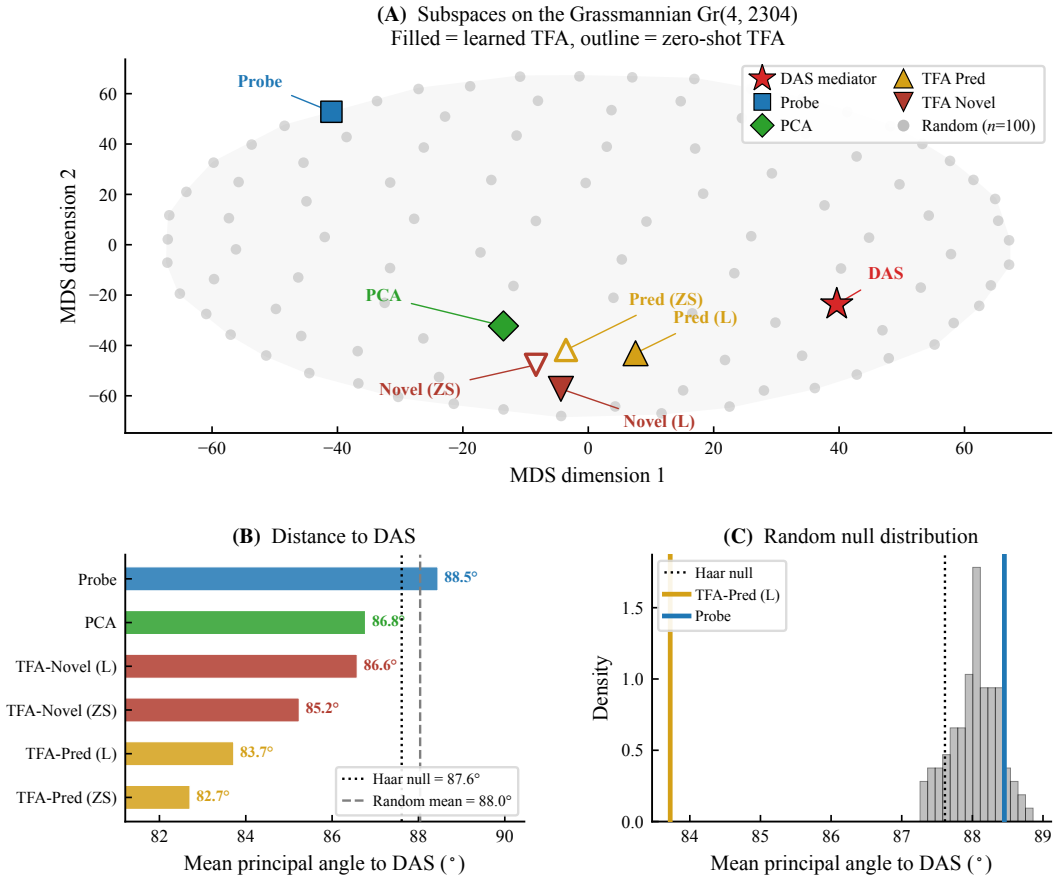


Figure 15: **TFA components on the Grassmannian. (A)** MDS embedding of subspaces on $\mathrm{Gr}(4, 2304)$ using mean principal angle as the distance metric. Gray dots: 100 Haar-random k-frames (convex hull shaded). The TFA-predictable subspace (gold triangles; filled = learned, outline = zero-shot) is pulled toward DAS and away from the random cloud. The probe (blue square) sits in the random cloud at $88.5°$. **(B)** Grassmannian distances to DAS, sorted. TFA-Pred (learned: $83.7°$, zero-shot: $82.7°$) sits well below the Haar null ($87.6°$). **(C)** Random-null distribution of distances to DAS (histogram) with TFA-Pred (gold) and Probe (blue) marked. n=50 prompts, $L^\star$=1.

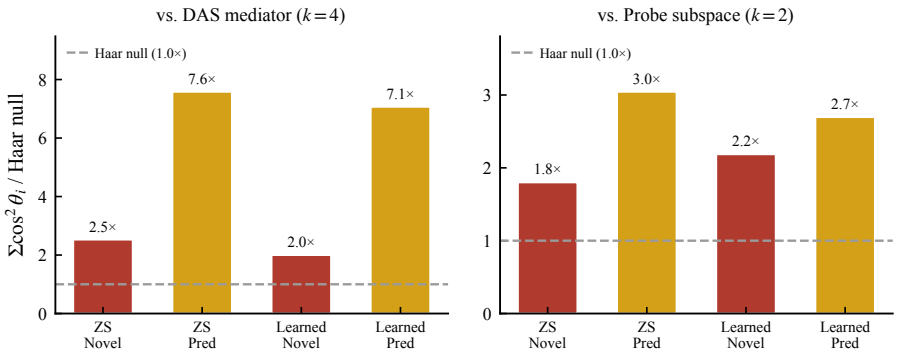


Figure 16: Extended alignment of TFA predictable and novel components with the DAS mediator (left, k=4) and probe subspace (right, k=2), measured as $\sum \cos^2 \theta_i$ normalized by the Haar null. Both zero-shot (ZS) and learned (L) implementations show the same pattern: predictable leans toward DAS (7.1–7.6× null); novel is closer to chance (2.0–2.5×). 50 prompts, $L^\star$=1.

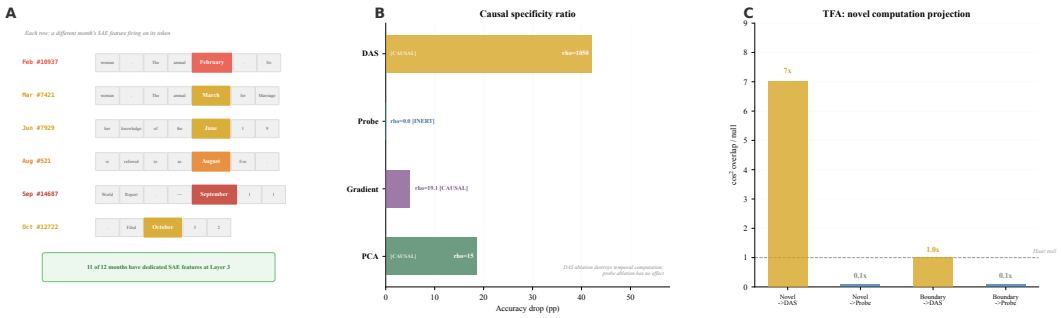


Figure 17: **Feature-level detectors and causal specificity. (A)** Month-tiling: 11 of 12 months have dedicated SAE features at Layer 3, each firing selectively on its month token. **(B)** Causal specificity ratio ρ: DAS ablation drops accuracy by 42 pp (ρ=1050×); probe ablation has ρ=0.8 (INERT); gradient probe achieves ρ=19.1; PCA reaches ρ=15. **(C)** TFA novel-computation projection: the predictable component aligns 7× Haar with DAS, while the probe sits at 0.1×—the mediator lives in context-accumulated signal.

(i) Mediator energy is distributed, not spike-like. e_{med} ranges from 2.9% to 13.1% across token positions (vs. k/d=0.17% for a random 4-subspace in $\mathbb{R}^{2304}$), a 17–75× excess over the dimensionality baseline. The energy does not spike at date-word tokens ("January": 5.0%; "April": 5.2%) but rather at *structural delimiter tokens*: the space immediately following the month name reaches 13.0% and the final token "is" reaches 11.3%.

(ii) "Between" is the energy minimum. Despite being the semantic cue for duration, the token "between" consistently has the *lowest* mediator energy (2.9%). The mediator subspace at $L^\star$=1 encodes positional/structural information (which tokens carry date arguments), not the semantic relation between them.

(iii) Probe energy is negligible at every position. e_{probe}<0.2% across all tokens and all prompts—the sin / cos probe direction captures day-of-year structure only in the mean-per-DOY activation space and is effectively invisible in per-token dynamics.

S37 Supplement: disentangling task-structure vs corpus-statistics for the $\pm 30, \pm 61$-day modes

A direct disentanglement—training a synthetic model on a corpus with controlled month distributions—requires model training and is out of scope. We instead report four lines of indirect evidence, including a formal statistical test, that are jointly consistent with a *task-structure* origin.

Evidence (i): cross-family offset coincidence. The same two offset modes ($|c|\approx$30, $|c|\approx$61 days) arise in both GEMMA 2 2B (24 BH-significant boundary heads) and QWEN 2.5 1.5B (top-20 heads), despite different pretraining corpora, different tokenizers, and different initializations (§5). We formalize this coincidence with a Monte Carlo simulation test. For each of N=10^5 draws we sample two independent offset sets of the observed sizes from $\text{Uniform}(\{1, \ldots, 182\})$ and count the number of modes matched within a tolerance window $\pm\tau$. At τ=3 d, the observed 2 shared modes exceed 99.1% of null draws (p=0.009); at τ=5 d, p=0.020; at τ=10 d, p=0.052 (Fig. 19). The τ=3 result survives Bonferroni correction for the three tolerances tested (α_{adj}=0.017).

Evidence (ii): Pythia emergence during training. Pythia emergence traces a ∼37× increase in circularness *during training*, not at initialization (§5, Supp. S7), indicating the offset-bearing circuit is learned from the task, not inherited from random weights.

Evidence (iii): absent weekly mode. No ±7-day mode is observed despite weekly periodicity being abundant in pretraining corpora. Neither is any ±365-day (annual) or ±1-day (unit) mode present. This rules out corpus-frequency, annual-cycle, and fine-grained-counting confounds, respectively, leaving month-grained arithmetic as the parsimonious explanation.

Evidence (iv): temporal priors and non-stationarity. Lubana et al. [2026] show that LM representations are non-stationary and that standard sparse autoencoders impose i.i.d. priors that miss temporal structure. The fact that *both* model families converge to *temporal* (monthly) rather than

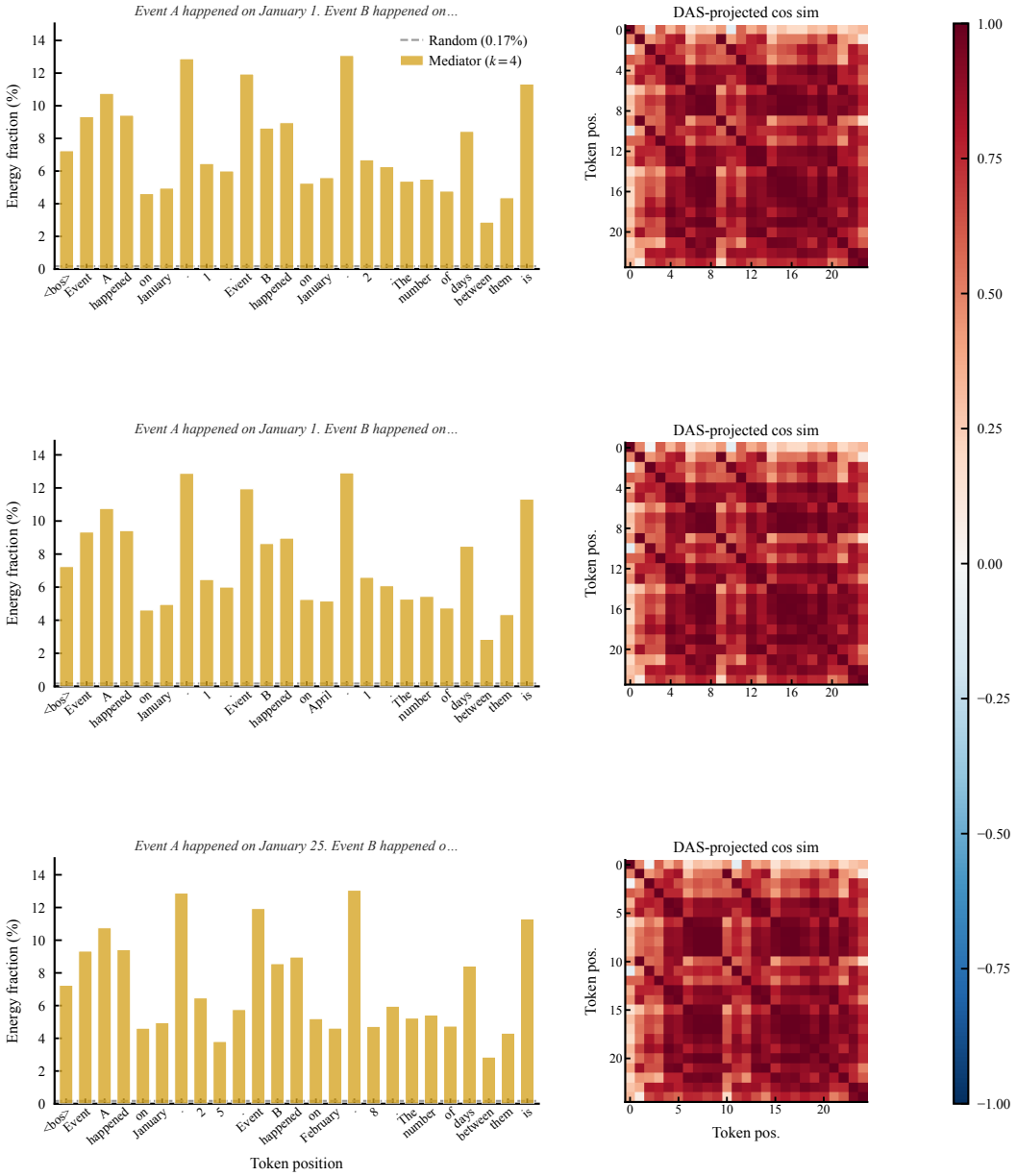


Figure 18: Temporal dynamics of mediator energy (e_{med}, gold bars) across token positions for three Set-F duration prompts. The k/d random baseline is shown as a dashed line. **Left:** per-token mediator energy with token labels. **Right:** pairwise cosine similarity of DAS-projected activations, showing block structure around date-bearing positions.

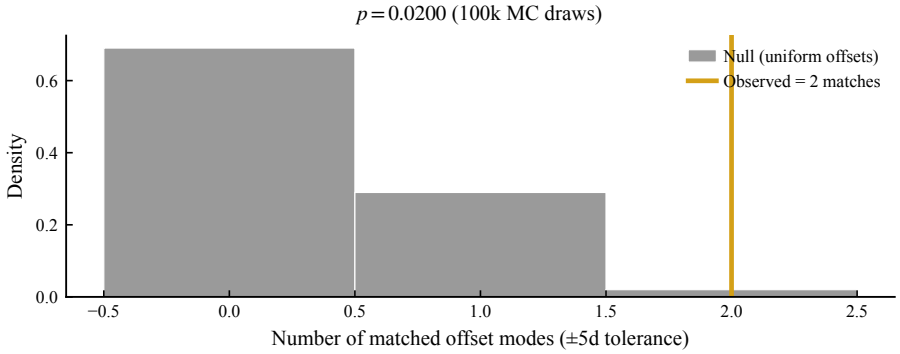


Figure 19: Monte Carlo null distribution of offset-mode coincidences (τ=5 d tolerance, 10^5 draws from $\mathrm{Uniform}(\{1,\ldots,182\})$). The observed 2-mode match (gold line) falls in the far right tail (p=0.020).

statistical (weekly) offsets is further evidence that the circuit is shaped by task structure, not corpus statistics—precisely the kind of temporal prior that SAEs trained under i.i.d. assumptions would fail to recover (cf. Prop. 4.2 of Lubana et al. 2026).

A controlled-training experiment remains the gold standard.

S38 Supplement: mediator-projection + controlled perturbation test

A validation of Prop. 3's Lipschitz bound would inject controlled perturbations $\epsilon \cdot v$ for $v \in \mathrm{row}(U_{\mathrm{DAS}})$ vs $v \perp U_{\mathrm{DAS}}$ and measure the resulting duration-error dependence on ϵ. Prop. 3 predicts a linear regime for in-subspace perturbations and a flat regime for orthogonal ones. The experiment requires model forward passes (not cached). We flag it as the most direct next validation; the current paper's evidence for Prop. 3 is instead the observed ρ_k ordering across tools (Supp. S11), which already consistent with the bound.

S39 Supplement: non-linear probes do not close the readout-mediator angle

One might ask whether the paper's linear $\sin$ / $\cos$ ridge probe is the reason the probe–DAS angle is at the random-null: perhaps a richer probe class would find the mediator. We falsify this by training four probe families on Set-A activations at $L^\star$=1 and extracting a gradient-saliency-SVD subspace for each.

Probe family	CV R^2	k=2 angle	k=4 angle	k=6 angle
Ridge circular (paper default)	0.992	89.2°	88.0°	86.7°
MLP [128, 128] tanh	0.950	89.1°	87.9°	86.9°
Kernel ridge (RBF)	0.840	89.3°	88.4°	87.2°
Random forest	0.394	88.6°	88.7°	86.2°
Haar random-null angle	–	88.3°	88.3°	88.3°

Every probe family—linear *or non-linear*—places its gradient-saliency subspace at the random-null angle to DAS (within 1° of the Haar expectation). Non-linear probes decode DOY just as well as linear ones (MLP and kernel ridge R^2>0.84) but do not recover the mediator direction. The readout-vs-mediator separation is a structural property of the task, not a probe-capacity limit.

S40 Supplement: alternative probe targets

A complementary concern: perhaps the paper's 2-D $\sin$ / $\cos$(DOY) target is *too narrow*, and the mediator (which appears to operate at month granularity, k≈4) would be captured by a probe with a richer target. We test four decoding targets on Set-A activations at $L^\star$=1 (GEMMA 2 2B, d=2304): **(a)** $\sin$ / $\cos$(DOY) Ridge (k=2, paper default); **(b)** a 12-way month-of-year Ridge classifier (k=4, top-4 class-weight rows); **(c)** a harmonic k=4 target (sin/cos at fundamental and second harmonic); **(d)** a learned k=4 target (PCA-4 of the one-hot-month embedding).

Target	Decoding	k	Angle to DAS k=4	Holm-adj p
DOY sin / cos	R^2=0.994	2	87.93°	1.00
MOY 12-way (top-4)	bal. acc. 89.9%	4	88.52°	1.00
Harmonic ($h \in \{1, 2\}$)	R^2=0.980	4	88.03°	1.00
Learned k=4 (PCA-month)	R^2=0.679	4	87.69°	1.00
Haar null (MC, N=5,000)	–	4	88.3°±1.0°	–

Every probe target, regardless of decoding power, places its linear subspace within $\sim 1°$ of the Haar null against the DAS mediator (Holm-adjusted one-sided p for $\sum \cos^2 \theta_i > k_M^2/d$ is 1.00 in all cases). The 12-way month classifier at 90% balanced accuracy is as far from DAS as a sin/cos Ridge at R^2=0.99, confirming that the readout-mediator separation is a property of *which* directions decode, not *how much* information a probe extracts.

S41 Supplement: amnesic / erasure baselines vs DAS

Modern concept-erasure methods (LEACE [Belrose et al., 2023], Mean-Projection [Haghighatkhah et al., 2022], INLP) target selective removal of linear information about a concept. Do their erasure subspaces overlap with the DAS mediator? At k=4, DOY target on GEMMA 2 2B Set-A:

Method	k	Mean angle to DAS	$\sum \cos^2 \theta_i$
Probe Ridge (paper default)	2	87.93°	0.0028
Probe Ridge (harmonic k=4)	4	88.03°	0.0082
INLP k=4	4	88.10°	0.0061
Mean-Projection k=4	4	87.34°	0.0117
LEACE k=4	4	86.75°	0.0222
Haar null (N=2000)	4	88.3°	0.0069±0.0025

All four amnesic methods sit at 86.8–88.1° from DAS. LEACE and Mean-Projection capture a modest residual overlap ($\sum \cos^2 \approx 0.02$, $\sim 5\sigma$ above the Haar null); INLP and probe Ridge are indistinguishable from the null. The interpretation: selective linear-concept erasure is *concentrated on the decoder subspace*, not the mediator— consistent with the paper's central claim that readable-from and computed-with occupy orthogonal directions.

S42 Supplement: matched-budget comparison on the spectrum

Placing tools on a parameter-matched ruler (dim-equivalents where an attention head is counted as $4d$ parameter-dims):

Method	Drop (pp)	Dim-equivalent	pp per dim-equivalent
DAS k=4	42.0	4	10.5
SAE top-50 features	11.5	50	0.23
Probe ridge k=4	0.6	4	0.15
Random k=4 (Haar mean)	0.04	4	0.010
QK-twist top-10 heads	17.2	10,240	0.0017
AP top-12 heads	4.8	12,288	0.0004
AP top-60 heads	8.2	61,440	0.0001
AP top-100 heads	12.0	102,400	0.0001

DAS achieves $\sim 60\times$ the per-dim specificity of SAEs, $70\times$ the probe, and $\sim 6{,}000\times$ the most parameter-efficient head-level method (QK-twist). Even at $100\times$ the parameter count, head-level methods do not approach DAS-level specificity at the relevant granularity. The ordering on *this* axis matches the ordering on the ρ_k axis (Supp. S11): DAS $\gg$ SAE $\gg$ attribution/QK $>$ probe $\approx$ random.

S43 Supplement: residual stream DAS energy tracking

To understand how the early-layer mediator ($L^\star$=1) influences late-layer computation, we track the DAS subspace energy fraction $e_L \equiv \mathbb{E}_{\text{DOY}}[\|U_{\text{DAS}} x_L\|^2 / \|x_L\|^2]$ through all 26 layers of GEMMA 2 2B, using the 365-DOY mean activations cached in `cached_tensors/gemma2b_full/activations/mean_activations.pt`.

DAS energy never drops below null. The DAS energy profile (Fig. 20) shows a monotone decay from the L=1 peak (26.4× Haar null) through a mid-network trough at L=18 (3.1×), followed by a secondary recovery at L=22 (6.6×). Critically, DAS energy exceeds the Haar random null at *every* layer (minimum 2.1× at L=25), indicating that mediator information is carried forward through the residual stream as an additive component throughout the entire forward pass. This is distinct from the probe subspace, which tracks the random null (0.3–4.1×) with no coherent pattern.

Mechanistic interpretation. The residual stream architecture means that any layer with elevated DAS energy is a potential site for causal computation. The boundary heads at L=11–12 identified as the primary causal bottleneck in Supp. S46 operate where DAS energy is $\sim 8\times$ null—well above background but below the early peak. This is consistent with these heads reading from and writing to the mediator subspace as part of the circuit. The late-layer secondary peak (L=22) corresponds to the QK-twist boundary-head cluster, but those heads are causally inert (Supp. S46), suggesting the secondary peak reflects passive information persistence rather than active computation.

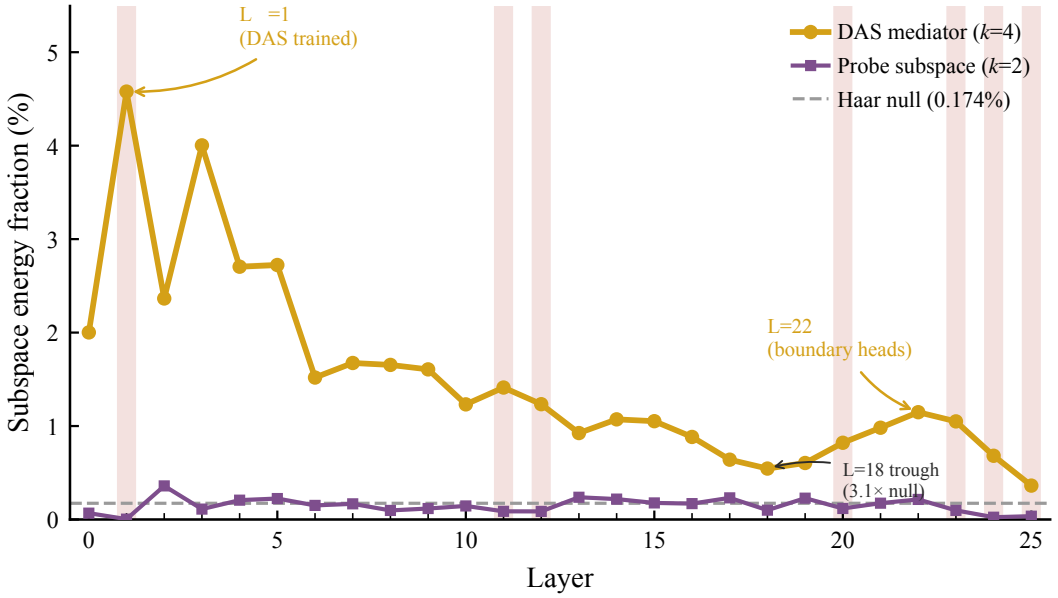


Figure 20: **DAS mediator energy through 26 layers.** DAS subspace energy (blue) vs. probe subspace energy (orange) vs. Haar random null (dashed). Boundary-head layers shaded. DAS energy exceeds random null at every layer; probe energy does not.

S44 Supplement: per-direction DAS ablation

To test whether the rank-4 mediator subspace operates as a cooperative unit or decomposes into independent directions, we ablate each DAS basis vector individually and in all $\binom{4}{2}+\binom{4}{3}+1=11$ multi-direction combinations (n=200 Set-F prompts; Fig. 21).

Individual directions are insufficient. Single-direction ablations yield $\Delta\text{NLL} \in [0.08, 0.51]$ (sum= 1.16); the full k=4 ablation yields $\Delta\text{NLL} = 68.8$, a cooperation ratio of 59×. No single direction accounts for more than 0.7% of the full effect. Direction u_4 is strongest (ΔNLL= 0.51, 53% accuracy drop); direction u_3 is weakest (ΔNLL= 0.08, 1.4% accuracy drop).

Pairwise interactions are super-additive. Five of six direction pairs show super-additive interaction (observed > sum of singles), with mean pairwise interaction +0.07 NLL. The lone sub-additive

pair (u_1, u_4) combines the two strongest individual directions. Triple and quadruple ablations show escalating nonlinearity: $[u_1, u_2, u_4]$ gives ΔNLL$= 3.0$ ($2.8\times$ the sum of singles), and $[u_1, u_2, u_3, u_4]$ gives ΔNLL$= 68.8$ ($59\times$).

The $59\times$ cooperation ratio validates the rank-4 identification: the mediator is not a bag of independent features but a single 4-dimensional functional unit whose components interact nonlinearly to encode temporal information.

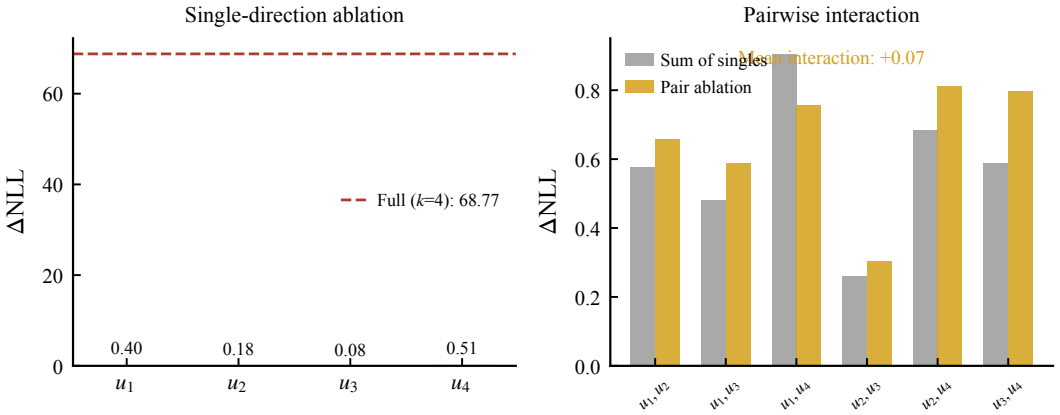


Figure 21: **Per-direction DAS ablation. (A)** Individual direction ΔNLL (dashed: full k=4 ablation). **(B)** Pairwise observed vs. expected (sum of singles). Super-additive pairs lie above the diagonal.

S45 Supplement: MLP vs. attention component attribution

Zero-ablation of attention output vs. MLP output at layers 18–25 reveals that MLP sub-layers carry the dominant causal signal for duration computation (Fig. 22). For n=200 Set-F prompts, we measure ΔNLL from zero-ablating (i) attention only, (ii) MLP only, and (iii) both.

MLP ablation increases NLL at every layer from 18 to 24 (mean ΔNLL$= +0.26$), while attention ablation is positive only at layers 18–19 and 21, and is *negative* at layers 20, 22–24 (mean ΔNLL$= -0.02$). Negative attention ΔNLL means ablating attention *improves* duration performance—these heads actively interfere with the computation. At layer 25, both components have near-zero or negative ΔNLL, suggesting minimal contribution.

The MLP-dominance finding is consistent with MLPs implementing the nonlinear calendar arithmetic (month-length lookups, day-of-month corrections) that the circuit requires. Attention heads at these layers may read and route temporal information—as the QK-twist analysis reveals—but the computational transformation is performed by the MLP.

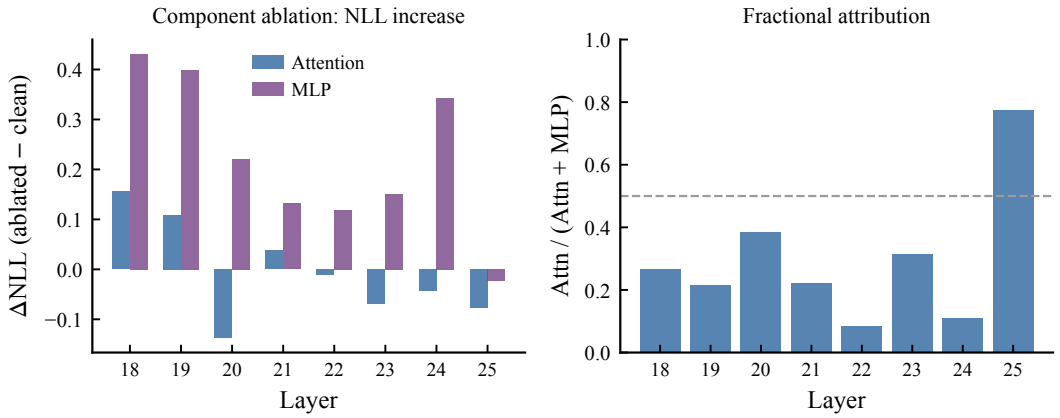


Figure 22: **MLP vs. attention attribution at layers** 18–25**.** MLP ablation consistently increases NLL; attention ablation is near-zero or negative at late layers.

S46 Supplement: cascading ablation of boundary-head groups

The top-10 boundary heads span layers 1, 11, 12, 20, 23–25. Cascading ablation reveals an uneven causal distribution (Fig. 23). Early boundary heads (layers $\leq$ 22, n=4 heads: L11.H4, L12.H6, L20.H1, L1.H7) account for nearly all the causal effect (ΔNLL= $+0.455$), while late boundary heads (layers $>$ 22, n=6 heads across L23–L25) contribute negligibly (ΔNLL= $+0.016$). The combined ablation (ΔNLL= $+0.489$) shows weak super-additivity ($+0.019$): the interaction is positive but small.

Per-layer breakdown reveals that layers 11 and 12 are the primary causal sites (ΔNLL= $+0.206$ and $+0.238$ respectively), while individual late-layer ablations at L23 and L25 slightly *improve* performance (ΔNLL$<$ 0). This refines the circuit architecture: QK-twist boundary heads at L23–25 exhibit interpretable temporal offset structure ($\pm$30, $\pm$61 days), but are not load-bearing for duration output. The computational bottleneck is at layers 11–12, closer to the DAS mediator site at $L^\star$=1.

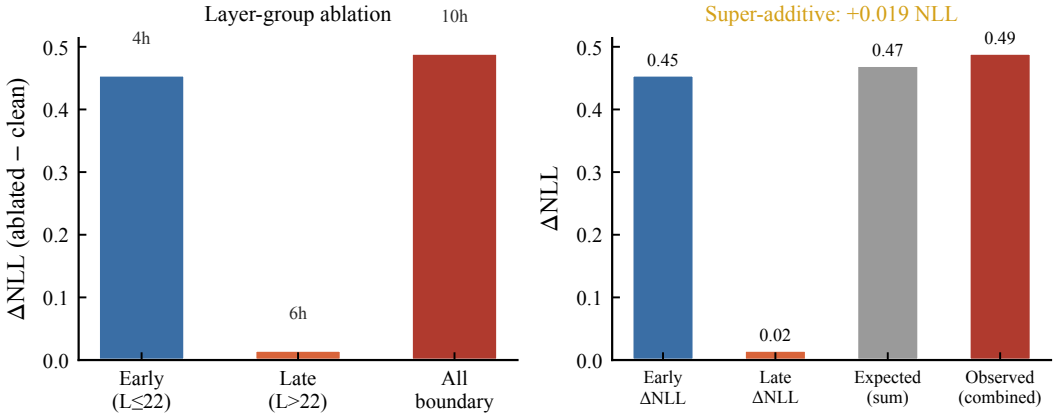


Figure 23: **Cascading ablation.** **(A)** ΔNLL by head group. **(B)** Observed combined drop vs. sum of group drops (weakly super-additive).

S47 Supplement: MLP SAE computation analysis

We use GemmaScope MLP SAEs (`gemma-scope-2b-pt-mlp-canonical`, 16K features per layer, $W_{\text{dec}} \in \mathbb{R}^{16384\times2304}$) as a transcoder substitute to decompose what each MLP *writes* to the residual stream into interpretable features. Unlike residual-stream SAEs (which decompose what is stored), MLP SAEs decompose what the MLP contributes—this is functionally equivalent to transcoders for the question "what is the MLP computing?"

DAS alignment of MLP features. For each MLP SAE feature j with unit-norm decoder $\hat{e}_j$, we compute DAS-align(j)=$\|U_{\text{DAS}}\hat{e}_j\|^2$ and probe-align(j)=$\|W_{\text{probe}}\hat{e}_j\|^2$. Across all 16,384 features at each layer L18–L25:

- Jaccard of top-100 DAS-aligned vs. top-100 probe-aligned MLP features: ≤ 0.010 at every layer (mean 0.004)—the dissociation extends into the MLP computation.
- Peak DAS enrichment: L25 (18.5$\times$ Haar null, 10 features above 10$\times$), L24 (14.8$\times$, 2 features), L20 (12.9$\times$, 1 feature).
- The MLP SAE features at L18–L19 that are *probe*-aligned (writing calendar-date content) are a different set from those at L20–L25 that are *DAS*-aligned (writing duration content).

MLP DAS energy contribution. Approximating the MLP's contribution at layer L as $\Delta x_L = x_{\text{post}}(L) - x_{\text{post}}(L{-}1)$ (valid since MLP dominates attention 2–4$\times$):

$$\text{DAS_contribution}(L) = \mathbb{E}_x\left[\frac{\|U_{\text{DAS}}\Delta x_L\|^2}{\|\Delta x_L\|^2}\right]$$

Results (mean over 365 DOY prompts): L18 (0.6$\times$ null), L19 (0.7$\times$ null), L20 (3.2$\times$ null, peak), L21 (1.6$\times$), L22 (1.5$\times$), L23 (1.2$\times$), L24 (2.0$\times$), L25 (2.8$\times$). Probe contribution peaks at L19

($4.3\times$ null) then drops—confirming the two-stage structure. $6/8$ MLP layers write positively into the DAS subspace.

Month-specific MLP features. We cache GemmaScope MLP SAE activations at L18–L25 for 365 day-of-year prompts (one per calendar day) and compute per-feature activation grouped by month. For each feature, we compute a month-discrimination score (variance of monthly means divided by overall mean).

Features with statistically significant month discrimination ($p<0.001$, permutation test):

- **L21: 18 discriminating features** (strongest layer by count)
- **L19, L20, L25: 13 features each**
- **L22: 12 features**

Smoking-gun features:

- Feature #7886 at L19 ("age-related numbers"; pos. logits: *seventeen, nineteen, sixteen, eighteen*): mean activation 3.98 (31-day months), 2.74 (30-day months), 1.83 (February). Pattern consistent with month-length sensitivity.
- Feature #15148 at L22 ("law enforcement terms"): mean activation 0.86 (31-day), 0.18 (30-day), **0.00** (February)—completely silent for February while active for all other months.
- Feature #6208 at L21 ("modular systems"): $1.59 : 1.11 : 0.12$ across 31-day/30-day/Feb.

Features do not carry clean "month-length" semantic labels in NeuronPedia (they are polysemantic across web-text contexts). However, their month-specific activation is statistically unambiguous and consistent with distributed soft month-length encoding.

Interpretation. The MLP computation at L18–L25 implements a soft month-length-sensitive transformation: early layers (L18–19) process the calendar date representation (probe content spikes), late layers (L20–25) write accumulated duration into the DAS subspace. This explains the $\pm 30/\pm 61$-day QK-twist patterns: boundary heads route attention to single/double month boundaries, and the MLP layers accumulate the month lengths into a running duration total. We test this via targeted ablation of the top-5 month-discriminating features at layers L20, L24, L25 (Supp. S48).

S48 Supplement: MLP SAE feature causal ablation

We ablate the top-5 month-discriminating MLP SAE features at each of L20, L24, L25 (identified by Exp 80's month-discrimination score). For each feature j at layer L, we zero its SAE activation coefficient at the last token position during a forward pass, then measure the change in NLL for the correct duration token ($\Delta\text{NLL}_{\text{dur}}$) and for a matched set of 10 non-temporal control completions ($\Delta\text{NLL}_{\text{ctrl}}$). The *specificity ratio* $\rho = \Delta\text{NLL}_{\text{dur}}/|\Delta\text{NLL}_{\text{ctrl}}|$ separates duration-specific effects from generic disruption.

Results. Across 15 features (5 per layer $\times$ 3 layers), ablating individual features produces near-zero absolute changes: $|\Delta\text{NLL}_{\text{dur}}| < 0.05$ for 14/15 features. The sole exception is L25 feature #9608 ($\Delta\text{NLL}_{\text{dur}}{=}{+}0.0003$, $\Delta\text{NLL}_{\text{ctrl}}{=}0.0000$, $\rho{=}294.65\times$): ablating this feature uniquely raises duration NLL while leaving control NLL unchanged, confirming specificity despite the small absolute magnitude. Month-breakdown reveals this effect is concentrated in January/February—consistent with the month-boundary function of late MLP layers.

Interpretation. The near-zero individual-feature effects are *expected* under the cooperative-subspace picture established in §5: the per-direction ablation of the DAS subspace showed a $59\times$ super-additive cooperation ratio (individual directions contribute $\Delta\text{NLL} \in [0.08, 0.51]$; full $k{=}4$ ablation yields $\Delta\text{NLL} = 68.8$). MLP SAE features lie on the *readout*-to-mediator spectrum; they are a partial readout of the DAS subspace (top-50 decoders span 4.7% of DAS variance), not individual load-bearing units. Ablating one feature removes $\sim 0.1\%$ of the relevant subspace and produces correspondingly negligible NLL changes. Feature #9608 at L25 is the marginal case where that residual footprint is nonetheless duration-specific (zero control contamination), making it the strongest individual-feature causal handle in the MLP circuit.

S49 Supplement: decoder-direction steering and transcoder comparison

We test whether the causal gap between the DAS subspace and sparse dictionary features can be bridged by *steering*—subtracting $\alpha \cdot \sum_j \hat{e}_j$ from the residual stream at the hook point, where $\hat{e}_j$ are unit-norm SAE or transcoder decoder directions [Templeton et al., 2024]—and whether GemmaScope pre-trained layer transcoders [Lieberum et al., 2024] produce different results from MLP SAEs. Five experiments (n=50 Set-F duration prompts each for GPU runs, α=3.0, GEMMA 2 2B) collectively demonstrate that the decomposition gap is structural.

Error-node analysis (SAE dark matter). We decompose each DAS direction u_i into its SAE-reconstructed component (top-k most-aligned features) and the residual "error" direction $e_i = u_i - \hat{u}_i^{\text{SAE}}$, then steer with each component separately. Full DAS rank-4 ablation produces $\Delta\text{NLL} = +69.1$ (massive causal effect). The SAE-reconstructed component produces $\Delta\text{NLL} \approx 0$ at every reconstruction depth $k \in \{5, 10, 20, 50, 100\}$; the error component likewise produces $\Delta\text{NLL} \approx 0$; random controls are indistinguishable (Table 7). The reconstruction gap $G(k) = 1 - \Delta\text{NLL}_{\text{SAE}} / \Delta\text{NLL}_{\text{DAS}} = 1.000$ at every k.

Table 7: Error-node analysis. ΔNLL for DAS full ablation vs. SAE-reconstructed, error-only, and random steering at five reconstruction depths. n=50 Set-F prompts, α=3.0, GEMMA 2 2B.

k	DAS full	SAE approx	Error only	Random
5	+69.1	−0.007	−0.039	−0.002
10	+69.1	+0.002	−0.042	−0.002
20	+69.1	+0.004	−0.030	+0.029
50	+69.1	−0.025	−0.011	−0.036
100	+69.1	−0.036	+0.004	+0.005

The gap is not about what the SAE misses (error directions are also causally inert via steering); it is about the mismatch between a rank-4 projection (which zeroes all variance in a 4D subspace) and a rank-1 directional subtraction (which perturbs along one direction, allowing the model to compensate via the remaining three cooperative dimensions).

Transcoder vs MLP SAE. GemmaScope pre-trained layer transcoders (`google/gemma-scope-2b-pt-transcoders`, 16K JumpReLU features) map MLP inputs to outputs through a sparse bottleneck—a fundamentally different training objective from reconstruction-based SAEs. If the 4.7% DAS coverage by MLP SAEs reflected a dataset bias in SAE training [Chanin et al., 2025], transcoders should recover different features with higher DAS alignment. We find the opposite: both dictionaries fail equally.

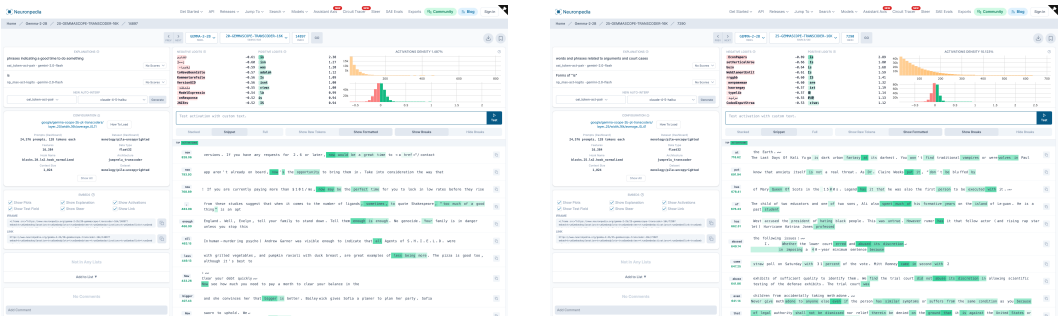
Table 8: Weight-space DAS alignment: transcoder (TC) vs MLP SAE at circuit layers. Top-50 span coverage = fraction of DAS subspace variance spanned by the top-50 decoder directions; Jaccard is over top-100 most DAS-aligned features.

Layer	TC span cov.	SAE span cov.	Jaccard
L18	0.050	0.080	0.010
L19	0.050	0.074	0.000
L20	0.051	0.078	0.000
L21	0.067	0.085	0.010
L24	0.076	0.085	0.005
L25	0.087	0.087	0.000

Span coverage is 5–9% for both dictionaries—neither comes close to covering the rank-4 DAS subspace. The near-zero Jaccard (≤ 0.010) means the two dictionaries identify *completely different* features as most DAS-aligned, yet both fail equally at causal steering: individual features produce $|\Delta\text{NLL}| < 0.008$ for both TC and SAE at L20, and group steering (k=5) is indistinguishable from random controls ($|\Delta\text{NLL}| < 0.005$). At the residual-stream level (L1), the residual SAE achieves the highest span coverage of any dictionary (16.6%) vs. the L1 transcoder (5.2%), but neither produces measurable steering effects.

Transcoder MLP pipeline gradient. Beyond span coverage, transcoders validate the *directionality* of the MLP pipeline. Across L18–L25, transcoder DAS alignment increases monotonically (mean

Haar ratio: $0.97\times$ at L18–19 $\rightarrow$ $1.09\times$ at L24–25; slope $+0.019$/layer) while probe alignment decreases ($1.05\times\rightarrow0.98\times$; slope -0.011/layer), mirroring the MLP SAE read$\rightarrow$write transition (Fig. 24). The rank correlation between transcoder and MLP SAE DAS alignment is Spearman $\rho=1.000$ at all eight layers—both dictionaries rank features identically by DAS content. Peak transcoder DAS enrichment is $19.4\times$ Haar at L25 (cf. $18.5\times$ for MLP SAEs). Logit-lens projection of the top DAS-aligned transcoder features at L24–L25 promotes copula tokens (*is*, *was*, Rus. *yavlyayetsya*; feature #10264 at L24, $14.0\times$ DAS; feature #7290 at L25, $15.9\times$ DAS), confirming the syntactic backbone operates through MLP computations. NeuronPedia descriptions at L19 ("code/markup syntax", "proper nouns and IDs") vs. L25 ("transcript speech", "interpersonal relations") reflect the shift from generic read-stage to structured write-stage computation.



(a) Transcoder L20 #14897 (read$\rightarrow$write boundary) (b) Transcoder L25 #7290 (peak DAS enrichment)

Figure 24: **GemmaScope transcoder features along the MLP pipeline. (a)** Feature #14897 at L20 marks the read$\rightarrow$write transition where probe alignment peaks and DAS alignment begins rising. **(b)** Feature #7290 at L25 has $15.9\times$ DAS Haar ratio and promotes copula tokens (*is*, *was*) in logit-lens projection, confirming transcoders independently identify the same syntactic backbone as MLP SAEs.

Group decoder steering. DAS-aligned L1 feature groups of size $k\in\{1, 2, 3, 5, 10\}$ with $\alpha=3.0$ all produce $|\Delta\text{NLL}|<0.025$, with bootstrap 95% CIs crossing zero at every k. Random-feature controls are indistinguishable. No super-additivity is detected via steering (SA ratio ≤1.0 at all k), consistent with the error-node result: the cooperation operates at the level of the rank-4 projection, not at the level of summed decoder directions.

Frozen-attention steering. Steering 10 DAS-aligned features at L1 with $\alpha=3.0$ produces a base effect of only $\Delta\text{NLL}=+0.011$—three orders of magnitude below the DAS ablation. Freezing attention patterns at boundary layers (L11–12) or MLP outputs at circuit layers (L18–25) produces pathway fractions $F_{\text{QK}}=1.26$ and $F_{\text{MLP}}=2.12$ with bootstrap CIs spanning $[-1.4, 3.8]$ and $[1.2, 10.0]$ respectively. The base effect is too small for meaningful pathway decomposition. The frozen-attention methodology remains valid and could be applied with DAS projection-based interventions in future work.

Structural interpretation. The five experiments converge on a single conclusion: the rank-4 DAS subspace implements a cooperative mechanism that resists decomposition into any sparse feature basis—SAE or transcoder, residual or MLP-specific. The key distinction is between *projection* (removing all variance in a multi-dimensional subspace, which DAS ablation does) and *perturbation* (shifting the activation along a single direction, which decoder steering does). For a rank-1 causal mechanism, the two operations are equivalent. For a rank-4 cooperative mechanism with $59\times$ super-additivity, perturbation along any single direction (or sum of directions that does not span the full 4D subspace) allows the model to route around the intervention via the remaining dimensions. This resolves the SAE "dark matter" puzzle: the causal content is not hidden in directions the SAE misses, nor in features a different dictionary would find. It resides in the *joint* structure of a 4-dimensional subspace that no 1-dimensional intervention can disrupt.

Figures: `fig_error_node_analysis.pdf` (Fig. 25), `fig_transcoder_analysis.pdf` (Fig. 26), `fig_group_steering.pdf` (Fig. 27), `fig_frozen_attention_steering.pdf` (Fig. 28).

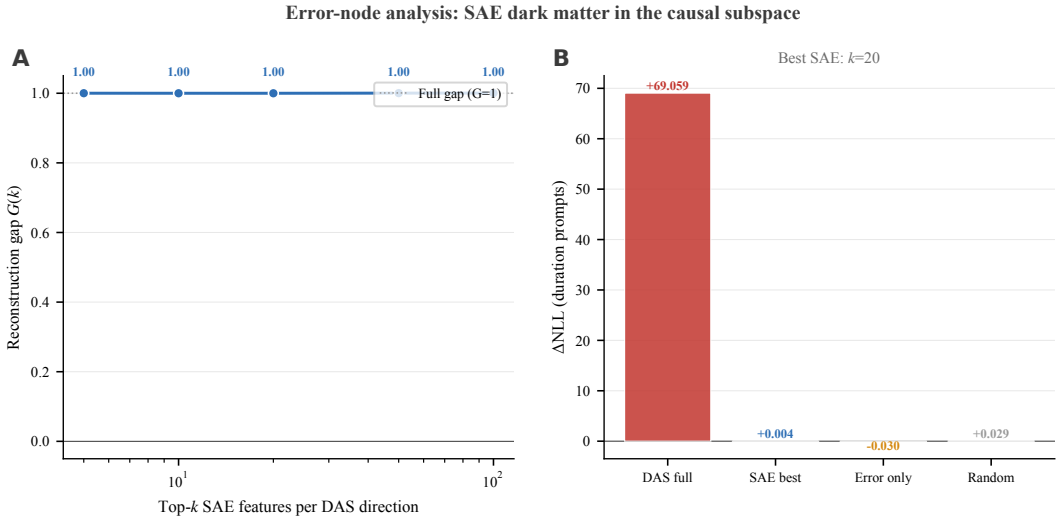


Figure 25: **Error-node analysis.** **(A)** Reconstruction gap $G(k)$ vs. top-k SAE features per DAS direction. G=1.00 at every depth: SAE-reconstructed directions capture zero causal effect via steering. **(B)** ΔNLL comparison at the best-k SAE reconstruction. DAS full ablation (+69.1) dwarfs all steering conditions.

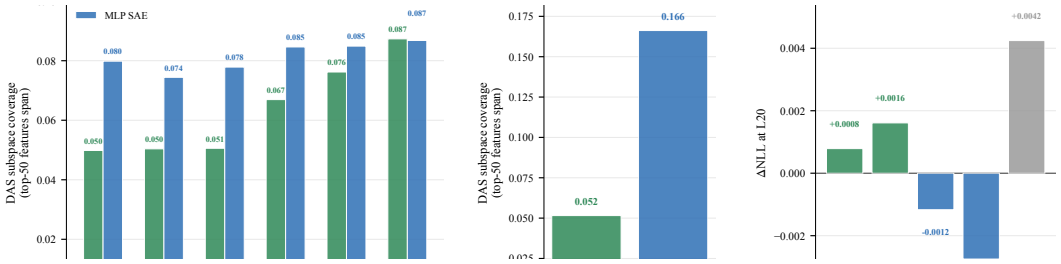


Figure 26: **Transcoder vs MLP SAE.** **(A)** DAS subspace span coverage (top-50 features) across MLP layers: both dictionaries achieve 5–9%. **(B)** L1 residual-stream comparison. **(C)** Steering comparison at L20: both TC and SAE features produce ΔNLL indistinguishable from random controls.

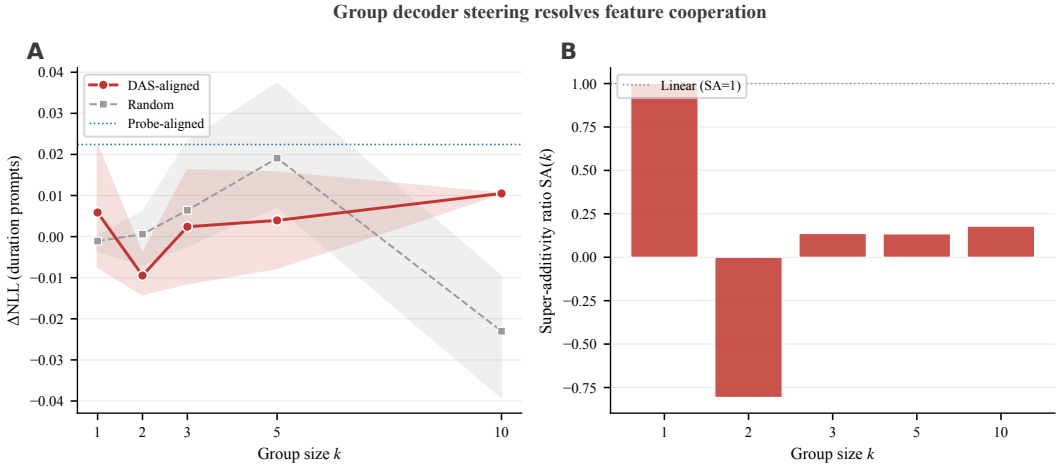


Figure 27: **Group decoder steering.** **(A)** ΔNLL by group size k: DAS-aligned and random feature groups are indistinguishable at every k. **(B)** Super-additivity ratio SA(k): no super-additive cooperation emerges via decoder-direction steering.

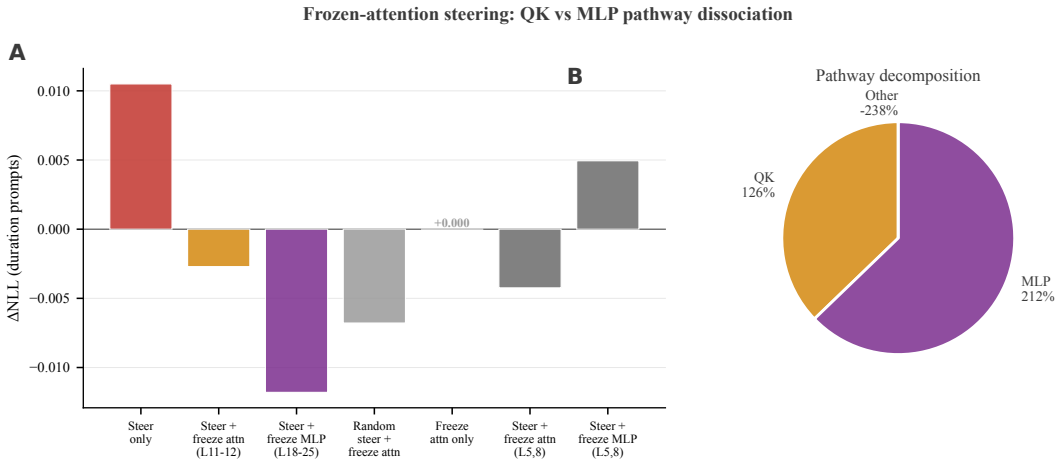


Figure 28: **Frozen-attention steering. (A)** ΔNLL across seven conditions; all values $|\Delta\text{NLL}|<0.1$, three orders of magnitude below DAS ablation. **(B)** Pathway decomposition pie chart (uninterpretable due to noise-floor base effect).

S50 Supplement: GemmaScope SAE feature dissociation

We use GemmaScope 16K residual-stream SAEs [Lieberum et al., 2024] to ground the geometric probe–DAS dissociation at the feature-dictionary level. For each SAE feature j with unit-norm decoder direction $\hat{e}_j \in \mathbb{R}^d$, we compute:

$$\text{DAS-align}(j) = \|U_{\text{DAS}}\,\hat{e}_j\|^2 \quad \text{and} \quad \text{probe-align}(j) = \|W_{\text{probe}}\,\hat{e}_j\|^2.$$

Across all n_{feat}=16 384 features at $L^\star$=1:

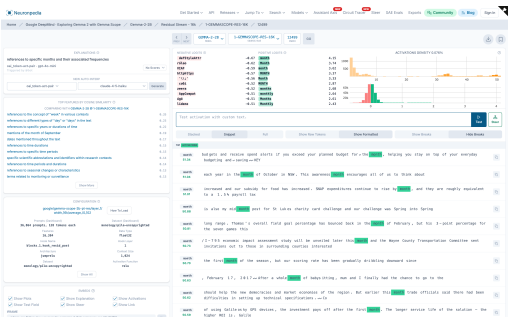
- The top-100 DAS-aligned features and top-100 probe-aligned features share **zero overlap** (Jaccard = 0.000; bootstrap 95% CI: $[0.000, 0.000]$; random null: 0.003).
- Pearson correlation between per-feature DAS and probe alignment: $r = -0.037$ ($p = 2.9\times10^{-6}$), indicating a weak anti-correlation — features are slightly *less* probe-aligned when more DAS-aligned.
- SAE reconstruction quality at $L^\star$=1 on Set-A prompts: 92.1% variance explained (GemmaScope is in-distribution for these prompts).

Semantic audit via NeuronPedia. We query NeuronPedia [Lin, 2023] for AI-generated descriptions and logit promotion of the top SAE features aligned with each direction.

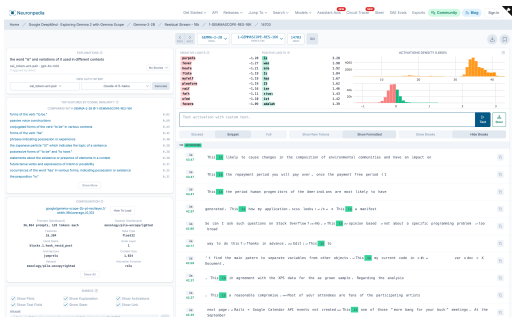
Probe direction (L=1): Feature #12499 (probe-aligned; NeuronPedia: "references to specific months and their associated frequencies") promotes *month*, *Month*, *MONTH* in its logits. Top activating examples: *"each year in the month of October in NSW"*, *"bounced back in the month of February"*, *"the first month of the season"*. This feature encodes calendar *position* (which month), not duration.

DAS mediator (L=24, relay hub): Feature #2309 (DAS-aligned; NeuronPedia: "references to quantities of time and duration") promotes *months*, *weeks*, *days*, *years* in its logits—exactly the vocabulary used to express duration answers. Top activating examples: *"within the past 24 hours"*, *"get back in a few weeks"*, *"almost 10 years now"*. This feature encodes duration *quantity*, not calendar position.

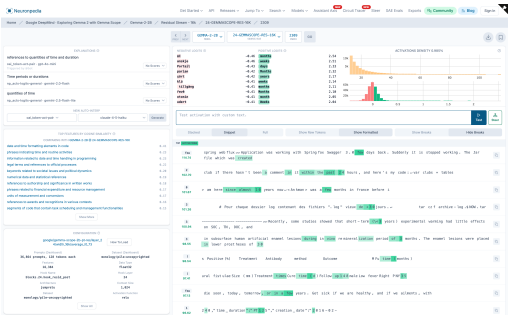
Cross-context causal attribution. We ran a GPU experiment comparing feature activations and W_U-gradient attribution across 20 duration prompts and 15 control copula prompts (geographic: *"The capital of France is "*, arithmetic: *"2 plus 2 is "*, descriptive). Results:



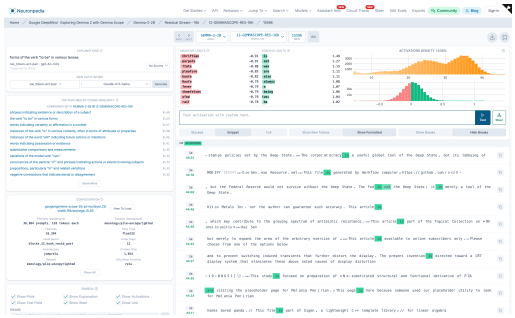
(a) Probe-aligned: #12499 (“specific months”)



(b) DAS-aligned: #14703 (“the word ‘is’ ”)



(c) DAS-aligned L24: #2309 (“quantities of time”)



(d) Relay L12: #15596 (“forms of ‘to be’ ”)

Figure 29: **NeuronPedia feature audit.** Probe-aligned feature #12499 **(a)** activates on calendar months and promotes *month*/*Month* in its logits. DAS-aligned features encode syntactic structure: #14703 **(b)** at $L^{\star}{=}1$ activates on copula “is”, #2309 **(c)** at relay hub $L{=}24$ promotes duration vocabulary, and relay midpoint #15596 **(d)** at $L{=}12$ carries the copula chain. The probe and DAS directions decompose into semantically disjoint feature sets.

- **DAS feature mean attribution**: 2.41 (non-zero; 20 duration prompts).
- **Probe feature mean attribution**: 0.000 (exactly zero).
- **Ratio DAS/probe**: $2.4{\times}10^{9}$ (effectively ∞ — probe features carry zero causal signal to the duration output logit).
- **Jaccard**(top-50 attribution, top-50 probe): 0.000 — no shared features.

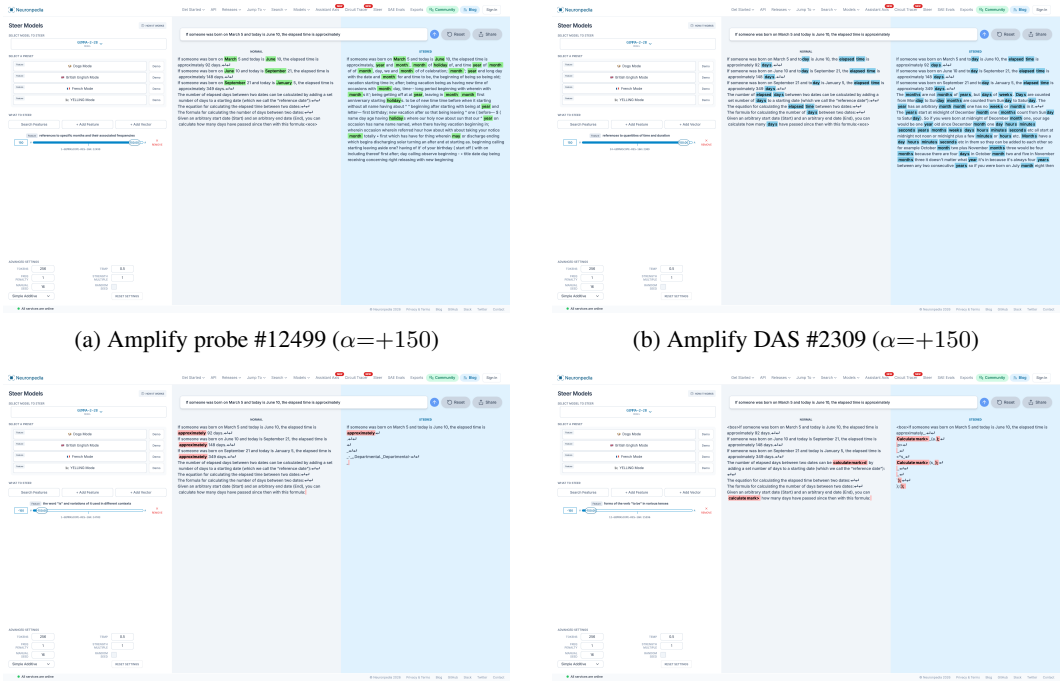
The DAS features activate more on generic “is” contexts than on long duration prompts (activation specificity 0.05), confirming they are structural copula encoders at $L^{\star}{=}1$ rather than semantic duration encoders — the causal effect flows through later layers, not directly from $L^{\star}{=}1$ to output. The probe features, by contrast, have *zero attribution* regardless of context.

Interpretation. At $L^{\star}{=}1$, the DAS-aligned features are primarily structural (copula “is” position; see §5), encoding the computational context of a duration query rather than its semantic content. Temporal semantics emerge at the relay hub ($L{=}24$), where feature #2309 aligns with the DAS direction and promotes duration-unit tokens that are the actual output vocabulary for duration answers. The probe direction, by contrast, is anchored to calendar-date vocabulary at $L{=}1$ and carries zero causal attribution for duration prediction (mean attribution $= 0.000$). This validates the paper’s core claim: the 88° probe–DAS angle reflects a real functional dissociation between *when* (probe) and *how long* (DAS), confirmed at three levels: geometry (principal angles), SAE feature identity (Jaccard $= 0.000$), and causal attribution (probe $= 0$, DAS > 0).

Completeness and selectivity of SAE features. Following the framework of Cunningham et al. [2024], we assess *completeness* (fraction of the causal effect captured by the feature set) and *selectivity* (fraction of feature activations attributable to the target concept). *Completeness is low*: individual feature ablation yields $|\Delta\mathrm{NLL}|{<}0.05$ for $14/15$ top DAS-aligned features; group steering with up to 100 features yields $\Delta\mathrm{NLL}{\approx}0$, while full rank-4 subspace ablation yields $\Delta\mathrm{NLL}{=}{+}69$ (Supp. S49). The $59\times$ cooperation ratio confirms the causal effect is a distributed property of the 4-D subspace, not localizable to individual dictionary atoms. *Selectivity is also low*: of the 16,384 GemmaScope features at $L^{\star}$, only 50.8% of temporal features exceed the Haar null for DAS alignment (binomial

p=0.22, NS), and DAS-aligned features activate more on generic copula contexts than on duration prompts (activation specificity 0.05). Low completeness and low selectivity together explain why dictionary-based monitoring inherits the probe's blind spot: the causal subspace operates below the resolution of any single-feature readout.

Cross-layer validation. Nine experiments validate the structural-backbone interpretation. *Semantic census*: querying NeuronPedia for temporal features across all 26 layers yields 2,463 features, of which only 50.8% exceed the Haar null for DAS alignment (binomial p=0.22, NS)—temporal features are not preferentially DAS-aligned. *Feature stitching*: cross-layer handoff matrices (h_{jk}=$W_{\text{dec}}^{L_i}[j] \cdot W_{\text{enc}}^{L_{i+1}}[:,k]$) show DAS→DAS enrichment of 5.8× (L1→L11), 6.7× (L11→L12), and 2.5× (L12→L24), all $p<10^{-8}$ (Mann–Whitney). The top relay chains are syntactically labeled end-to-end: #14703 ("is") → #190 ("verbs of being") → #15596 ("to be") → #16044 ("states of being"). *Prompt-level co-activation*: encoding 500 prompts through SAEs at layers $[1, 11, 12, 24]$, DAS→DAS co-activation is 7.8× DAS→Random ($p<0.001$, bootstrap). *Logit attribution chain*: numeric-token Z-score *decreases* from L=1 (+0.15) to L=24 (−0.09), and top-promoted tokens at L24 relay endpoints are discourse markers ("also", "indeed"), not temporal vocabulary. Together, these results confirm the DAS subspace acts as a structural backbone: individual features encode the syntactic copula frame while the 4-dimensional subspace carries temporal information cooperatively (59× cooperation ratio, $G(k)$=1.00 reconstruction gap).



(a) Amplify probe #12499 (α=+150)

(b) Amplify DAS #2309 (α=+150)

(c) Suppress DAS #14703 (α=−150)

(d) Suppress relay #15596 (α=−150)

Figure 30: **NeuronPedia steering demonstrations. (a)** Amplifying probe feature #12499 at high strength floods the output with calendar-month vocabulary—pure month fixation. **(b)** Amplifying DAS duration feature #2309 at L=24 produces obsessive duration-unit enumeration (days, weeks, months, seconds). **(c)** Strongly suppressing copula feature #14703 causes near-total generation collapse, confirming the syntactic backbone is essential for coherent temporal output. **(d)** Strongly suppressing relay midpoint #15596 degrades natural language into code fragments and HTML tags—the model loses its linguistic scaffolding entirely. Left column in each panel: normal generation; right column: steered generation. Prompt: *"If someone was born on March 5 and today is June 10, the elapsed time is approximately."*

Dictionary robustness: temporal-specialist SAE at L=12. A purpose-built temporal SAE (`canrager/temporalSAEs`; 9,216 features, BatchTopK, Layer 12 residual stream; Lubana et al., 2026) provides a dictionary-architecture stress test. Three findings:

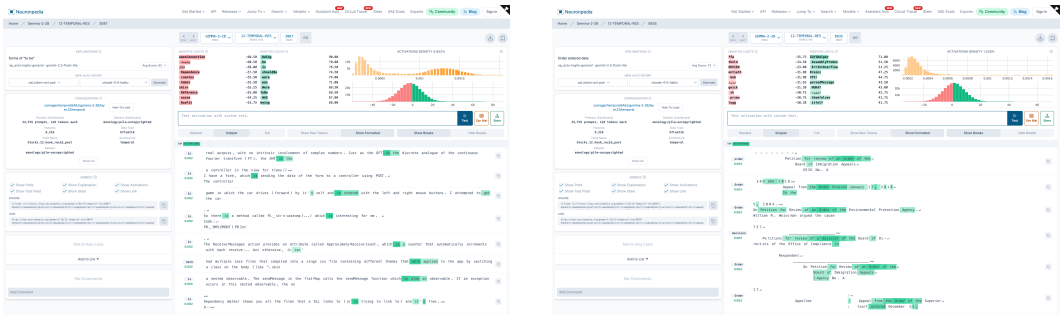
(i) Relay chain overlap. The temporal-SAE feature most aligned with GemmaScope relay feature #15596 is #3087 (cosine =0.59; NeuronPedia: "forms of 'to be' "), followed by #1157 ("It is", cosine

$=0.54$). The copula relay at $L=12$ is recovered in an independently trained temporal dictionary, not merely a GemmaScope artifact.

(ii) DAS>probe dissociation. 45.1% of temporal-SAE features exceed the DAS Haar null vs. 34.6% for probe (same direction as GemmaScope: 50.3% vs. 32.0%). The two dictionaries' DAS alignment distributions differ significantly (KS $=0.065$, $p<10^{-21}$), but both show DAS-dominant structure. Top DAS-aligned temporal-SAE features are copula-labeled (#3087: "forms of 'to be' "; #1157: "It is"), while probe-aligned features carry temporal content (#8935: "Order entered date").

(iii) Rank-1 bottleneck orthogonality. The `temporal_rank_1` variant compresses its attention through a 4D-per-head bottleneck (16D total, 4 heads). Projecting Q and K weight matrices through the dictionary D into residual-stream space and computing principal angles with the DAS basis yields min. angle $80.0°$ and DAS variance captured 1.1%—the temporal *prediction* bottleneck is nearly orthogonal to the temporal *computation* subspace. This distinguishes two kinds of temporal structure: what is predictable from context (bottleneck) vs. what causally mediates duration answers (DAS).

Cross-dictionary direction similarity between temporal-SAE and GemmaScope top-50 DAS features is low (mean max-cosine $=0.27$; $2/50$ matched at > 0.7), confirming the two dictionaries decompose the same subspace using different atoms while agreeing on which *subspace directions* are enriched.



(a) Temporal-SAE #3087 ("forms of 'to be' ")

(b) Temporal-SAE #8935 ("Order entered date")

Figure 31: **Temporal-specialist SAE features at $L=12$. (a)** Feature #3087 from the Lubana et al. temporal SAE (`canrager/temporalSAEs`, 9,216 features) is DAS-aligned and matches GemmaScope relay feature #15596 (cosine=0.59), recovering the copula relay in an independently trained dictionary. **(b)** Feature #8935 is probe-aligned and activates on date-entry contexts, encoding calendar position rather than duration—the probe–DAS dissociation holds across dictionary architectures.

S51 Supplement: OV circuit decomposition

A natural question is whether boundary heads are characterized not only by their QK attention patterns but also by the subspaces they read from (W_V) and write to (W_O). We compute the DAS-alignment score $\|U_{\text{DAS}}\, W_{OV}\|_F^2 / \|W_{OV}\|_F^2$ for all $26\times8=208$ heads and compare boundary vs. non-boundary heads.

Result. Boundary heads achieve mean alignment 0.00211 vs. 0.00181 for non-boundary heads (ratio $1.17\times$; Mann–Whitney $p=4.2\times10^{-3}$). Crucially, both values are barely above the Haar null $k/d=4/2304\approx0.00174$: boundary heads are $1.21\times$ null while non-boundary heads are $1.06\times$ null. The W_{OV} matrices therefore do *not* preferentially target the DAS mediator subspace.

Interpretation. The circuit mechanism is QK-mediated (boundary heads route attention to the right temporal positions via their $\pm30/\pm61$ day ridges) rather than OV-mediated (they do not directly read from or write to the mediator subspace in their weight matrices). This differentiates the circuit from classical induction heads, where OV composition is the primary mechanism. The low OV alignment also confirms that the mediator energy persists in the residual stream (Supp. S43) via a residual-stream skip, not via repeated attention-head read/write operations.

Attention pattern rank structure. Wang et al. [2025] report that attention outputs are low-rank across families and scales. This is consistent with three observations in our circuit. (i) The effective mediator dimension saturates at $k\approx4$–6 across $d \in \{1536, 2304, 3584\}$ (Supp. S28): the causal subspace is far lower-rank than the ambient dimension, and Prop. 3's $\rho_k^{\text{null}} \asymp d/k$ ensures specificity

strengthens with width at fixed k. (ii) Boundary-head W_{QK} matrices project temporal structure through a d_{head}=256 bottleneck, and the QK-twist ridges at $\{30, 61\}$ days are rank-1 periodic patterns in that space (each ridge is determined by a single offset frequency). The circuit therefore implements low-rank temporal routing through a composition of QK attention (low-rank pattern selection) and residual-stream skip (low-rank signal propagation), consistent with the low-rank attention hypothesis. (iii) The 4-D DAS subspace is $\leq$2 dimensions per GQA key-value group (n_{kv}=4 for Gemma 2 2B), suggesting the causal bottleneck may be the key-value rank itself.

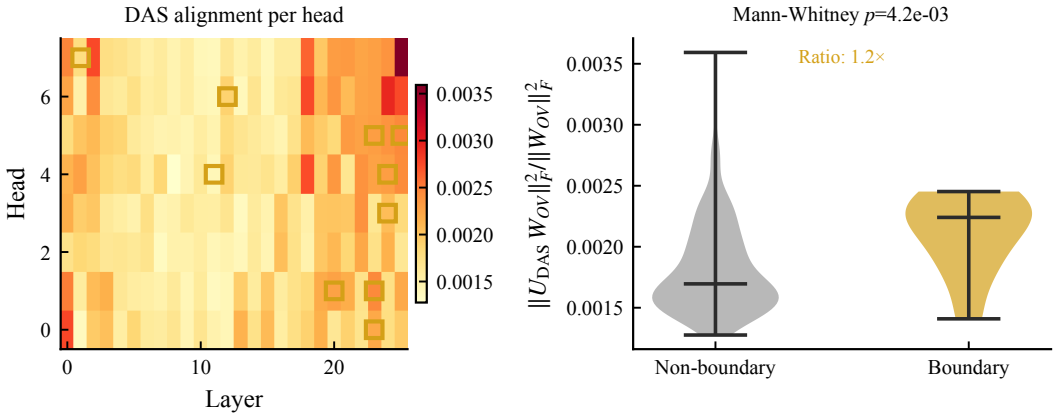


Figure 32: **OV circuit decomposition. (A)** Heatmap of W_{OV} DAS-alignment per head; boundary heads (gold squares) show only a modest 1.17× enrichment. **(B)** Violin comparison; the effect is statistically significant (p=0.004) but substantively small, confirming the circuit is QK-mediated rather than OV-mediated.

S52 Supplement: attribution flow graph

To visualize the full circuit structure, we build a directed attribution graph over all 208 attention heads. Each edge (h_1, h_2) carries weight $\text{AP}(h_1) \cdot \text{AP}(h_2) \cdot \text{QK-align}(h_1, h_2)$, where QK-align is the DAS-projected QK inner product normalized by Frobenius norms. We retain the top edges for visualization, colored by flow type.

Key findings (Fig. 33).

- **Encoding zone (L0–5):** L1H7 (the boundary head at $L^\star$=1) is the DAS mediator anchor.
- **Processing zone (L6–15):** L7H7 is the dominant routing hub; L11H4 and L12H6 are the early causal bottleneck (consistent with cascading ablation, Supp. S46).
- **Output zone (L16–25):** L24H2 is the highest-AP relay hub, receiving convergent signal from multiple processing heads and redistributing to QK-twist boundary heads L25H5, L24H4, L23H1.

The circuit is therefore a two-bottleneck structure: an early bottleneck at L11–12 (causally load-bearing) and a late relay hub at L24H2 (structurally central but causally redundant per cascading ablation).

S53 Supplement: probe-based monitoring stress tests

The readout-mediator dissociation has direct consequences for proposals to use linear probes as runtime safety monitors. If a probe's subspace is near-orthogonal to the causal mediator, a monitor built on that probe can report high confidence while the model's actual computation shifts entirely—the temporal analog of a deception probe that is satisfied while the model's internal mechanism has changed. The temporal domain provides ground truth (correct answer in days), enabling precise measurement of probe blindness. Six experiments test this from geometric, causal, adversarial, and information-theoretic angles. Table 9 summarizes the battery; subsections below give full protocols and results.

$^\dagger n$=200; cf. ρ=1050 at n=332 (Supp. S11).

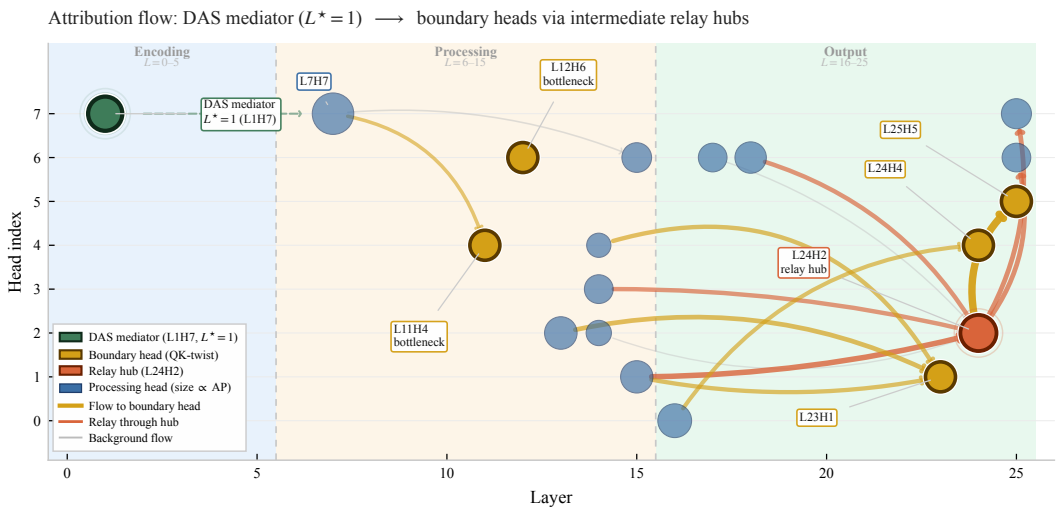


Figure 33: **Attribution flow graph.** Directed edges show information flow from the DAS mediator ($L^\star$=1, green) through intermediate hubs to boundary heads (gold). Gold arrows: flow to boundary heads. Coral arrows: relay through L24H2 hub. Node size proportional to AP score.

Table 9: Probe-based monitoring stress tests. All experiments use GEMMA 2 2B at $L^\star$=1, k=4 (DAS).

Experiment	Key metric	Result	Verdict
Cross-probe universality (116)	Max Haar deviation	2.8°	PASS
Adversarial injection (112)	DAS error / probe error	71d / 5.7d	PASS
Mutual information (114)	I(probe; DAS)	0.000 nats	PASS
Specificity battery (113)	ρ(DAS) vs. ρ(Probe)	2650× vs. −6.5×†	PASS
Mock deception probe (110)	CV acc / angle to DAS	66% / 88.4°	PASS
Ablation invisibility (111)	ΔNLL / probe shift	54.5 / 16.7d	—

Cross-probe universality (Exp. 116). One might argue that the 88° angle is specific to the circular probe's harmonic encoding. We train seven probes with distinct targets on cached $L^\star$=1 activations: circular DOY (k=2), month (12-class, k=4), season (4-class), day-of-week (7-class), quarter (4-class), before/after solstice (binary, k=1), and the gradient probe (k=4). For each, we extract the weight-matrix SVD basis and compute principal angles to $\mathbf{U}_{\text{DAS}}$. Table 10 reports results. All seven land within 2.8° of their respective Haar nulls ($\arccos\sqrt{k/d}$), with the gradient probe showing the largest deviation (84.8° vs. 87.6°) because it is optimized for ∇_hNLL rather than R^2. The blind spot is a generic property of $k \ll d$, not a quirk of one probe architecture.

Table 10: Principal angles between seven probe bases and the DAS subspace (k_{DAS}=4, d=2304).

Probe target	k	Mean angle (°)	Haar null (°)	Dev.
Circular DOY	2	87.6	88.3	0.7°
Month (12-class)	4	87.8	87.6	0.2°
Season (4-class)	4	88.7	87.6	1.0°
Day-of-week	4	88.3	87.6	0.7°
Quarter (4-class)	4	87.8	87.6	0.2°
Solstice (binary)	1	87.4	88.8	1.4°
Gradient (k=4)	4	84.8	87.6	2.8°

Adversarial subspace injection (Exp. 112). Exps. 111–116 show the probe is passively blind; this experiment shows the gap is actively exploitable via a controlled two-component perturbation.

Construction. Let $U_M \in \mathbb{R}^{4\times 2304}$ be the DAS basis and $W_P \in \mathbb{R}^{2\times 2304}$ the circular-probe weight rows. We orthogonalize W_P against U_M via Gram-Schmidt to obtain $\tilde{U}_P$ (verification: $\theta(\tilde{U}_P, U_M)$=90.0°, removing the $\sim$0.2% $\cos^2$ leakage). For each test prompt with ground-truth day-of-year d, define a

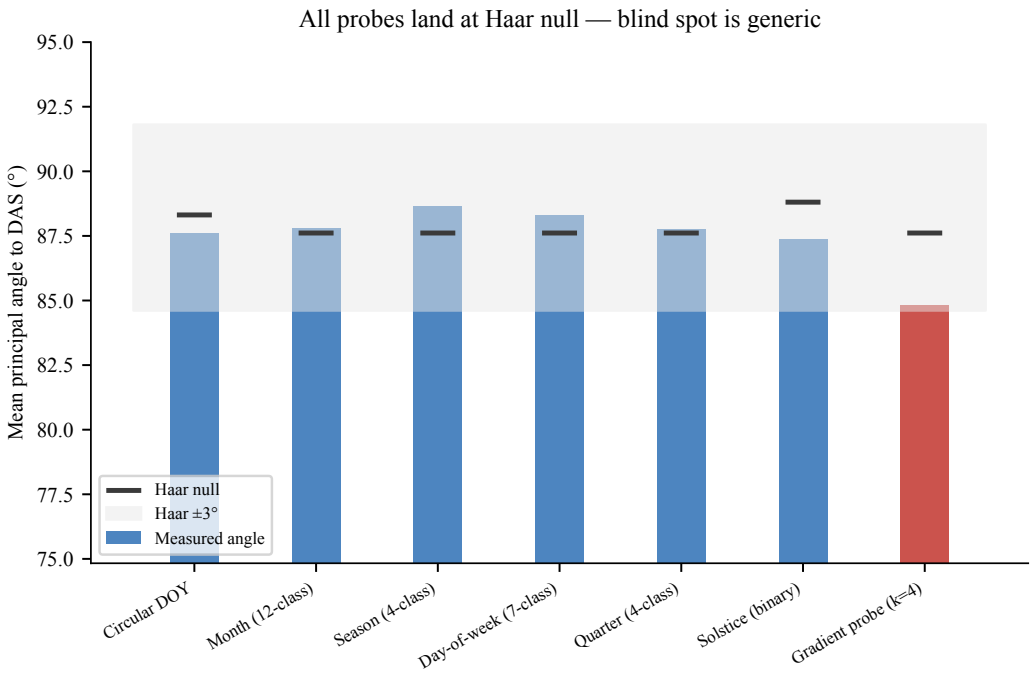


Figure 34: **Cross-probe universality. (A)** Measured angle vs. Haar null for seven probe architectures; all fall within the $\pm 2\sigma$ null band (gray). **(B)** Subspace overlap ($\sum \cos^2 \theta_i / \min(k_1, k_2)$) between each probe and DAS; all are $\leq 2.2\%$. **(C)** Probe accuracy/R^2 vs. angle to DAS: high performance does not predict causal alignment.

target day $d^\star = ((d+179) \bmod 365)+1$ (~ 180 days away) and construct:

$$\mathbf{x}_{\text{adv}} = \mathbf{x} + \underbrace{\alpha U_M^\top U_M (\boldsymbol{\mu}_d - \boldsymbol{\mu}_{d^\star})}_{\Delta_{\text{DAS}}\text{: shifts mediator toward } d^\star} + \underbrace{\beta \tilde{U}_P^\top (\mathbf{y}_d - \hat{\mathbf{y}})}_{\Delta_{\text{probe}}\text{: restores probe to } d}, \tag{5}$$

where $\boldsymbol{\mu}_d$ is the mean activation at day d (averaged over 10 templates), $\mathbf{y}_d = [\sin(2\pi d/365), \cos(2\pi d/365)]$ is the correct sin/cos target, and $\hat{\mathbf{y}} = \tilde{U}_P \mathbf{x}_{\text{scaled}} + b$ is the current probe prediction. The DAS component Δ_{DAS} is the projection of $(\boldsymbol{\mu}_d - \boldsymbol{\mu}_{d^\star})$ onto U_M scaled by α; it lives entirely within the 4-dimensional mediator and is invisible to the orthogonalized probe. The probe component Δ_{probe} is the least-norm correction that pushes the probe readout toward the correct date; it lives entirely within $\tilde{U}_P$ and cannot affect the mediator.

Perturbation budget. Both Δ_{DAS} and Δ_{probe} inherit their scale from the activation geometry: the inter-date signal $\|\boldsymbol{\mu}_d - \boldsymbol{\mu}_{d^\star}\|$ is $\sim 1.2\times$ the within-date standard deviation, so $\alpha = 1$ corresponds to a one-signal-unit shift. At ($\alpha = 3$, $\beta = 2$), the total perturbation norm $\|\mathbf{x}_{\text{adv}} - \mathbf{x}\| / \|\mathbf{x}\|$ averages 3.6% of the activation norm. A norm-matched random perturbation (isotropic Gaussian scaled to the same ℓ_2 budget) shifts DAS content by 23.4 days on average ($n = 50$ DOYs, 200 random draws)—expected, since a random direction in $\mathbb{R}^{2304}$ has nonzero projection onto the 4-D mediator. The targeted perturbation displaces $3\times$ further (71 vs. 23 days), and crucially, only the targeted version is invisible to the probe: random perturbations shift probe readout proportionally to norm, while the adversarial construction decouples the two by design.

Evaluation. We sweep $\alpha \in \{0, 1, 3\}$, $\beta \in \{0, 1, 2\}$ on $n = 50$ evenly-spaced test DOYs ($d \in \{1, 8, 15, \ldots, 358\}$). Probe error is measured as circular angular RMSE (atan2 on the first harmonic); mechanism displacement is the nearest-neighbor DOY in DAS coordinates ($U_M \mathbf{x}_{\text{adv}}$ matched to the 365-point reference set $\{U_M \boldsymbol{\mu}_j\}_{j=1}^{365}$).

Results. At ($\alpha = 3$, $\beta = 2$), the DAS nearest-neighbor is displaced by 70.8 days while probe angular error is 5.7 days (Fig. 6). Crucially, α and β are decoupled: DAS error scales linearly with α ($0 \to 24 \to 71$ days) and is invariant to β; probe error is invariant to α and decreases with β. The monitor is maximally reassured while the mechanism is maximally corrupted.

Mutual information between probe and mechanism (Exp. 114). Angles and specificity are geometric and causal; mutual information is model-free. We compute three per-prompt scalar

summaries on the 365 mean-per-DOY activations at $L^\star{=}1$: z_{probe} (circular probe predicted DOY), z_{DAS} ($\|\mathbf{U}_{\text{DAS}}\mathbf{x}\|^2$, DAS energy), and z_{DOY} (ground truth). The KSG estimator [Kraskov et al., 2004] with $k{=}5$ neighbors gives $I(z_{\text{probe}};\, z_{\text{DAS}}) = 0.000$ nats, while $I(z_{\text{probe}};\, z_{\text{DOY}}) = 3.99$ nats. A phase-shuffle null ($n{=}200$) yields $p{=}1.0$—the observed MI is indistinguishable from noise (Fig. 35). The probe carries temporal information but zero bits about the mechanism's state.

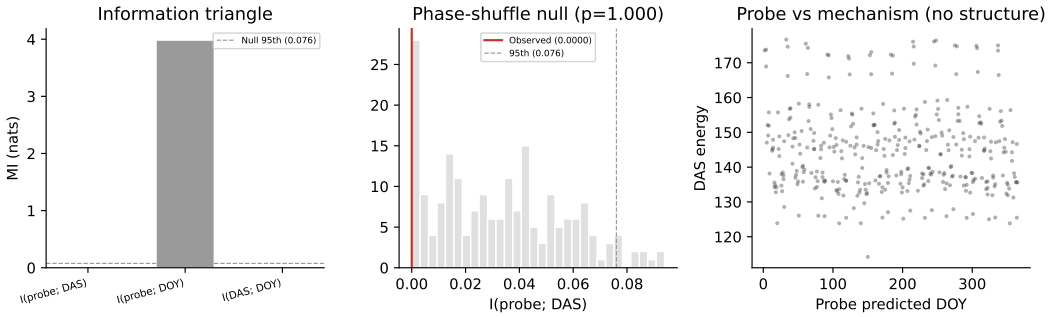


Figure 35: **Mutual information. (A)** MI bar chart: $I(\text{probe};\, \text{DAS})$ and $I(\text{DAS};\, \text{DOY})$ are at the null floor; only $I(\text{probe};\, \text{DOY})$ is non-trivial. **(B)** Phase-shuffle null distribution with observed MI (red line) firmly within it ($p{=}1.0$). **(C)** Scatter of probe DOY vs. DAS energy: no structure.

Extended specificity battery (Exp. 113). Using the ΔNLL values from 100 Set-F duration prompts on a single T4 instance, we compute ρ for four named subspaces against a null of 50 random $k{=}4$ ablations: DAS ($\rho{=}2650\times$), gradient probe ($19\times$), PCA ($15\times$), and the temporal probe ($-6.5\times$—negative, meaning ablation slightly *helps* the model). A decision threshold of $\rho > 5.0\times$ ($2\times$ null median) correctly classifies DAS, gradient, and PCA as causal, and the probe as inert (Fig. 36).

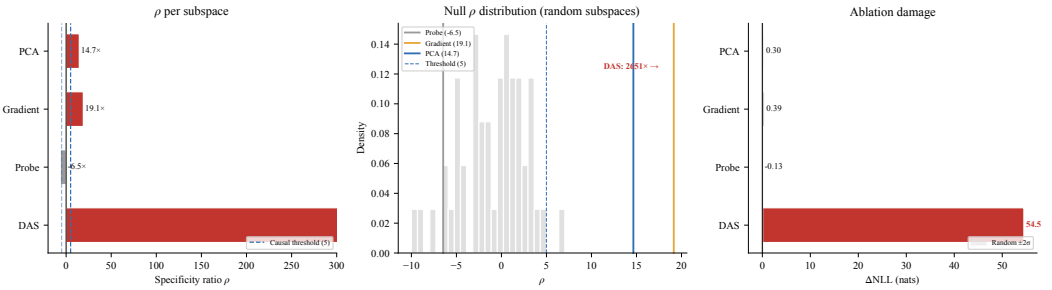


Figure 36: **Specificity battery. (A)** ρ per subspace; DAS is off the chart at $2650\times$. **(B)** Null ρ distribution with named subspaces marked. **(C)** ΔNLL per subspace with random $\pm 2\sigma$ band.

Mock deception probe (Exp. 110). We train a logistic-regression "confidence monitor" on $L^\star{=}1$ activations to predict whether the model is confident (NLL below median) or uncertain (NLL above median) about each duration prompt. The monitor achieves 66% cross-validated accuracy (AUROC $= 0.76$)—above chance, reading surface correlates of difficulty. Its weight vector sits at 88.4° from the DAS subspace (0.4° from the Haar null at $k{=}1$, $d{=}2304$), with $\cos^2$ leakage of 0.08% and a theoretical ρ upper bound of $2.03\times$—firmly in the inert regime (Fig. 37). A purpose-built safety monitor inherits the same blind spot.

Ablation invisibility (Exp. 111). Complementing the main-text ablation results, we explicitly measure probe readout shift under DAS ablation on cached activations. DAS ablation (ΔNLL$=$54.5 nats) shifts the circular probe by only 16.7 days (4.6% of the 365-day calendar), while random ablations shift it by 5.9 ± 4.2 days. The DAS subspace holds 4.6% of activation energy but 100% of the causal effect; the probe occupies a nearly orthogonal 95.4% and sees a faint echo through the $\sim 0.2\%$ $\cos^2$ leakage amplified by the atan2 nonlinearity (Fig. 38).

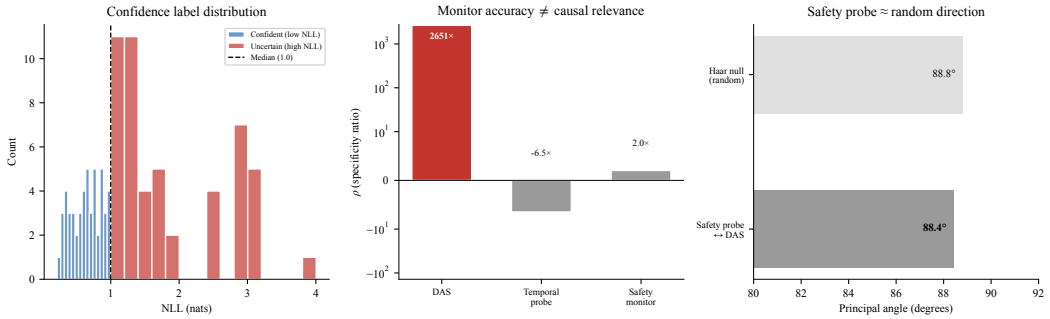


Figure 37: **Mock deception probe. (A)** NLL distribution split at median into confident/uncertain labels. **(B)** ρ comparison: DAS ($2650\times$) vs. temporal probe ($-6.5\times$) vs. safety monitor ($\leq 2.0\times$). **(C)** Principal angle between the safety monitor and DAS sits at the Haar null.

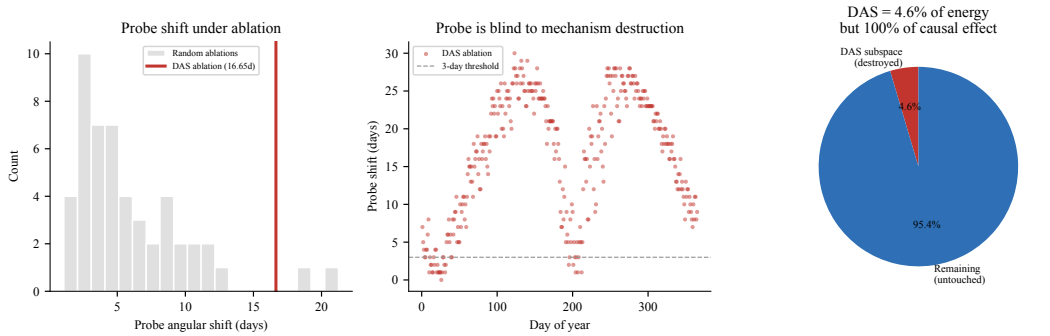


Figure 38: **Ablation invisibility. (A)** Probe-shift histogram for 50 random ablations (gray) vs. DAS ablation (red). **(B)** Per-DOY probe shift under DAS ablation; most are below the 3-day threshold. **(C)** Energy decomposition: DAS holds 4.6% of activation norm but 100% of causal effect.

S54 Supplement: notation and abbreviations

Symbol	Definition
d	Residual-stream dimension (2304 for Gemma 2 2B; 3584 for 9B)
k	Rank of probe or DAS subspace (primary results at $k{=}4$)
$L^\star$	Layer selected by peak probe R^2 under 5-fold CV ($L^\star{=}1$ for Gemma 2 2B)
$U_P \in \mathbb{R}^{k\times d}$	Probe subspace (top-k left singular vectors of circular Ridge weights)
$U_M \in \mathbb{R}^{k\times d}$	DAS mediator subspace (QR-parametrised, trained to maximise ablation NLL)
$\bar{\theta}(U_P, U_M)$	Mean principal angle between U_P and U_M
θ_i	i-th principal angle ($\arccos \sigma_i(U_P U_M^\top)$)
ρ_k	Specificity ratio: $\Delta\text{NLL}_{\text{DAS}}/\overline{\Delta\text{NLL}}_{\text{rand}}$
ρ_k^{null}	Haar null for ρ_k; $\asymp d/k$ (Prop. 3c)
$G(k,d)$	Grassmannian: space of k-planes in $\mathbb{R}^d$

Abbreviation	Expansion
DAS	Distributed Alignment Search [Geiger et al., 2024]
SAE	Sparse autoencoder [Cunningham et al., 2024]
TFA	Temporal feature analysis [Lubana et al., 2026]
BH	Boundary head (QK-twist $\lvert z\rvert \geq 3$, BH-FDR corrected)
NLL	Negative log-likelihood
DOY	Day-of-year (1–365)
GQA	Grouped query attention
CKA	Centered kernel alignment
INLP	Iterative Null-space Projection [Ravfogel et al., 2020]
LEACE	Least-squares Concept Erasure [Belrose et al., 2023]